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Surgical robot platform

Abstract

A medical robot system, including a robot coupled to an effectuator element with the robot configured for controlled movement and positioning. The system may include a transmitter configured to emit one or more signals, and the transmitter is coupled to an instrument coupled to the effectuator element. The system may further include a motor assembly coupled to the robot and a plurality of receivers configured to receive the one or more signals emitted by the transmitter. A control unit is coupled to the motor assembly and the plurality of receivers, and the control unit is configured to supply one or more instruction signals to the motor assembly. The instruction signals can be configured to cause the motor assembly to selectively move the effectuator element and is further configured to (i) calculate a position of the at least one transmitter by analysis of the signals received by the plurality of receivers; (ii) display the position of the at least one transmitter with respect to the body of the patient; and (iii) selectively control actuation of the motor assembly in response to the signals received by the plurality of receivers.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4150293	12/1978	Franke	N/A	N/A
5246010	12/1992	Gazzara et al.	N/A	N/A
5354314	12/1993	Hardy et al.	N/A	N/A
5397323	12/1994	Taylor et al.	N/A	N/A
5598453	12/1996	Baba et al.	N/A	N/A
5772594	12/1997	Barrick	N/A	N/A
5791908	12/1997	Gillio	N/A	N/A
5820559	12/1997	Ng et al.	N/A	N/A

5825982	12/1997	Wright et al.	N/A	N/A
5887121	12/1998	Funda et al.	N/A	N/A
5911449	12/1998	Daniele et al.	N/A	N/A
5951475	12/1998	Gueziec et al.	N/A	N/A
5987960	12/1998	Messner et al.	N/A	N/A
6012216	12/1999	Esteves et al.	N/A	N/A
6031888	12/1999	Ivan et al.	N/A	N/A
6033415	12/1999	Mittelstadt et al.	N/A	N/A
6080181	12/1999	Jensen et al.	N/A	N/A
6106511	12/1999	Jensen	N/A	N/A
6122541	12/1999	Cosman et al.	N/A	N/A
6144875	12/1999	Schweikard et al.	N/A	N/A
6157853	12/1999	Blume et al.	N/A	N/A
6167145	12/1999	Foley et al.	N/A	N/A
6167292	12/1999	Badano et al.	N/A	N/A
6201984	12/2000	Funda et al.	N/A	N/A
6203196	12/2000	Meyer et al.	N/A	N/A
6205411	12/2000	DiGioia, III et al.	N/A	N/A
6212419	12/2000	Blume et al.	N/A	N/A
6231565	12/2000	Tovey et al.	N/A	N/A
6236875	12/2000	Bucholz et al.	N/A	N/A
6246900	12/2000	Cosman et al.	N/A	N/A
6276471	12/2000	Kratzenberg et al.	N/A	N/A
6301495	12/2000	Gueziec et al.	N/A	N/A
6306126	12/2000	Montezuma	N/A	N/A
6312435	12/2000	Wallace et al.	N/A	N/A
6314311	12/2000	Williams et al.	N/A	N/A
6320929	12/2000	Von Der Haar	N/A	N/A
6322567	12/2000	Mittelstadt et al.	N/A	N/A
6325808	12/2000	Bernard et al.	N/A	N/A
6340363	12/2001	Bolger et al.	N/A	N/A
6377011	12/2001	Ben-Ur	N/A	N/A
6379302	12/2001	Kessman et al.	N/A	N/A
6402762	12/2001	Hunter et al.	N/A	N/A
6424885	12/2001	Niemeyer et al.	N/A	N/A
6447503	12/2001	Wynne et al.	N/A	N/A
6451027	12/2001	Cooper et al.	N/A	N/A
6477400	12/2001	Barrick	N/A	N/A
6484049	12/2001	Seeley et al.	N/A	N/A
6487267	12/2001	Wolter	N/A	N/A
6490467	12/2001	Bucholz et al.	N/A	N/A
6490475	12/2001	Seeley et al.	N/A	N/A
6499488	12/2001	Hunter et al.	N/A	N/A
6501981	12/2001	Schweikard et al.	N/A	N/A
6507751	12/2002	Blume et al.	N/A	N/A
6535756	12/2002	Simon et al.	N/A	N/A
6560354	12/2002	Maurer, Jr. et al.	N/A	N/A
6565554	12/2002	Niemeyer	N/A	N/A
6587750	12/2002	Gerbi et al.	N/A	N/A
6614453	12/2002	Suri et al.	N/A	N/A
6614871	12/2002	Kobiki et al.	N/A	N/A
6619840	12/2002	Rasche et al.	N/A	N/A
6636757	12/2002	Jascob et al.	N/A	N/A
6645196	12/2002	Nixon et al.	N/A	N/A
6666579	12/2002	Jensen	N/A	N/A
6669635	12/2002	Kessman et al.	N/A	N/A
6701173	12/2003	Nowinski et al.	N/A	N/A

6757068	12/2003	Foxlin	N/A	N/A
6782287	12/2003	Grzeszczuk et al.	N/A	N/A
6783524	12/2003	Anderson et al.	N/A	N/A
6786896	12/2003	Madhani et al.	N/A	N/A
6788018	12/2003	Blumenkranz	N/A	N/A
6804581	12/2003	Wang et al.	N/A	N/A
6823207	12/2003	Jensen et al.	N/A	N/A
6827351	12/2003	Graziani et al.	N/A	N/A
6837892	12/2004	Shoham	N/A	N/A
6839612	12/2004	Sanchez et al.	N/A	N/A
6856826	12/2004	Seeley et al.	N/A	N/A
6856827	12/2004	Seeley et al.	N/A	N/A
6879880	12/2004	Nowlin et al.	N/A	N/A
6892090	12/2004	Verard et al.	N/A	N/A
6920347	12/2004	Simon et al.	N/A	N/A
6922632	12/2004	Foxlin	N/A	N/A
6968224	12/2004	Kessman et al.	N/A	N/A
6978166	12/2004	Foley et al.	N/A	N/A
6988009	12/2005	Grimm et al.	N/A	N/A
6991627	12/2005	Madhani et al.	N/A	N/A
6996487	12/2005	Jutras et al.	N/A	N/A
6999852	12/2005	Green	N/A	N/A
7007699	12/2005	Martinelli et al.	N/A	N/A
7016457	12/2005	Senzig et al.	N/A	N/A
7043961	12/2005	Pandey et al.	N/A	N/A
7062006	12/2005	Pelc et al.	N/A	N/A
7063705	12/2005	Young et al.	N/A	N/A
7072707	12/2005	Galloway, Jr. et al.	N/A	N/A
7083615	12/2005	Peterson et al.	N/A	N/A
7097640	12/2005	Wang et al.	N/A	N/A
7099428	12/2005	Clinthorne et al.	N/A	N/A
7108421	12/2005	Gregerson et al.	N/A	N/A
7130676	12/2005	Barrick	N/A	N/A
7139418	12/2005	Abovitz et al.	N/A	N/A
7139601	12/2005	Bucholz et al.	N/A	N/A
7155316	12/2005	Sutherland et al.	N/A	N/A
7164968	12/2006	Treat et al.	N/A	N/A
7167738	12/2006	Schweikard et al.	N/A	N/A
7169141	12/2006	Brock et al.	N/A	N/A
7172627	12/2006	Fiere et al.	N/A	N/A
7194120	12/2006	Wicker et al.	N/A	N/A
7197107	12/2006	Arai et al.	N/A	N/A
7231014	12/2006	Levy	N/A	N/A
7231063	12/2006	Naimark et al.	N/A	N/A
7239940	12/2006	Wang et al.	N/A	N/A
7248914	12/2006	Hastings et al.	N/A	N/A
7301648	12/2006	Foxlin	N/A	N/A
7302288	12/2006	Schellenberg	N/A	N/A
7313430	12/2006	Urquhart et al.	N/A	N/A
7318805	12/2007	Schweikard et al.	N/A	N/A
7318827	12/2007	Leitner et al.	N/A	N/A
7319897	12/2007	Leitner et al.	N/A	N/A
7324623	12/2007	Heuscher et al.	N/A	N/A
7327865	12/2007	Fu et al.	N/A	N/A
7331967	12/2007	Lee et al.	N/A	N/A
7333642	12/2007	Green	N/A	N/A
7339341	12/2007	Oleynikov et al.	N/A	N/A

7366562	12/2007	Dukesherer et al.	N/A	N/A
7379790	12/2007	Toth et al.	N/A	N/A
7386365	12/2007	Nixon	N/A	N/A
7422592	12/2007	Morley et al.	N/A	N/A
7435216	12/2007	Kwon et al.	N/A	N/A
7440793	12/2007	Chauhan et al.	N/A	N/A
7460637	12/2007	Clinthorne et al.	N/A	N/A
7466303	12/2007	Yi et al.	N/A	N/A
7493153	12/2008	Ahmed et al.	N/A	N/A
7505617	12/2008	Fu et al.	N/A	N/A
7533892	12/2008	Schena et al.	N/A	N/A
7542791	12/2008	Mire et al.	N/A	N/A
7555331	12/2008	Viswanathan	N/A	N/A
7567834	12/2008	Clayton et al.	N/A	N/A
7594912	12/2008	Cooper et al.	N/A	N/A
7606613	12/2008	Simon et al.	N/A	N/A
7607440	12/2008	Coste-Maniere et al.	N/A	N/A
7623902	12/2008	Pacheco	N/A	N/A
7630752	12/2008	Viswanathan	N/A	N/A
7630753	12/2008	Simon et al.	N/A	N/A
7643862	12/2009	Schoenefeld	N/A	N/A
7660623	12/2009	Hunter et al.	N/A	N/A
7661881	12/2009	Gregerson et al.	N/A	N/A
7683331	12/2009	Chang	N/A	N/A
7683332	12/2009	Chang	N/A	N/A
7689320	12/2009	Prisco et al.	N/A	N/A
7691098	12/2009	Wallace et al.	N/A	N/A
7702379	12/2009	Avinash et al.	N/A	N/A
7702477	12/2009	Tuemmler et al.	N/A	N/A
7711083	12/2009	Heigl et al.	N/A	N/A
7711406	12/2009	Kuhn et al.	N/A	N/A
7720523	12/2009	Omernick et al.	N/A	N/A
7725253	12/2009	Foxlin	N/A	N/A
7726171	12/2009	Langlotz et al.	N/A	N/A
7742801	12/2009	Neubauer et al.	N/A	N/A
7751865	12/2009	Jascob et al.	N/A	N/A
7760849	12/2009	Zhang	N/A	N/A
7762825	12/2009	Burbank et al.	N/A	N/A
7763015	12/2009	Cooper et al.	N/A	N/A
7787699	12/2009	Mahesh et al.	N/A	N/A
7796728	12/2009	Bergfjord	N/A	N/A
7813838	12/2009	Sommer	N/A	N/A
7818044	12/2009	Dukesherer et al.	N/A	N/A
7819859	12/2009	Prisco et al.	N/A	N/A
7824401	12/2009	Manzo et al.	N/A	N/A
7831294	12/2009	Viswanathan	N/A	N/A
7834484	12/2009	Sartor	N/A	N/A
7835557	12/2009	Kendrick et al.	N/A	N/A
7835778	12/2009	Foley et al.	N/A	N/A
7835784	12/2009	Mire et al.	N/A	N/A
7840253	12/2009	Tremblay et al.	N/A	N/A
7840256	12/2009	Lakin et al.	N/A	N/A
7843158	12/2009	Prisco	N/A	N/A
7844320	12/2009	Shahidi	N/A	N/A
7853305	12/2009	Simon et al.	N/A	N/A
7853313	12/2009	Thompson	N/A	N/A
7865269	12/2010	Prisco et al.	N/A	N/A

D631966	12/2010	Perloff et al.	N/A	N/A
7879045	12/2010	Gielen et al.	N/A	N/A
7881767	12/2010	Strommer et al.	N/A	N/A
7881770	12/2010	Melkent et al.	N/A	N/A
7886743	12/2010	Cooper et al.	N/A	N/A
RE42194	12/2010	Foley et al.	N/A	N/A
RE42226	12/2010	Foley et al.	N/A	N/A
7900524	12/2010	Calloway et al.	N/A	N/A
7907166	12/2010	Lamprecht et al.	N/A	N/A
7909122	12/2010	Schena et al.	N/A	N/A
7925653	12/2010	Saptharishi	N/A	N/A
7930065	12/2010	Larkin et al.	N/A	N/A
7935130	12/2010	Willliams	N/A	N/A
7940999	12/2010	Liao et al.	N/A	N/A
7945012	12/2010	Ye et al.	N/A	N/A
7945021	12/2010	Shapiro et al.	N/A	N/A
7953470	12/2010	Vetter et al.	N/A	N/A
7954397	12/2010	Choi et al.	N/A	N/A
7971341	12/2010	Dukesherer et al.	N/A	N/A
7974674	12/2010	Hauck et al.	N/A	N/A
7974677	12/2010	Mire et al.	N/A	N/A
7974681	12/2010	Wallace et al.	N/A	N/A
7979157	12/2010	Anvari	N/A	N/A
7983733	12/2010	Viswanathan	N/A	N/A
7988215	12/2010	Seibold	N/A	N/A
7996110	12/2010	Lipow et al.	N/A	N/A
8004121	12/2010	Sartor	N/A	N/A
8004229	12/2010	Nowlin et al.	N/A	N/A
8010177	12/2010	Csavoy et al.	N/A	N/A
8019045	12/2010	Kato	N/A	N/A
8021310	12/2010	Sanborn et al.	N/A	N/A
8035685	12/2010	Jensen	N/A	N/A
8046054	12/2010	Kim et al.	N/A	N/A
8046057	12/2010	Clarke	N/A	N/A
8052688	12/2010	Wolf, II	N/A	N/A
8054184	12/2010	Cline et al.	N/A	N/A
8054752	12/2010	Druke et al.	N/A	N/A
8057397	12/2010	Li et al.	N/A	N/A
8057407	12/2010	Martinelli et al.	N/A	N/A
8062288	12/2010	Cooper et al.	N/A	N/A
8062375	12/2010	Glerum et al.	N/A	N/A
8066524	12/2010	Burbank et al.	N/A	N/A
8073335	12/2010	Labonville et al.	N/A	N/A
8079950	12/2010	Stern et al.	N/A	N/A
8086299	12/2010	Adler et al.	N/A	N/A
8092370	12/2011	Roberts et al.	N/A	N/A
8098914	12/2011	Liao et al.	N/A	N/A
8100950	12/2011	St. Clair et al.	N/A	N/A
8105320	12/2011	Manzo	N/A	N/A
8108025	12/2011	Csavoy et al.	N/A	N/A
8109877	12/2011	Moctezuma de la Barrera et al.	N/A	N/A
8112292	12/2011	Simon	N/A	N/A
8116430	12/2011	Shapiro et al.	N/A	N/A
8120301	12/2011	Goldberg et al.	N/A	N/A
8121249	12/2011	Wang et al.	N/A	N/A
8123675	12/2011	Funda et al.	N/A	N/A
8133229	12/2011	Bonutti	N/A	N/A

8142420	12/2011	Schena	N/A	N/A
8147494	12/2011	Leitner et al.	N/A	N/A
8150494	12/2011	Simon et al.	N/A	N/A
8150497	12/2011	Gielen et al.	N/A	N/A
8150498	12/2011	Gielen et al.	N/A	N/A
8165658	12/2011	Waynik et al.	N/A	N/A
8170313	12/2011	Kendrick et al.	N/A	N/A
8179073	12/2011	Farritor et al.	N/A	N/A
8182476	12/2011	Julian et al.	N/A	N/A
8184880	12/2011	Zhao et al.	N/A	N/A
8202278	12/2011	Orban, III et al.	N/A	N/A
8208708	12/2011	Homan et al.	N/A	N/A
8208988	12/2011	Jensen	N/A	N/A
8219177	12/2011	Smith et al.	N/A	N/A
8219178	12/2011	Smith et al.	N/A	N/A
8220468	12/2011	Cooper et al.	N/A	N/A
8224024	12/2011	Foxlin et al.	N/A	N/A
8224484	12/2011	Swarup et al.	N/A	N/A
8225798	12/2011	Baldwin et al.	N/A	N/A
8228368	12/2011	Zhao et al.	N/A	N/A
8231610	12/2011	Jo et al.	N/A	N/A
8239001	12/2011	Verard et al.	N/A	N/A
8241271	12/2011	Millman et al.	N/A	N/A
8248413	12/2011	Gattani et al.	N/A	N/A
8256319	12/2011	Cooper et al.	N/A	N/A
8263933	12/2011	Zeile	N/A	N/A
8271069	12/2011	Jascob et al.	N/A	N/A
8271130	12/2011	Hourtash	N/A	N/A
8281670	12/2011	Larkin et al.	N/A	N/A
8282653	12/2011	Nelson et al.	N/A	N/A
8301226	12/2011	Csavoy et al.	N/A	N/A
8311611	12/2011	Csavoy et al.	N/A	N/A
8320991	12/2011	Jascob et al.	N/A	N/A
8332012	12/2011	Kienzle, III	N/A	N/A
8333755	12/2011	Cooper et al.	N/A	N/A
8335552	12/2011	Stiles	N/A	N/A
8335557	12/2011	Maschke	N/A	N/A
8348931	12/2012	Cooper et al.	N/A	N/A
8353963	12/2012	Glerum	N/A	N/A
8358818	12/2012	Miga et al.	N/A	N/A
8359730	12/2012	Burg et al.	N/A	N/A
8374673	12/2012	Adcox et al.	N/A	N/A
8374723	12/2012	Zhao et al.	N/A	N/A
8379791	12/2012	Forthmann et al.	N/A	N/A
8386019	12/2012	Camus et al.	N/A	N/A
8392022	12/2012	Ortmaier et al.	N/A	N/A
8394099	12/2012	Patwardhan	N/A	N/A
8395342	12/2012	Prisco	N/A	N/A
8398634	12/2012	Manzo et al.	N/A	N/A
8400094	12/2012	Schena	N/A	N/A
8414957	12/2012	Enzerink et al.	N/A	N/A
8418073	12/2012	Mohr et al.	N/A	N/A
8450694	12/2012	Baviera et al.	N/A	N/A
8452447	12/2012	Nixon	N/A	N/A
RE44305	12/2012	Foley et al.	N/A	N/A
8462911	12/2012	Vesel et al.	N/A	N/A
8465476	12/2012	Rogers et al.	N/A	N/A

8465771	12/2012	Wan et al.	N/A	N/A
8467851	12/2012	Mire et al.	N/A	N/A
8467852	12/2012	Csavoy et al.	N/A	N/A
8469947	12/2012	Devengenzo et al.	N/A	N/A
RE44392	12/2012	Hynes	N/A	N/A
8483434	12/2012	Buehner et al.	N/A	N/A
8483800	12/2012	Jensen et al.	N/A	N/A
8486532	12/2012	Enzerink et al.	N/A	N/A
8489235	12/2012	Moll et al.	N/A	N/A
8500722	12/2012	Cooper	N/A	N/A
8500728	12/2012	Newton et al.	N/A	N/A
8504201	12/2012	Moll et al.	N/A	N/A
8506555	12/2012	Ruiz Morales	N/A	N/A
8506556	12/2012	Schena	N/A	N/A
8508173	12/2012	Goldberg et al.	N/A	N/A
8512318	12/2012	Tovey et al.	N/A	N/A
8515576	12/2012	Lipow et al.	N/A	N/A
8518120	12/2012	Glerum et al.	N/A	N/A
8521331	12/2012	Itkowitz	N/A	N/A
8526688	12/2012	Groszmann et al.	N/A	N/A
8526700	12/2012	Isaacs	N/A	N/A
8527094	12/2012	Kumar et al.	N/A	N/A
8528440	12/2012	Morley et al.	N/A	N/A
8532741	12/2012	Heruth et al.	N/A	N/A
8541970	12/2012	Nowlin et al.	N/A	N/A
8548563	12/2012	Simon et al.	N/A	N/A
8549732	12/2012	Burg et al.	N/A	N/A
8551114	12/2012	Ramos de la Pena	N/A	N/A
8551116	12/2012	Julian et al.	N/A	N/A
8556807	12/2012	Scott et al.	N/A	N/A
8556979	12/2012	Glerum et al.	N/A	N/A
8560118	12/2012	Green et al.	N/A	N/A
8561473	12/2012	Blumenkranz	N/A	N/A
8562594	12/2012	Cooper et al.	N/A	N/A
8571638	12/2012	Shoham	N/A	N/A
8571710	12/2012	Coste-Maniere et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8574303	12/2012	Sharkey et al.	N/A	N/A
8585420	12/2012	Burbank et al.	N/A	N/A
8594841	12/2012	Zhao et al.	N/A	N/A
8597198	12/2012	Sanborn et al.	N/A	N/A
8600478	12/2012	Verard et al.	N/A	N/A
8603077	12/2012	Cooper et al.	N/A	N/A
8611985	12/2012	Lavallee et al.	N/A	N/A
8613230	12/2012	Blumenkranz et al.	N/A	N/A
8621939	12/2013	Blumenkranz et al.	N/A	N/A
8624537	12/2013	Nowlin et al.	N/A	N/A
8630389	12/2013	Kato	N/A	N/A
8634897	12/2013	Simon et al.	N/A	N/A
8634957	12/2013	Toth et al.	N/A	N/A
8638056	12/2013	Goldberg et al.	N/A	N/A
8638057	12/2013	Goldberg et al.	N/A	N/A
8639000	12/2013	Zhao et al.	N/A	N/A
8641726	12/2013	Bonutti	N/A	N/A
8644907	12/2013	Hartmann et al.	N/A	N/A
8657809	12/2013	Schoepp	N/A	N/A
8660635	12/2013	Simon et al.	N/A	N/A

8666544	12/2013	Moll et al.	N/A	N/A
8675939	12/2013	Moctezuma de la Barrera	N/A	N/A
8678647	12/2013	Gregerson et al.	N/A	N/A
8679125	12/2013	Smith et al.	N/A	N/A
8679183	12/2013	Glerum et al.	N/A	N/A
8682413	12/2013	Lloyd	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8685098	12/2013	Glerum et al.	N/A	N/A
8693730	12/2013	Umasuthan et al.	N/A	N/A
8694075	12/2013	Groszmann et al.	N/A	N/A
8696458	12/2013	Foxlin et al.	N/A	N/A
8700123	12/2013	Okamura et al.	N/A	N/A
8706086	12/2013	Glerum	N/A	N/A
8706185	12/2013	Foley et al.	N/A	N/A
8706301	12/2013	Zhao et al.	N/A	N/A
8717430	12/2013	Simon et al.	N/A	N/A
8727618	12/2013	Maschke et al.	N/A	N/A
8734432	12/2013	Tuma et al.	N/A	N/A
8738115	12/2013	Amberg et al.	N/A	N/A
8738181	12/2013	Greer et al.	N/A	N/A
8740882	12/2013	Jun et al.	N/A	N/A
8746252	12/2013	McGrogan et al.	N/A	N/A
8749189	12/2013	Nowlin et al.	N/A	N/A
8749190	12/2013	Nowlin et al.	N/A	N/A
8761930	12/2013	Nixon	N/A	N/A
8764448	12/2013	Yang et al.	N/A	N/A
8771170	12/2013	Mesallum et al.	N/A	N/A
8781186	12/2013	Clements et al.	N/A	N/A
8781630	12/2013	Banks et al.	N/A	N/A
8784385	12/2013	Boyden et al.	N/A	N/A
8786241	12/2013	Nowlin et al.	N/A	N/A
8787520	12/2013	Baba	N/A	N/A
8792704	12/2013	Isaacs	N/A	N/A
8798231	12/2013	Notohara et al.	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8808164	12/2013	Hoffman et al.	N/A	N/A
8812077	12/2013	Dempsey	N/A	N/A
8814793	12/2013	Brabrand	N/A	N/A
8816628	12/2013	Nowlin et al.	N/A	N/A
8818105	12/2013	Myronenko et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8821511	12/2013	Von Jako et al.	N/A	N/A
8823308	12/2013	Nowlin et al.	N/A	N/A
8827996	12/2013	Scott et al.	N/A	N/A
8828024	12/2013	Farritor et al.	N/A	N/A
8830224	12/2013	Zhao et al.	N/A	N/A
8834489	12/2013	Cooper et al.	N/A	N/A
8834490	12/2013	Bonutti	N/A	N/A
8838270	12/2013	Druke et al.	N/A	N/A
8844789	12/2013	Shelton, IV et al.	N/A	N/A
8855822	12/2013	Bartol et al.	N/A	N/A
8858598	12/2013	Seifert et al.	N/A	N/A
8860753	12/2013	Bhandarkar et al.	N/A	N/A
8864751	12/2013	Prisco et al.	N/A	N/A
8864798	12/2013	Weiman et al.	N/A	N/A
8864833	12/2013	Glerum et al.	N/A	N/A
8867703	12/2013	Shapiro et al.	N/A	N/A

8870880	12/2013	Himmelberger et al.	N/A	N/A
8876866	12/2013	Zappacosta et al.	N/A	N/A
8880223	12/2013	Raj et al.	N/A	N/A
8882803	12/2013	Iott et al.	N/A	N/A
8883210	12/2013	Truncale et al.	N/A	N/A
8888821	12/2013	Rezach et al.	N/A	N/A
8888853	12/2013	Glerum et al.	N/A	N/A
8888854	12/2013	Glerum et al.	N/A	N/A
8894652	12/2013	Seifert et al.	N/A	N/A
8894688	12/2013	Suh	N/A	N/A
8894691	12/2013	Iott et al.	N/A	N/A
8906069	12/2013	Hansell et al.	N/A	N/A
8964934	12/2014	Ein-Gal	N/A	N/A
8992580	12/2014	Bar et al.	N/A	N/A
8996169	12/2014	Lightcap et al.	N/A	N/A
9001963	12/2014	Sowards-Emmerd et al.	N/A	N/A
9002076	12/2014	Khadem et al.	N/A	N/A
9044190	12/2014	Rubner et al.	N/A	N/A
9107683	12/2014	Hourtash et al.	N/A	N/A
9125556	12/2014	Zehavi et al.	N/A	N/A
9131986	12/2014	Greer et al.	N/A	N/A
9215968	12/2014	Schostek et al.	N/A	N/A
9308050	12/2015	Kostrzewski et al.	N/A	N/A
9380984	12/2015	Li et al.	N/A	N/A
9393039	12/2015	Lechner et al.	N/A	N/A
9398886	12/2015	Gregerson et al.	N/A	N/A
9398890	12/2015	Dong et al.	N/A	N/A
9414859	12/2015	Ballard et al.	N/A	N/A
9420975	12/2015	Gutfleisch et al.	N/A	N/A
9492235	12/2015	Hourtash et al.	N/A	N/A
9592096	12/2016	Maillet et al.	N/A	N/A
9750465	12/2016	Engel et al.	N/A	N/A
9757203	12/2016	Hourtash et al.	N/A	N/A
9795354	12/2016	Menegaz et al.	N/A	N/A
9814535	12/2016	Bar et al.	N/A	N/A
9820783	12/2016	Donner et al.	N/A	N/A
9833265	12/2016	Donner et al.	N/A	N/A
9848922	12/2016	Tohmeh et al.	N/A	N/A
9925011	12/2017	Gombert et al.	N/A	N/A
9931025	12/2017	Graetzel et al.	N/A	N/A
10034717	12/2017	Miller et al.	N/A	N/A
2001/0036302	12/2000	Miller	N/A	N/A
2002/0035321	12/2001	Bucholz et al.	N/A	N/A
2003/0055049	12/2002	Brock	N/A	N/A
2003/0153829	12/2002	Sarin et al.	N/A	N/A
2004/0024311	12/2003	Quaid	N/A	N/A
2004/0068172	12/2003	Nowinski et al.	N/A	N/A
2004/0076259	12/2003	Jensen et al.	N/A	N/A
2005/0096502	12/2004	Khalili	N/A	N/A
2005/0143651	12/2004	Verard et al.	N/A	N/A
2005/0171558	12/2004	Abovitz et al.	N/A	N/A
2005/0245820	12/2004	Sarin	N/A	N/A
2006/0100610	12/2005	Wallace et al.	N/A	N/A
2006/0173329	12/2005	Marquart et al.	N/A	N/A
2006/0184396	12/2005	Dennis et al.	N/A	N/A
2006/0241416	12/2005	Marquart et al.	N/A	N/A
2006/0291612	12/2005	Nishide et al.	N/A	N/A

2007/0015987	12/2006	Benlloch Baviera et al.	N/A	N/A
2007/0021738	12/2006	Hasser et al.	N/A	N/A
2007/0038059	12/2006	Sheffer et al.	N/A	N/A
2007/0073133	12/2006	Schoenefeld	N/A	N/A
2007/0156121	12/2006	Millman et al.	N/A	N/A
2007/0156157	12/2006	Nahum et al.	N/A	N/A
2007/0167712	12/2006	Keglovich et al.	N/A	N/A
2007/0233238	12/2006	Huynh et al.	N/A	N/A
2008/0004523	12/2007	Jensen	N/A	N/A
2008/0013809	12/2007	Zhu et al.	N/A	N/A
2008/0033283	12/2007	Dellaca et al.	N/A	N/A
2008/0046122	12/2007	Manzo et al.	N/A	N/A
2008/0082109	12/2007	Moll et al.	N/A	N/A
2008/0108912	12/2007	Node-Langlois	N/A	N/A
2008/0108991	12/2007	Von Jako	N/A	N/A
2008/0109012	12/2007	Falco et al.	N/A	N/A
2008/0144906	12/2007	Allred et al.	N/A	N/A
2008/0161680	12/2007	Von Jako et al.	N/A	N/A
2008/0161682	12/2007	Kendrick et al.	N/A	N/A
2008/0177203	12/2007	von Jako	N/A	N/A
2008/0214922	12/2007	Hartmann et al.	N/A	N/A
2008/0228068	12/2007	Viswanathan et al.	N/A	N/A
2008/0228196	12/2007	Wang et al.	N/A	N/A
2008/0235052	12/2007	Node-Langlois et al.	N/A	N/A
2008/0269596	12/2007	Revie et al.	N/A	N/A
2008/0287771	12/2007	Anderson	N/A	N/A
2008/0287781	12/2007	Revie et al.	N/A	N/A
2008/0300477	12/2007	Lloyd et al.	N/A	N/A
2008/0300478	12/2007	Zuhars et al.	N/A	N/A
2008/0302950	12/2007	Park et al.	N/A	N/A
2008/0306490	12/2007	Lakin et al.	N/A	N/A
2008/0319311	12/2007	Hamadeh	N/A	N/A
2009/0012509	12/2008	Csavoy et al.	N/A	N/A
2009/0030428	12/2008	Omori et al.	N/A	N/A
2009/0080737	12/2008	Battle et al.	N/A	N/A
2009/0185655	12/2008	Koken et al.	N/A	N/A
2009/0198121	12/2008	Hoheisel	N/A	N/A
2009/0216113	12/2008	Meier et al.	N/A	N/A
2009/0228019	12/2008	Gross et al.	N/A	N/A
2009/0259123	12/2008	Navab et al.	N/A	N/A
2009/0259230	12/2008	Khadem et al.	N/A	N/A
2009/0264899	12/2008	Appenrodt et al.	N/A	N/A
2009/0281417	12/2008	Hartmann et al.	N/A	N/A
2010/0022874	12/2009	Wang et al.	N/A	N/A
2010/0039506	12/2009	Sarvestani et al.	N/A	N/A
2010/0125286	12/2009	Wang et al.	N/A	N/A
2010/0130986	12/2009	Mailloux et al.	N/A	N/A
2010/0228117	12/2009	Hartmann	N/A	N/A
2010/0228265	12/2009	Prisco	N/A	N/A
2010/0249571	12/2009	Jensen et al.	N/A	N/A
2010/0274120	12/2009	Heuscher	N/A	N/A
2010/0280363	12/2009	Skarda et al.	N/A	N/A
2010/0331858	12/2009	Simaan et al.	N/A	N/A
2011/0022229	12/2010	Jang et al.	N/A	N/A
2011/0077504	12/2010	Fischer et al.	N/A	N/A
2011/0098553	12/2010	Robbins et al.	N/A	N/A
2011/0137152	12/2010	Li	N/A	N/A

2011/0213384	12/2010	Jeong	N/A	N/A
2011/0224684	12/2010	Larkin et al.	N/A	N/A
2011/0224685	12/2010	Larkin et al.	N/A	N/A
2011/0224686	12/2010	Larkin et al.	N/A	N/A
2011/0224687	12/2010	Larkin et al.	N/A	N/A
2011/0224688	12/2010	Larkin et al.	N/A	N/A
2011/0224689	12/2010	Larkin et al.	N/A	N/A
2011/0224825	12/2010	Larkin et al.	N/A	N/A
2011/0230967	12/2010	O'Halloran et al.	N/A	N/A
2011/0238080	12/2010	Ranjit et al.	N/A	N/A
2011/0276058	12/2010	Choi et al.	N/A	N/A
2011/0282189	12/2010	Graumann	N/A	N/A
2011/0286573	12/2010	Schretter et al.	N/A	N/A
2011/0295062	12/2010	Gratacos Solsona et al.	N/A	N/A
2011/0295370	12/2010	Suh et al.	N/A	N/A
2011/0306986	12/2010	Lee et al.	N/A	N/A
2012/0035507	12/2011	George et al.	N/A	N/A
2012/0046668	12/2011	Gantes	N/A	N/A
2012/0051498	12/2011	Koishi	N/A	N/A
2012/0053597	12/2011	Anvari et al.	N/A	N/A
2012/0059248	12/2011	Holsing et al.	N/A	N/A
2012/0071753	12/2011	Hunter et al.	N/A	N/A
2012/0108954	12/2011	Schulhauser et al.	N/A	N/A
2012/0136372	12/2011	Amat Girbau et al.	N/A	N/A
2012/0143084	12/2011	Shoham	N/A	N/A
2012/0184839	12/2011	Woerlein	N/A	N/A
2012/0197182	12/2011	Millman et al.	N/A	N/A
2012/0209290	12/2011	Selover et al.	N/A	N/A
2012/0226145	12/2011	Chang et al.	N/A	N/A
2012/0235909	12/2011	Birkenbach et al.	N/A	N/A
2012/0245596	12/2011	Meenink	N/A	N/A
2012/0253332	12/2011	Moll	N/A	N/A
2012/0253360	12/2011	White et al.	N/A	N/A
2012/0256092	12/2011	Zingerman	N/A	N/A
2012/0294498	12/2011	Popovic	N/A	N/A
2012/0296203	12/2011	Hartmann et al.	N/A	N/A
2013/0006267	12/2012	Odermatt et al.	N/A	N/A
2013/0016889	12/2012	Myronenko et al.	N/A	N/A
2013/0030571	12/2012	Ruiz Morales et al.	N/A	N/A
2013/0035583	12/2012	Park et al.	N/A	N/A
2013/0060146	12/2012	Yang et al.	N/A	N/A
2013/0060337	12/2012	Petersheim et al.	N/A	N/A
2013/0094742	12/2012	Feilkas	N/A	N/A
2013/0096574	12/2012	Kang et al.	N/A	N/A
2013/0113791	12/2012	Isaacs et al.	N/A	N/A
2013/0116706	12/2012	Lee et al.	N/A	N/A
2013/0131695	12/2012	Scarfogliero et al.	N/A	N/A
2013/0144307	12/2012	Jeong et al.	N/A	N/A
2013/0158542	12/2012	Manzo et al.	N/A	N/A
2013/0165937	12/2012	Patwardhan	N/A	N/A
2013/0178867	12/2012	Farritor et al.	N/A	N/A
2013/0178868	12/2012	Roh	N/A	N/A
2013/0178870	12/2012	Schena	N/A	N/A
2013/0204271	12/2012	Brisson et al.	N/A	N/A
2013/0211419	12/2012	Jensen	N/A	N/A
2013/0211420	12/2012	Jensen	N/A	N/A
2013/0218142	12/2012	Tuma et al.	N/A	N/A

2013/0223702	12/2012	Holsing et al.	N/A	N/A
2013/0225942	12/2012	Holsing et al.	N/A	N/A
2013/0225943	12/2012	Holsing et al.	N/A	N/A
2013/0231556	12/2012	Holsing et al.	N/A	N/A
2013/0237995	12/2012	Lee et al.	N/A	N/A
2013/0245375	12/2012	DiMaio et al.	N/A	N/A
2013/0261640	12/2012	Kim et al.	N/A	N/A
2013/0272488	12/2012	Bailey et al.	N/A	N/A
2013/0272489	12/2012	Dickman et al.	N/A	N/A
2013/0274761	12/2012	Devengenzo et al.	N/A	N/A
2013/0281821	12/2012	Liu et al.	N/A	N/A
2013/0296884	12/2012	Taylor et al.	N/A	N/A
2013/0303887	12/2012	Holsing et al.	N/A	N/A
2013/0307955	12/2012	Deitz et al.	N/A	N/A
2013/0317521	12/2012	Choi et al.	N/A	N/A
2013/0325033	12/2012	Schena et al.	N/A	N/A
2013/0325035	12/2012	Hauck et al.	N/A	N/A
2013/0331686	12/2012	Freysinger et al.	N/A	N/A
2013/0331858	12/2012	Devengenzo et al.	N/A	N/A
2013/0331861	12/2012	Yoon	N/A	N/A
2013/0342578	12/2012	Isaacs	N/A	N/A
2013/0345717	12/2012	Markvicka et al.	N/A	N/A
2013/0345718	12/2012	Crawford et al.	N/A	N/A
2013/0345757	12/2012	Stad	N/A	N/A
2014/0001235	12/2013	Shelton, IV	N/A	N/A
2014/0012131	12/2013	Heruth et al.	N/A	N/A
2014/0031664	12/2013	Kang et al.	N/A	N/A
2014/0046128	12/2013	Lee et al.	N/A	N/A
2014/0046132	12/2013	Hoeg et al.	N/A	N/A
2014/0046340	12/2013	Wilson et al.	N/A	N/A
2014/0049629	12/2013	Siewerdsen et al.	N/A	N/A
2014/0058406	12/2013	Tsekos	N/A	N/A
2014/0066944	12/2013	Taylor et al.	N/A	N/A
2014/0073914	12/2013	Lavallee et al.	N/A	N/A
2014/0080086	12/2013	Chen	N/A	N/A
2014/0081128	12/2013	Verard et al.	N/A	N/A
2014/0088612	12/2013	Bartol et al.	N/A	N/A
2014/0094694	12/2013	Moctezuma de la Barrera	N/A	N/A
2014/0094851	12/2013	Gordon	N/A	N/A
2014/0096369	12/2013	Matsumoto et al.	N/A	N/A
2014/0100587	12/2013	Farritor et al.	N/A	N/A
2014/0121676	12/2013	Kostrzewski et al.	N/A	N/A
2014/0128882	12/2013	Kwak et al.	N/A	N/A
2014/0130810	12/2013	Azizian et al.	N/A	N/A
2014/0135796	12/2013	Simon et al.	N/A	N/A
2014/0142591	12/2013	Alvarez et al.	N/A	N/A
2014/0142592	12/2013	Moon et al.	N/A	N/A
2014/0148692	12/2013	Hartmann et al.	N/A	N/A
2014/0163581	12/2013	Devengenzo et al.	N/A	N/A
2014/0171781	12/2013	Stiles	N/A	N/A
2014/0171900	12/2013	Stiles	N/A	N/A
2014/0171965	12/2013	Loh et al.	N/A	N/A
2014/0180308	12/2013	vun Grunberg	N/A	N/A
2014/0180309	12/2013	Seeber et al.	N/A	N/A
2014/0187915	12/2013	Yaroshenko et al.	N/A	N/A
2014/0188132	12/2013	Kang	N/A	N/A
2014/0194699	12/2013	Roh et al.	N/A	N/A

2014/0221819	12/2013	Sarment	N/A	N/A
2014/0222023	12/2013	Kim et al.	N/A	N/A
2014/0228631	12/2013	Kwak et al.	N/A	N/A
2014/0234804	12/2013	Huang et al.	N/A	N/A
2014/0257328	12/2013	Kim et al.	N/A	N/A
2014/0257329	12/2013	Jang et al.	N/A	N/A
2014/0257330	12/2013	Choi et al.	N/A	N/A
2014/0275760	12/2013	Lee et al.	N/A	N/A
2014/0275955	12/2013	Crawford et al.	N/A	N/A
2014/0275985	12/2013	Walker et al.	N/A	N/A
2014/0276931	12/2013	Parihar et al.	N/A	N/A
2014/0276940	12/2013	Seo	N/A	N/A
2014/0276944	12/2013	Farritor et al.	N/A	N/A
2014/0288413	12/2013	Hwang et al.	N/A	N/A
2014/0299648	12/2013	Shelton, IV et al.	N/A	N/A
2014/0303434	12/2013	Farritor et al.	N/A	N/A
2014/0303643	12/2013	Ha et al.	N/A	N/A
2014/0305995	12/2013	Shelton, IV et al.	N/A	N/A
2014/0309659	12/2013	Roh et al.	N/A	N/A
2014/0316436	12/2013	Bar et al.	N/A	N/A
2014/0323803	12/2013	Hoffman et al.	N/A	N/A
2014/0324070	12/2013	Min et al.	N/A	N/A
2014/0330288	12/2013	Date et al.	N/A	N/A
2014/0364720	12/2013	Darrow et al.	N/A	N/A
2014/0371577	12/2013	Maillet et al.	N/A	N/A
2014/0379130	12/2013	Lee et al.	N/A	N/A
2015/0039034	12/2014	Frankel et al.	N/A	N/A
2015/0085970	12/2014	Bouhnik et al.	N/A	N/A
2015/0146847	12/2014	Liu	N/A	N/A
2015/0150524	12/2014	Yorkston et al.	N/A	N/A
2015/0196261	12/2014	Funk	N/A	N/A
2015/0213633	12/2014	Chang et al.	N/A	N/A
2015/0335480	12/2014	Alvarez et al.	N/A	N/A
2015/0342647	12/2014	Frankel et al.	N/A	N/A
2016/0005194	12/2015	Schretter et al.	N/A	N/A
2016/0166329	12/2015	Langan et al.	N/A	N/A
2016/0235480	12/2015	Scholl et al.	N/A	N/A
2016/0249990	12/2015	Glozman et al.	N/A	N/A
2016/0302871	12/2015	Gregerson et al.	N/A	N/A
2016/0320322	12/2015	Suzuki	N/A	N/A
2016/0331335	12/2015	Gregerson et al.	N/A	N/A
2017/0135770	12/2016	Scholl et al.	N/A	N/A
2017/0143284	12/2016	Sehnert et al.	N/A	N/A
2017/0143426	12/2016	Isaacs et al.	N/A	N/A
2017/0156816	12/2016	Ibrahim	N/A	N/A
2017/0202629	12/2016	Maillet et al.	N/A	N/A
2017/0212723	12/2016	Atarot et al.	N/A	N/A
2017/0215825	12/2016	Johnson et al.	N/A	N/A
2017/0215826	12/2016	Johnson et al.	N/A	N/A
2017/0215827	12/2016	Johnson et al.	N/A	N/A
2017/0231710	12/2016	Scholl et al.	N/A	N/A
2017/0258426	12/2016	Risher-Kelly et al.	N/A	N/A
2017/0273748	12/2016	Hourtash et al.	N/A	N/A
2017/0296277	12/2016	Hourtash et al.	N/A	N/A
2017/0360493	12/2016	Zucher et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

2286729	12/2010	EP	N/A
898843	12/1995	JP	N/A
8313304	12/1995	JP	N/A
2008538184	12/2007	JP	N/A
02071369	12/2001	WO	N/A

OTHER PUBLICATIONS

US 8,231,638 B2, 07/2012, Swarup et al. (withdrawn) cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) This application a continuation of U.S. patent application Ser. No. 17/290,423 filed on Sep. 30, 2021 which is a continuation of is U.S. patent application Ser. No. 15/449,260, filed on Mar. 3, 2017, which is a continuation of U.S. patent application Ser. No. 13/924,505 filed on Jun. 21, 2013 which claims priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 61/662,702 filed on Jun. 21, 2012 and U.S. Provisional Patent Application No. 61/800,527 filed on Mar. 15, 2013, all of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Various medical procedures require the precise localization of a three-dimensional position of a surgical instrument within the body in order to effect optimized treatment. For example, some surgical procedures to fuse vertebrae require that a surgeon drill multiple holes into the bone structure at specific locations. To achieve high levels of mechanical integrity in the fusing system, and to balance the forces created in the bone structure, it is necessary that the holes are drilled at the correct location. Vertebrae, like most bone structures, have complex shapes made up of non-planar curved surfaces making precise and perpendicular drilling difficult. Conventionally, a surgeon manually holds and positions a drill guide tube by using a guidance system to overlay the drill tube's position onto a three dimensional image of the bone structure. This manual process is both tedious and time consuming. The success of the surgery is largely dependent upon the dexterity of the surgeon who performs it.

(2) Limited robotic assistance for surgical procedures is currently available. For example, the da Vinci® medical robot system (da Vinci® is a registered trademark of Intuitive Surgical) is a robot used in certain surgical applications. In the da Vinci® system, the user controls manipulators that control a robotic actuator. The system converts the surgeon's gross movements into micro-movements of the robotic actuator. Although the da Vinci® system eliminates hand tremor and provides the user with the ability to work through a small opening, like many of the robots commercially available today, it is expensive, obtrusive, and the setup is cumbersome. Further, for procedures such as thoracolumbar pedicle screw insertion, these conventional methods are known to be error-prone and tedious.

(3) One of the characteristics of many of the current robots used in surgical applications which make them error prone is that they use an articular arm based on a series of rotational joints. The use of an articular system may create difficulties in arriving at an accurately targeted location because the level of any error is increased over each joint in the articular system.

SUMMARY

(4) Some embodiments of the invention provide a surgical robot (and optionally an imaging system) that utilizes a Cartesian positioning system that allows movement of a surgical instrument to be individually controlled in an x-axis, y-axis and z-axis. In some embodiments, the surgical robot can include a base, a robot arm coupled to and configured for articulation relative to the base, as well as an end-effectuator coupled to a distal end of the robot arm. The effectuator element can include the surgical instrument or can be configured for operative coupling to the surgical instrument. Some embodiments of the invention allow the roll, pitch and yaw rotation of the end-effectuator and/or surgical instrument to be controlled without creating movement along the x-axis, y-axis, or z-axis.

(5) In some embodiments, the end-effectuator can include a guide tube, a tool, and/or a penetrating shaft with a leading edge that is either beveled (shaft cross-cut at an angle) or non-beveled (shaft ending in a pointed tip). In some embodiments, a non-beveled end-effectuator element can be employed to ablate a pathway through tissue to reach the target position while avoiding the mechanical forces and deflection created by a typical bevel tissue cutting system.

- (6) Some embodiments of the surgical robot can include a motor assembly comprising three linear motors that separately control movement of the effector element and/or surgical instrument on the respective x-, y- and z-axes. These separate motors can provide a degree of accuracy that is not provided by conventional surgical robots, thereby giving the surgeon the capability of more exactly determining position and strike angles on a three dimensional image.
- (7) In some embodiments, at least one RF transmitter can be mounted on the effector element and/or the surgical instrument. Three or more RF receivers can be mounted in the vicinity of the surgical robot. The location of the RF transmitter and, therefore, the surgical instrument, can be accurately determined by analyzing the RF signals that are emitted from the RF transmitter. For example, by measuring the time of flight of the RF signal from the transmitter to the RF receivers that are positioned at known locations, the position of the end-effector element with respect to a patient can be determined. In some embodiments, a physician or surgeon can perform epidural injections of steroids into a patient to alleviate back pain without the use of x-rays as is currently required with x-ray fluoroscopic techniques.
- (8) Some embodiments of the invention use RF feedback to actively control the movement of the surgical robot. For example, RF signals can be sent by the RF transmitter on an iterative basis and then analyzed in an iterative process to allow the surgical robot to automatically move the effector element and/or surgical instrument to a desired location within a patient's body. The location of the effector element and/or surgical instrument can be dynamically updated and, optionally, can be displayed to a user in real-time.
- (9) In some embodiments, at least one RF transmitter can be disposed on other elements of the surgical robot, or anywhere within the room where an invasive procedure is taking place, in order to track other devices.
- (10) Some embodiments of the invention dispose one or more RF transmitters on the anatomical part of the patient that is the target of the invasive procedure. This system can be used to correct the movement of the surgical robot in the event the anatomical target moves during the procedure.
- (11) In some embodiments, the system can be configured to automatically position and rigidly hold the end-effector and/or the surgical instrument in accurate alignment with a required trajectory, such as, for example, a selected trajectory of a pedicle screw during pedicle screw insertion procedures. In case of movement of the patient, the system can be configured to automatically adjust the position of the robot to maintain desired alignment relative to an anatomical region of interest.
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Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a partial perspective view of a room in which a medical procedure is taking place by using a surgical robot, the movement of which is controlled by analysis of RF signals that are emitted from an inside the patient and received by RF receivers mounted therein.
- (2) FIG. 2 is a perspective view of a surgical robot according to an embodiment of the invention.
- (3) FIGS. 3A-3B are perspective views of the surgical robot illustrated in FIG. 2, which show the movement of the base of the surgical robot in the z-axis direction in accordance with an embodiment of the invention.
- (4) FIG. 4 is a partial perspective view of the surgical robot of FIG. 2 which shows how the robot arm can be moved in the x-axis direction.
- (5) FIGS. 5A-5B are partial perspective views of the surgical robot of FIG. 2, which show how the robot arm can be moved in the y-axis direction.
- (6) FIG. 6 is a perspective view of a portion of the robot arm of FIG. 2 showing how an effector element can be twisted about a y-axis.
- (7) FIG. 7 is a perspective view of a portion of a robot arm of FIG. 2 showing how an effector element can be pivoted about a pivot axis that is perpendicular to the y-axis.
- (8) FIGS. 8A-8B are partial perspective views of the surgical robot of FIG. 2, which show the movement of a surgical instrument 35 along the z-axis from an effector element.
- (9) FIG. 9 is a system diagram which shows local positioning sensors, a controlling PC, and a Radiofrequency (RF) transmitter in accordance with an embodiment of the invention.
- (10) FIG. 10 is a system diagram of the controlling PC, user input, and motors for controlling the robot in accordance with an embodiment of the invention.
- (11) FIG. 11 is a flow chart diagram for general operation of a surgical robot in accordance with one embodiment of the invention.
- (12) FIG. 12 is a flow chart diagram for a closed screw/needle insertion performed using a surgical robot in accordance with one embodiment of the invention.
- (13) FIG. 13 is a flow chart diagram of a safe zone surgery performed using a surgical robot as described herein in

accordance with one embodiment of the invention.

(14) FIG. 14 is a flow chart diagram of a flexible catheter insertion procedure performed using a surgical robot as described herein in accordance with one embodiment of the invention.

(15) FIG. 15A shows a screenshot of a monitor display showing a set up of the anatomy in X, Y and Z views in accordance with one embodiment of the invention.

(16) FIG. 15B shows a screenshot of a monitor display showing what the user views during an invasive procedure in accordance with one embodiment of the invention.

(17) FIG. 16 depicts a surgical robot having a plurality of optical markers mounted for tracking movement in an x-direction in accordance with one embodiment of the invention.

(18) FIGS. 17A-17B depict surgical instruments having a stop mechanism in accordance with one embodiment of the invention.

(19) FIGS. 17C-17E illustrate tools for manually adjusting a drill stop with reference to drill bit markings in accordance with one embodiment of the invention.

(20) FIGS. 17F-J illustrate tools for locking and holding a drill bit in a set position in accordance with one embodiment of the invention.

(21) FIGS. 18A-18B depicts an end-effector having a clearance mechanism in accordance with one embodiment of the invention.

(22) FIG. 19A-19B depicts an end-effector having an attachment element for applying distraction and/or compression forces in accordance with one embodiment of the invention.

(23) FIGS. 20A-20E show the use of calibration frames with the guidance system in accordance with one embodiment of the invention.

(24) FIG. 21A depicts flexible roll configurations of a targeting fixture in accordance with one embodiment of the invention.

(25) FIG. 21B shows possible positions of markers along a line in space in accordance with one embodiment of the invention.

(26) FIG. 21C depicts flexible roll configurations of a targeting fixture in accordance with one embodiment of the invention.

(27) FIG. 21D shows a fixture that can be employed to provide desired stiffness to the unrolled fixture such that it maintains its position after unrolling occurs in accordance with one embodiment of the invention.

(28) FIGS. 22A-22D depict a targeting fixture and method configured for application to the skull of a patient in accordance with one embodiment of the invention.

(29) FIG. 23 depicts a dynamic tracking device mounted to the spinous process of the lumbar spine of a patient in accordance with one embodiment of the invention.

(30) FIGS. 24-33 illustrate methods in accordance with one embodiment of the invention.

(31) FIG. 34 illustrates a computing platform that enables implementation of various embodiments of the invention.

(32) FIGS. 35A-35B display a surgical robot in accordance with one embodiment of the invention.

(33) FIG. 36 illustrates a surgical robot system having a surveillance marker in accordance with one or more embodiments described herein.

(34) FIG. 37 illustrates an example of a methodology for tracking a visual point on a rigid body using an array of three attached markers in accordance with one embodiment of the invention.

(35) FIG. 38 illustrates a procedure for monitoring the location of a point of interest relative to three markers based on images received from the methodology illustrated in FIG. 37.

(36) FIGS. 39A-F illustrates examples of tracking methodology based on an array of three attached markers in accordance with one embodiment of the invention.

(37) FIG. 40 illustrates an example of a two dimensional representation for rotation about the Y-axis in accordance with one embodiment of the invention.

(38) FIG. 41A illustrates an alternative representation of a two dimensional representation for rotation about an X-axis in accordance with one embodiment of the invention.

(39) FIG. 41B illustrates an alternative representation of a two dimensional representation for rotation about a Y-axis in accordance with one embodiment of the invention.

(40) FIG. 41C illustrates an alternative representation of a two dimensional representation for rotation about a Z-axis in accordance with one embodiment of the invention.

(41) FIG. 42 provides a depiction of a noise within a frame of data.

(42) FIG. 43 illustrates the depiction of a noise within a frame of data as shown in FIG. 42 with a stored point of interest.

(43) FIG. 44 illustrates a depiction of results of applying a least squares fitting algorithm for establishing a

reference frame and transforming markers in accordance with one embodiment of the invention.

(44) FIG. 45 illustrates a depiction of results of applying a least squares fitting algorithm for establishing a reference frame and transforming markers as shown in FIG. 44 including noise.

(45) FIG. 46 illustrates a depiction of error calculation for reference frame markers in accordance with one embodiment of the invention.

(46) FIG. 47 illustrates a graphical representation of methods of tracking three dimensional movement of a rigid body.

(47) FIG. 48 shows a perspective view illustrating a bayonet mount used to removably couple the surgical instrument to the end-effectuator in accordance with one embodiment of the invention.

(48) FIGS. 49A-49F depict illustrations of targeting fixtures in accordance with one embodiment of the invention.

(49) FIG. 50A shows an example illustration of one portion of a spine with markers in accordance with one embodiment of the invention.

(50) FIGS. 50B-50D show various illustrations of one portion of a spine with two independent trackers with markers in accordance with one embodiment of the invention.

(51) FIG. 50E illustrates a representation of a display of a portion of a spine based on the location of a tracker in accordance with one embodiment of the invention.

(52) FIG. 50F illustrates a representation of a display of a portion of a spine based on the location of a tracker in accordance with one embodiment of the invention.

(53) FIGS. 50G-50H represent images of segmented CT scans in accordance with one embodiment of the invention.

(54) FIG. 51 shows an example of a fixture for use with fluoroscopic views in accordance with one embodiment of the invention.

(55) FIGS. 52A-52B illustrates expected images on anteroposterior and lateral x-rays of the spine with a misaligned fluoroscopy (x-ray) machine in accordance with one embodiment of the invention.

(56) FIGS. 53A-53B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine in accordance with one embodiment of the invention.

(57) FIGS. 54A-54B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine including overlaid computer-generated graphical images showing the planned trajectory and the current actual position of the robot end-effectuator in accordance with one embodiment of the invention.

(58) FIGS. 55A-55B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine showing a feature on the targeting fixture designed to eliminate ambiguity about directionality in accordance with one embodiment of the invention.

(59) FIG. 56 illustrates an axial view of a spine showing how a cartoonish axial approximation of the spine can be constructed based on lateral and anteroposterior x-rays in accordance with one embodiment of the invention.

(60) FIGS. 57A-57B illustrates examples of targeting fixtures that facilitate desired alignment of the targeting fixture relative to the x-ray image plane in accordance with one embodiment of the invention.

(61) FIGS. 58A-58B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine when parallax is present in accordance with one embodiment of the invention.

(62) FIG. 59A illustrates two parallel plates with identically positioned radio-opaque markers in accordance with one embodiment of the invention.

(63) FIG. 59B illustrates resulting expected x-ray demonstrating how marker overlay is affected due to parallax using the two parallel plates as shown in FIG. 59A in accordance with one embodiment of the invention.

(64) FIG. 60 shows a representation of the rendering of a computer screen with an x-ray image that is affected by parallax overlaid by graphical markers over the radio-opaque markers on two plates that have the same geometry in accordance with one embodiment of the invention.

(65) FIG. 61 shows a graphical overlay for the x-ray image screen intended to help the user physically line up the x-ray machine to get a view in which the markers on the two calibration plates shown in FIG. 59A in the case where parallax complicates the view in accordance with one embodiment of the invention.

(66) FIG. 62 illustrates a method in accordance with at least one embodiment of the invention.

(67) FIGS. 63A-63C illustrates various embodiments of an end-effectuator including a modified mount with a clamping piece in accordance with at least one embodiment of the invention.

(68) FIGS. 64-65 illustrate embodiments of clamping piece actuation on a spinous process in accordance with some embodiments of the invention.

(69) FIG. 66A illustrates a clamping piece modified with a targeting fixture including a temporary marker skirt in accordance with at least one embodiment of the invention.

(70) FIG. **66B** illustrates a clamping piece modified with a targeting fixture as shown in FIG. **66A** with the temporary marker skirt detached in accordance with at least one embodiment of the invention.

(71) FIG. **67** shows a modified Mayfield frame **6700** including one possible configuration for active and radio-opaque markers in accordance with one embodiment of the invention.

(72) FIG. **68** shows end-effectuator **30** that includes nested dilators in accordance with at least one embodiment of the invention.

(73) FIGS. **69A-69C** illustrates various embodiments of an end-effectuator including cylindrical dilator tubes in accordance with at least one embodiment of the invention.

(74) FIG. **70** illustrates a method in accordance with at least one embodiment of the invention.

(75) FIG. **71A** illustrates a robot end-effectuator coupled with a curved guide tube for use with a curved or straight wire or tool in accordance with at least one embodiment of the invention.

(76) FIG. **71B** illustrates a robot end-effectuator coupled with a straight guide tube for use with a curved or straight wire or tool in accordance with at least one embodiment of the invention.

(77) FIG. **72** illustrates a guide tube in accordance with at least one embodiment of the invention.

(78) FIG. **73** illustrates a steerable and trackable needle in accordance with at least one embodiment of the invention.

(79) FIG. **74** illustrates one embodiment of intersecting and interlocking bone screws in accordance with at least one embodiment of the invention.

(80) FIG. **75A-75B** illustrates configurations of a robot for positioning alongside a bed of a patient that includes a targeting fixture coupled to an end-effectuator using a snap-in post.

(81) FIG. **76** illustrates a surgical robot having a plurality of optical markers mounted for calibration and tracking movement in accordance with one embodiment of the invention.

(82) FIG. **77** illustrates a CT scan and methods in accordance with one embodiment of the invention.

(83) FIG. **78** illustrates a biopsy tool in accordance with one embodiment of the invention.

(84) FIG. **79** illustrates a deep brain stimulation electrode placement method performed by the robot system in accordance with one embodiment of the invention.

(85) FIG. **80** illustrates a partial view of a surgical robot system including a visual indicator comprising lights projected on the surgical field in accordance with one embodiment of the invention.

(86) FIG. **81** illustrates a perspective view of a robot system including a camera arm in accordance with one embodiment of the invention.

(87) FIG. **82A** illustrates a front-side perspective view of a robot system including a camera arm in a stored position in accordance with one embodiment of the invention.

(88) FIG. **82B** illustrates a rear-side perspective view of a robot system including a camera arm in a stored position in accordance with one embodiment of the invention.

(89) FIG. **83** shows a lateral illustration of a patient lying supine, showing the normal relative positions of the prostate, rectum, bladder, and pubic bone.

(90) FIG. **84A** shows a lateral illustration of a patient lying supine, showing how inflation of a balloon can cause anterior displacement of the prostate toward the pubic bone, and a controllable amount of compression against the pubic bone in accordance with one embodiment of the invention.

(91) FIG. **84B** shows a lateral illustration of a patient lying supine, showing how shifting of a paddle in the rectum can cause anterior displacement of the prostate toward the pubic bone, and a controllable amount of compression against the pubic bone in accordance with one embodiment of the invention.

(92) FIG. **85** shows a sketch of a targeting fixture and immobilization device to be used for tracking the prostate during image-guided surgical procedures in accordance with one embodiment of the invention.

(93) FIG. **86** shows an illustration of the device as illustrated in FIG. **85**, in place in the rectum with prostate compressed and immobilized and tracking markers visible protruding caudal to the rectum in accordance with one embodiment of the invention.

(94) FIG. **87** illustrates a demonstration of a fibre Bragg grating (“FBG”) interrogation technology with a flexible fiber optic cable in accordance with one embodiment of the invention.

(95) FIG. **88** illustrates a tracker attached to the surface of the skin of a patient and rigidly interconnected to a fiber optic probe to allow accurate tracking of the prostate in accordance with one embodiment of the invention.

(96) FIG. **89** illustrates the fiber optic probe as depicted in FIG. **88** with optically visible and MRI visible markings in accordance with one embodiment of the invention.

(97) FIGS. **90-93** illustrate various embodiments of a fiber optic probe tracking system to allow accurate tracking of the prostate for image-guided therapy in accordance with one embodiment of the invention.

(98) FIG. **94** illustrates one embodiment of a nerve sensing probe.

DETAILED DESCRIPTION

(99) Before any embodiments are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

(100) The following discussion is presented to enable a person skilled in the art to make and use embodiments of the invention. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other embodiments and applications without departing from embodiments of the invention. Thus, embodiments of the invention are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of embodiments of the invention. Skilled artisans will recognize the examples provided herein have many useful alternatives and fall within the scope of embodiments of the invention.

(101) The present invention can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and their previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this invention is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(102) The following description is provided as an enabling teaching of the invention in its best, currently known embodiment. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the invention described herein, while still obtaining the beneficial results of the present invention. It will also be apparent that some of the desired benefits of the present invention can be obtained by selecting some of the features of the present invention without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present invention are possible and can even be desirable in certain circumstances and are a part of the present invention. Thus, the following description is provided as illustrative of the principles of the present invention and not in limitation thereof.

(103) As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a delivery conduit” can include two or more such delivery conduits unless the context indicates otherwise.

(104) As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance may or may not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

(105) In some embodiments, the disclosed devices and systems can comprise elements of the devices and systems described in U.S. Patent Publication Nos. 2007/0238985, 2008/0154389, and 2008/0215181, the disclosures of which are incorporated herein by reference in their entireties.

(106) As employed in this specification and annexed drawings, the terms “unit,” “component,” “interface,” “system,” “platform,” and the like are intended to include a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the computer-related entity or the entity related to the operational apparatus can be either hardware, a combination of hardware and software, software, or software in execution. One or more of such entities are also referred to as “functional elements.” As an example, a unit may be, but is not limited to being, a process running on a processor, a processor, an object, an executable computer program, a thread of execution, a program, a memory (e.g., a hard disc drive), and/or a computer. As another example, a unit can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry which is operated by a software application or a firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. In addition or in the alternative, a unit can provide specific functionality based on physical structure or specific arrangement of hardware elements. As yet another example, a unit can be an apparatus that provides specific functionality through electronic functional elements without mechanical parts, the

electronic functional elements can include a processor therein to execute software or firmware that provides at least in part the functionality of the electronic functional elements. An illustration of such apparatus can be control circuitry, such as a programmable logic controller. The foregoing example and related illustrations are but a few examples and are not intended to be limiting. Moreover, while such illustrations are presented for a unit, the foregoing examples also apply to a component, a system, a platform, and the like. It is noted that in certain embodiments, or in connection with certain aspects or features thereof, the terms “unit,” “component,” “system,” “interface,” “platform” can be utilized interchangeably.

(107) Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude, for example, other additives, components, integers or steps.

(108) Referring now to FIGS. 1 and 35A, some embodiments include a surgical robot system **1** is disclosed in a room **10** where a medical procedure is occurring. In some embodiments, the surgical robot system **1** can comprise a surgical robot **15** and one or more positioning sensors **12**. In this aspect, the surgical robot **15** can comprise a display means **29** (including for example a display **150** shown in FIG. **10**), and a housing **27**. In some embodiments a display **150** can be attached to the surgical robot **15**, whereas in other embodiments, a display means **29** can be detached from surgical robot **15**, either within surgical room **10** or in a remote location. In some embodiments, the housing **27** can comprise a robot arm **23**, and an end-effectuator **30** coupled to the robot arm **23** controlled by at least one motor **160**. For example, in some embodiments, the surgical robot system **1** can include a motor assembly **155** comprising at least one motor (represented as **160** in FIG. **10**). In some embodiments, the end-effectuator **30** can comprise a surgical instrument **35**. In other embodiments, the end-effectuator **30** can be coupled to the surgical instrument **35**. As used herein, the term “end-effectuator” is used interchangeably with the terms “end-effectuator,” “effectuator element,” and “effectuator element.” In some embodiments, the end-effectuator **30** can comprise any known structure for effecting the movement of the surgical instrument **35** in a desired manner.

(109) In some embodiments, prior to performance of an invasive procedure, a three-dimensional (“3D”) image scan can be taken of a desired surgical area of the patient **18** and sent to a computer platform in communication with surgical robot **15** as described herein (see for example the platform **3400** including the computing device **3401** shown in FIG. **34**). In some embodiments, a physician can then program a desired point of insertion and trajectory for surgical instrument **35** to reach a desired anatomical target within or upon the body of patient **18**. In some embodiments, the desired point of insertion and trajectory can be planned on the 3D image scan, which in some embodiments, can be displayed on display means **29**. In some embodiments, a physician can plan the trajectory and desired insertion point (if any) on a computed tomography scan (hereinafter referred to as “CT scan”) of a patient **18**. In some embodiments, the CT scan can be an isocentric C-arm type scan, an O-arm type scan, or intraoperative CT scan as is known in the art. However, in some embodiments, any known 3D image scan can be used in accordance with the embodiments of the invention described herein.

(110) In some embodiments, the surgical robot system **1** can comprise a local positioning system (“LPS”) subassembly to track the position of surgical instrument **35**. The LPS subassembly can comprise at least one radio-frequency (RF) transmitter **120** that is coupled were affixed to the end-effectuator **30** or the surgical instrument **35** at a desired location. In some embodiments, the at least one RF transmitter **120** can comprise a plurality of transmitters **120**, such as, for example, at least three RF transmitters **120**. In another embodiment, the LPS subassembly can comprise at least one RF receiver **110** configured to receive one or more RF signals produced by the at least one RF transmitter **120**. In some embodiments, the at least one RF receiver **110** can comprise a plurality of RF receivers **110**, such as, for example, at least three RF receivers **110**. In these embodiments, the RF receivers **110** can be positioned at known locations within the room **10** where the medical procedure is to take place. In some embodiments, the RF receivers **110** can be positioned at known locations within the room **10** such that the RF receivers **110** are not coplanar within a plane that is parallel to the floor of the room **10**.

(111) In some embodiments, during use, the time of flight of an RF signal from each RF transmitter **120** of the at least one RF transmitter **120** to each RF receiver **110** of the at least one RF receiver **110** (e.g., one RF receiver, two RF receivers, three RF receivers, etc.) can be measured to calculate the position of each RF transmitter **120**. Because the velocity of the RF signal is known, the time of flight measurements result in at least three distance measurements for each RF transmitter **120** (one to each RF receiver **110**).

(112) In some embodiments, the surgical robot system **1** can comprise a control device (for example a computer **100** having a processor and a memory coupled to the processor). In some embodiments, the processor of the control device **100** can be configured to perform time of flight calculations as described herein. Further, in some embodiments, can be configured to provide a geometrical description of the location of the at least one RF transmitter **120** with respect to an operative end of the surgical instrument **35** or end-effectuator **30** that is utilized

to perform or assist in performing an invasive procedure. In some further embodiments, the position of the RF transmitter **120**, as well as the dimensional profile of the surgical instrument **35** or the effectuator element **30** can be displayed on a monitor (for example on a display means **29** such as the display **150** shown in FIG. **10**). In one embodiment, the end-effectuator **30** can be a tubular element (for example a guide tube **50**) that is positioned at a desired location with respect to, for example, a patient's **18** spine to facilitate the performance of a spinal surgery. In some embodiments, the guide tube **50** can be aligned with the z axis **70** defined by a corresponding robot motor **160** or, for example, can be disposed at a selected angle relative to the z-axis **70**. In either case, the processor of the control device (i.e. the computer **100**) can be configured to account for the orientation of the tubular element and the position of the RF transmitter **120**. As further described herein, in some embodiments, the memory of the control device (computer **100** for example) can store software for performing the calculations and/or analyses required to perform many of the surgical method steps set forth herein.

(113) Another embodiment of the disclosed surgical robot system **1** involves the utilization of a robot **15** that is capable of moving the end-effectuator **30** along x-, y-, and z-axes (see **66, 68, 70** in FIG. **35B**). In this embodiment, the x-axis **66** can be orthogonal to the y-axis **68** and z-axis **70**, the y-axis **68** can be orthogonal to the x-axis **66** and z-axis **70**, and the z-axis **70** can be orthogonal to the x-axis **66** and the y-axis **68**. In some embodiments, the robot **15** can be configured to effect movement of the end-effectuator **30** along one axis independently of the other axes. For example, in some embodiments, the robot **15** can cause the end-effectuator **30** to move a given distance along the x-axis **66** without causing any significant movement of the end-effectuator **30** along the y-axis **68** or z-axis **70**.

(114) In some further embodiments, the end-effectuator **30** can be configured for selective rotation about one or more of the x-axis **66**, y-axis **68**, and z-axis **70** (such that one or more of the Cardanic Euler Angles (e.g., roll, pitch, and/or yaw) associated with the end-effectuator **30** can be selectively controlled). In some embodiments, during operation, the end-effectuator **30** and/or surgical instrument **35** can be aligned with a selected orientation axis (labeled “Z Tube” in FIG. **35B**) that can be selectively varied and monitored by an agent (for example computer **100** and platform **3400**) that can operate the surgical robot system **1**. In some embodiments, selective control of the axial rotation and orientation of the end-effectuator **30** can permit performance of medical procedures with significantly improved accuracy compared to conventional robots that utilize, for example, a six degree of freedom robot arm **23** comprising only rotational axes.

(115) In some embodiments, as shown in FIG. **1**, the robot arm **23** that can be positioned above the body of the patient **18**, with the end-effectuator **30** selectively angled relative to the z-axis toward the body of the patient **18**. In this aspect, in some embodiments, the robotic surgical system **1** can comprise systems for stabilizing the robotic arm **23**, the end-effectuator **30**, and/or the surgical instrument **35** at their respective positions in the event of power failure. In some embodiments, the robotic arm **23**, end-effectuator **30**, and/or surgical instrument **35** can comprise a conventional worm-drive mechanism (not shown) coupled to the robotic arm **23**, configured to effect movement of the robotic arm along the z-axis **70**. In some embodiments, the system for stabilizing the robotic arm **23**, end-effectuator **30**, and/or surgical instrument **35** can comprise a counterbalance coupled to the robotic arm **23**. In another embodiment, the means for maintaining the robotic arm **23**, end-effectuator **30**, and/or surgical instrument **35** can comprise a conventional brake mechanism (not shown) that is coupled to at least a portion of the robotic arm **23**, such as, for example, the end-effectuator **30**, and that is configured for activation in response to a loss of power or “power off” condition of the surgical robot **15**.

(116) Referring to FIG. **1**, in some embodiments, the surgical robot system **1** can comprise a plurality of positioning sensors **12** configured to receive RF signals from the at least one conventional RF transmitter (not shown) located within room **10**. In some embodiments, the at least one RF transmitter **120** can be disposed on various points on the surgical robot **15** and/or on patient **18**. For example, in some embodiments, the at least one RF transmitter **120** can be attached to one or more of the housing **27**, robot arm **23**, end-effectuator **30**, and surgical instrument **35**. Some embodiments include positioning sensors **12** that in some embodiments comprise RF receivers **110**. In some embodiments, RF receivers **110** are in communication with a computer platform as described herein (see for example **3400** comprising a computing device **3401** FIG. **34**) that receives the signal from the RF transmitters **120**. In some embodiments, each transmitter **120** of the at least one RF transmitter **120** can transmit RF energy on a different frequency so that the identity of each transmitter **120** in the room **10** can be determined. In some embodiments, the location of the at least one RF transmitter **120**, and, consequently, the objects to which the transmitters **120** are attached, are calculated by the computer (e.g., computing device **3401** in FIG. **34**) using time-of-flight processes.

(117) In some embodiments, the computer (not shown in FIG. **1**) is also in communication with surgical robot **15**. In some embodiments, a conventional processor (not shown) of the computer **100** of the computing device **3401** can be configured to effect movement of the surgical robot **15** according to a preplanned trajectory selected prior to the procedure. For example, in some embodiments, the computer **100** of the computing device **3401** can use

robotic guidance software **3406** and robotic guidance data storage **3407** (shown in FIG. **34**) to effect movement of the surgical robot **15**.

(118) In some embodiments, the position of surgical instrument **35** can be dynamically updated so that surgical robot **15** is aware of the location of surgical instrument **35** at all times during the procedure. Consequently, in some embodiments, the surgical robot **15** can move the surgical instrument **35** to the desired position quickly, with minimal damage to patient **18**, and without any further assistance from a physician (unless the physician so desires). In some further embodiments, the surgical robot **15** can be configured to correct the path of surgical instrument **35** if the surgical instrument **35** strays from the selected, preplanned trajectory.

(119) In some embodiments, the surgical robot **15** can be configured to permit stoppage, modification, and/or manual control of the movement of the end-effector **30** and/or surgical instrument **35**. Thus, in use, in some embodiments, an agent (e.g., a physician or other user) that can operate the system **1** has the option to stop, modify, or manually control the autonomous movement of end-effector **30** and/or surgical instrument **35**. Further, in some embodiments, tolerance controls can be preprogrammed into the surgical robot **15** and/or processor of the computer platform **3400** (such that the movement of the end-effector **30** and/or surgical instrument **35** is adjusted in response to specified conditions being met). For example, in some embodiments, if the surgical robot **15** cannot detect the position of surgical instrument **35** because of a malfunction in the at least one RF transmitter **120**, then the surgical robot **15** can be configured to stop movement of end-effector **30** and/or surgical instrument **35**. In some embodiments, if surgical robot **15** detects a resistance, such as a force resistance or a torque resistance above a tolerance level, then the surgical robot **15** can be configured to stop movement of end-effector **30** and/or surgical instrument **35**.

(120) In some embodiments, the computer **100** for use in the system (for example represented by computing device **3401**), as further described herein, can be located within surgical robot **15**, or, alternatively, in another location within surgical room **10** or in a remote location. In some embodiments, the computer **100** can be positioned in operative communication with positioning sensors **12** and surgical robot **15**.

(121) In some further embodiments, the surgical robot **15** can also be used with existing conventional guidance systems. Thus, alternative conventional guidance systems beyond those specifically disclosed herein are within the scope and spirit of the invention. For instance, a conventional optical tracking system **3417** for tracking the location of the surgical device, or a commercially available infrared optical tracking system **3417**, such as Optotrak® (Optotrak® is a registered trademark of Northern Digital Inc. Northern Digital, Waterloo, Ontario, Canada), can be used to track the patient **18** movement and the robot's base **25** location and/or intermediate axis location, and used with the surgical robot system **1**. In some embodiments in which the surgical robot system **1** comprises a conventional infrared optical tracking system **3417**, the surgical robot system **1** can comprise conventional optical markers attached to selected locations on the end-effector **30** and/or the surgical instrument **35** that are configured to emit or reflect light. In some embodiments, the light emitted from and/or reflected by the markers can be read by cameras (for example with cameras **8200** shown in FIG. **81**) and/or optical sensors and the location of the object can be calculated through triangulation methods (such as stereo-photogrammetry).

(122) Referring now to FIG. **2**, it is seen that, in some embodiments, the surgical robot **15** can comprise a base **25** connected to wheels **31**. The size and mobility of these embodiments can enable the surgical robot to be readily moved from patient to patient and room to room as desired. As shown, in some embodiments, the surgical robot **15** can further comprise a case **40** that is slidably attached to base **25** such that the case **40** can slide up and down along the z-axis **70** substantially perpendicular to the surface on which base **25** sits. In some embodiments, the surgical robot **15** can include a display means **29**, and a housing **27** which contains robot arm **23**.

(123) As described earlier, the end-effector **30** can comprise a surgical instrument **35**, whereas in other embodiments, the end-effector **30** can be coupled to the surgical instrument **35**. In some embodiments, it is arm **23** can be connected to the end-effector **30**, with surgical instrument **35** being removably attached to the end-effector **30**.

(124) Referring now to FIGS. **2**, **3A-3B**, **4**, **5A-5B**, **6**, **7**, and **8A-8B**, in some embodiments, the effector element **30** can include an outer surface **30d**, and can comprise a distal end **30a** defining a beveled leading edge **30b** and a non-beveled leading edge **30c**. In some embodiments, the surgical instrument **35** can be any known conventional instrument, device, hardware component, and/or attachment that is used during performance of an invasive or non-invasive medical procedure (including surgical, therapeutic, and diagnostic procedures). For example and without limitation, in some embodiments, the surgical instrument **35** can be embodied in or can comprise a needle **7405**, **7410**, a conventional probe, a conventional screw, a conventional drill, a conventional tap, a conventional catheter, a conventional scalpel forceps, or the like. In addition or in the alternative, in some embodiments, the surgical instrument **35** can be a biological delivery device, such as, for example and without limitation, a conventional syringe, which can distribute biologically acting compounds throughout the body of a

patient **18**. In some embodiments, the surgical instrument **35** can comprise a guide tube **50** (also referred to herein as a “Z-tube **50**”) that defines a central bore configured for receipt of one or more additional surgical instruments **35**.

(125) In some embodiments, the surgical robot **15** is moveable in a plurality of axes (for instance x-axis **66**, y-axis **68**, and z-axis **70**) in order to improve the ability to accurately and precisely reach a target location. Some embodiments include a robot **15** that moves on a Cartesian positioning system; that is, movements in different axes can occur relatively independently of one another instead of at the end of a series of joints.

(126) Referring now to FIGS. **3A** and **3B**, the movement of case **40** relative to base **25** of surgical robot **15** is represented as a change of height of the system **1** and the position of the case **40** with respect to the base **25**. As illustrated, in some embodiments, case **40** can be configured to be raised and lowered relative to the base **25** along the z-axis. Some embodiments include a housing **27** that can be attached to case **40** and be configured to move in the z-direction (defined by z-frame **72**) with case **40** when case **40** is raised and lowered. Consequently, in some embodiments, arm **23**, the end-effector **30**, and surgical instrument **35** can be configured to move with case **40** as case **40** is raised and lowered relative to base **25**.

(127) In a further embodiment, referring now to FIG. **4**, housing **27** can be slidably attached to case **40** so that it can extend and retract along the x-axis **66** relative to case **40** and substantially perpendicularly to the direction case **40** moves relative to base **25**. Consequently, in some embodiments, the robot arm **23**, the end-effector **30**, and surgical instrument **35** can be configured to move with housing **27** as housing **27** is extended and retracted relative to case **40**.

(128) Referring now to FIGS. **5A** and **5B**, the extension of arm **23** along the y-axis **68** is shown. In some embodiments, robot arm **23** can be extendable along the y-axis **68** relative to case **40**, base **25**, and housing **27**. Consequently, in some embodiments, the end-effector **30** and surgical instrument **35** can be configured to move with arm **23** as arm **23** is extended and retracted relative to housing **27**. In some embodiments, arm **23** can be attached to a low profile rail system (not shown) which is encased by housing **27**.

(129) Referring now to FIGS. **6**, **7** and FIGS. **8A-B**, the movement of the end-effector **30** is shown. FIG. **6** shows an embodiment of an end-effector **30** that is configured to rotate about the y-axis **68**, performing a rotation having a specific roll **62**. FIG. **7** shows an embodiment of an end-effector **30** that is configured to rotate about the x-axis **66**, performing a rotation having a specific pitch **60**. FIG. **8** shows an embodiment of an end-effector **30** that is configured to raise and lower surgical instrument **35** along a substantially vertical axis, which can be a secondary movable axis **64**, referred to as “Z-tube axis **64**”. In some embodiments, the orientation of the guide tube **50** can be initially aligned with z-axis **70**, but such orientation can change in response to changes in roll **62** and/or pitch **60**.

(130) FIG. **9** shows a system diagram of the 3D positioning sensors **110**, computer **100**, and RF transmitters **120** in accordance with some embodiments of the invention is provided. As shown, computer **100** is in communication with positioning sensors **110**. In some embodiments, during operation, RF transmitters **120** are attached to various points on the surgical robot **15**. In some embodiments, the RF transmitters **120** can also be attached to various points on or around an anatomical target of a patient **18**. In some embodiments, computer **100** can be configured to send a signal to the RF transmitters **120**, prompting the RF transmitters **120** to transmit RF signals that are read by the positioning sensors **110**. In some embodiments, the computer **100** can be coupled to the RF transmitters **120** using any conventional communication means, whether wired or wireless. In some embodiments, the positioning sensors **110** can be in communication with computer **100**, which can be configured to calculate the location of the positions of all the RF transmitters **120** based on time-of-flight information received from the positioning sensors **110**. In some embodiments, computer **100** can be configured to dynamically update the calculated location of the surgical instrument **35** and/or end-effector **30** being used in the procedure, which can be displayed to the agent.

(131) Some embodiments can include a system diagram of surgical robot system **1** having a computer **100**, a display means **29** comprising a display **150**, user input **170**, and motors **160**, provided as illustrated in FIG. **10**. In some embodiments, motors **160** can be installed in the surgical robot **15** and control the movement of the end-effector **30** and/or surgical instrument **35** as described above. In some embodiments, computer **100** can be configured to dynamically update the location of the surgical instrument **35** being used in the procedure, and can be configured to send appropriate signals to the motors **160** such that the surgical robot **15** has a corresponding response to the information received by computer **100**. For example, in some embodiments, in response to information received by computer **100**, the computer **100** can be configured to prompt the motors **160** to move the surgical instrument **35** along a preplanned trajectory.

(132) In some embodiments, prior to performance of a medical procedure, such as, for example, an invasive surgical procedure, user input **170** can be used to plan the trajectory for a desired navigation. After the medical procedure has commenced, if changes in the trajectory and/or movement of the end-effector **30** and/or surgical

instrument **35** is desired, a user can use the user input **170** to input the desired changes, and the computer **100** can be configured to transmit corresponding signals to the motors **160** in response to the user input **170**.

(133) In some embodiments, the motors **160** can be or can comprise conventional pulse motors. In this aspect, in some embodiments, the pulse motors can be in a conventional direct drive configuration or a belt drive and pulley combination attached to the surgical instrument **35**. Alternatively, in other embodiments, the motors **160** can be conventional pulse motors that are attached to a conventional belt drive rack-and-pinion system or equivalent conventional power transmission component.

(134) In some embodiments, the use of conventional linear pulse motors within the surgical robot **15** can permit establishment of a non-rigid position for the end-effector **30** and/or surgical instrument **35**. Thus, in some embodiments, the end-effector **30** and/or surgical instrument **35** will not be fixed in a completely rigid position, but rather the end-effector **30** and/or the surgical instrument **35** can be configured such that an agent (e.g., a surgeon or other user) can overcome the x-axis **66** and y-axis **68**, and force the end-effector **30** and/or surgical instrument **35** from its current position. For example, in some embodiments, the amount of force necessary to overcome such axes can be adjusted and configured automatically or by an agent. In some embodiments, the surgical robot **15** can comprise circuitry configured to monitor one or more of: (a) the position of the robot arm **23**, the end-effector **30**, and/or the surgical instrument **35** along the x-axis **66**, y-axis **68**, and z-axis **70**; (b) the rotational position (e.g., roll **62** and pitch **60**) of the robot arm **23**, the end-effector **30**, and/or the surgical instrument **35** relative to the x-(**66**), y-(**68**), and z-(**70**) axes; and (c) the position of the end-effector **30**, and/or the surgical instrument **35** along the travel of the re-orientable axis that is parallel at all times to the end-effector **30** and surgical instrument **35** (the Z-tube axis **64**).

(135) In one embodiment, circuitry for monitoring the positions of the x-axis **66**, y-axis **68**, z-axis **70**, Z-tube axis **64**, roll **62**, and/or pitch **60** can comprise relative or absolute conventional encoder units (also referred to as encoders) embedded within or functionally coupled to conventional actuators and/or bearings of at least one of the motors **160**. Optionally, in some embodiments, the circuitry of the surgical robot **15** can be configured to provide auditory, visual, and/or tactile feedback to the surgeon or other user when the desired amount of positional tolerance (e.g., rotational tolerance, translational tolerance, a combination thereof, or the like) for the trajectory has been exceeded. In some embodiments, the positional tolerance can be configurable and defined, for example, in units of degrees and/or millimeters.

(136) In some embodiments, the robot **15** moves into a selected position, ready for the surgeon to deliver a selected surgical instrument **35**, such as, for example and without limitation, a conventional screw, a biopsy needle **8110**, and the like. In some embodiments, as the surgeon works, if the surgeon inadvertently forces the end-effector **30** and/or surgical instrument **35** off of the desired trajectory, then the system **1** can be configured to provide an audible warning and/or a visual warning. For example, in some embodiments, the system **1** can produce audible beeps and/or display a warning message on the display means **29**, such as “Warning: Off Trajectory,” while also displaying the axes for which an acceptable tolerance has been exceeded.

(137) In some embodiments, in addition to, or in place of the audible warning, a light illumination may be directed to the end-effector **30**, the guide tube **50**, the operation area (i.e. the surgical field **17**) of the patient **18**, or a combination of these regions. For example, some embodiments include at least one visual indication **900** capable of illuminating a surgical field **17** of a patient **18**. Some embodiments include at least one visual indication **900** capable of indicating a target lock by projecting an illumination on a surgical field **17**. In some embodiments, the system **1** can provide feedback to the user regarding whether the robot **15** is locked on target. In some other embodiments, the system **1** can provide an alert to the user regarding whether at least one marker **720** is blocked, or whether the system **1** is actively seeking one or more markers **720**.

(138) In some embodiments, the visual indication **900** can be projected by one or more conventional light emitting diodes mounted on or near the robot end-effector **30**. In some embodiments, the visual indication can comprise lights projected on the surgical field **17** including a color indicative of the current situation (see for example, FIG. **80**). In some embodiments, a green projected light could represent a locked-on-target situation, whereas in some embodiments, a red illumination could indicate a trajectory error, or obscured markers **720**. In some other embodiments, a yellow illumination could indicate the system **1** is actively seeking one or more markers **720**.

(139) In some embodiments, if the surgeon attempts to exceed the acceptable tolerances, the robot **15** can be configured to provide mechanical resistance (“push back” or haptic feedback) to the movement of the end-effector **30** and/or surgical instrument **35** in this manner, thereby promoting movement of the end-effector **30** and/or surgical instrument **35** back to the correct, selected orientation. In some embodiments, when the surgeon then begins to correct the improper position, the robot **15** can be configured to substantially immediately return the end-effector **30** and/or surgical instrument **35** back to the desired trajectory, at which time the audible and visual warnings and alerts can be configured to cease. For example, in some embodiments, the visual warning could include a visual indication **900** that may include a green light if no tolerances have been exceeded, or a red

light if tolerances are about to, or have been exceeded.

(140) As one will appreciate, a conventional worm-drive system would be absolutely rigid, and a robot **15** having such a worm-drive system would be unable to be passively moved (without breaking the robot **15**) no matter how hard the surgeon pushed. Furthermore, a completely rigid articulation system can be inherently unsafe to a patient **18**. For example, if such a robot **15** were moving toward the patient **18** and inadvertently collided with tissues, then these tissues could be damaged. Although conventional sensors can be placed on the surface of such a robot **15** to compensate for these risks, such sensors can add considerable complexity to the overall system **1** and would be difficult to operate in a fail-safe mode. In contrast, during use of the robot **15** described herein, if the end-effector **30** and/or surgical instrument **35** inadvertently collides with tissues of the patient **18**, a collision would occur with a more tolerable force that would be unlikely to damage such tissues. Additionally, in some embodiments, auditory and/or visual feedback as described above can be provided to indicate an increase in the current required to overcome the obstacle. Furthermore, in some embodiments, the end-effector **30** of the robot **15** can be configured to displace itself (move away) from the inadvertently contacted tissue if a threshold required motor **160** current is encountered. In some embodiments, this threshold could be configured (by a control component, for example) for each axis such that the moderate forces associated with engagement between the tissue and the end-effector **30** can be recognized and/or avoided.

(141) In some embodiments, the amount of rigidity associated with the positioning and orientation of the end-effector **30** and/or the surgical instrument **35** can be selectively varied. For example, in some embodiments, the robot **15** can be configured to shift between a high-rigidity mode and a low-rigidity mode. In some embodiments, the robot **15** can be programmed so that it automatically shifts to the low-rigidity mode as the end-effector **30** and surgical instrument **35** are shifted from one trajectory to another, from a starting position as they approach a target trajectory and/or target position. Moreover, in some embodiment, once the end-effector **30** and/or surgical instrument **35** is within a selected distance of the target trajectory and/or target position, such as, for example, within about 1° and about 1 mm of the target, the robot **15** can be configured to shift to the high-rigidity mode. In some embodiments, this mechanism may improve safety because the robot **15** would be unlikely to cause injury if it inadvertently collided with the patient **18** while in the low-rigidity mode.

(142) Some embodiments include a robot **15** that can be configured to effect movement of the end-effector **30** and/or surgical instrument **35** in a selected sequence of distinct movements. In some embodiments, during movement of the end-effector **30** and/or surgical instrument **35** from one trajectory to another trajectory, the x-axis **66**, y-axis **68**, roll **62**, and pitch **60** orientations are all changed simultaneously, and the speed of movement of the end-effector **30** can be increased. Consequently, because of the range of positions through which the end-effector **30** travels, the likelihood of a collision with the tissue of the patient **18** can also be increased. Hence, in some embodiments, the robot **15** can be configured to effect movement of the end-effector **30** and/or surgical instrument **35** such that the position of the end-effector **30** and/or surgical instrument **35** within the x-axis **66** and the y-axis **68** are adjusted before the roll **62** and pitch **60** of the end-effector **30** and/or surgical instrument **35** are adjusted. In some alternative embodiments, the robot **15** can be configured to effect movement of the end-effector **30** and/or surgical instrument **35** so that the roll **62** and pitch **60** are shifted to 0°. The position of the end-effector **30** and/or surgical instrument **35** within the x-axis **66** and the y-axis **68** are adjusted, and then the roll **62** and pitch **60** of the end-effector **30** and/or surgical instrument **35** are adjusted.

(143) Some embodiments include a robot **15** that can be optionally configured to ensure that the end-effector **30** and/or surgical instrument **35** are moved vertically along the z-axis **70** (away from the patient **18**) by a selected amount before a change in the position and/or trajectory of the end-effector **30** and/or surgical instrument **35** is effected. For example, in some embodiments, when an agent (for example, a surgeon or other user, or equipment) changes the trajectory of the end-effector **30** and/or surgical instrument **35** from a first trajectory to a second trajectory, the robot **15** can be configured to vertically displace the end-effector **30** and/or surgical instrument **35** from the body of the patient **18** along the z-axis **70** by the selected amount (while adjusting x-axis **66** and y-axis **68** configurations to remain on the first trajectory vector, for example), and then effecting the change in position and/or orientation of the end-effector **30** and/or surgical instrument **35**. This ensures that the end-effector **30** and/or surgical instrument **35** do not move laterally while embedded within the tissue of the patient **18**. Optionally, in some embodiments, the robot **15** can be configured to produce a warning message that seeks confirmation from the agent (for example, a surgeon or other user, or equipment) that it is safe to proceed with a change in the trajectory of the end-effector **30** and/or surgical instrument **35** without first displacing the end-effector **30** and/or surgical instrument **35** along the z-axis.

(144) In some embodiments, at least one conventional force sensor (not shown) can be coupled to the end-effector **30** and/or surgical instrument **35** such that the at least one force sensor receives forces applied along the orientation axis (Z-tube axis **64**) to the surgical instrument **35**. In some embodiments, the at least one force sensor can be configured to produce a digital signal. In some embodiments for example, the digital signal can be

indicative of the force that is applied in the direction of the Z-tube axis **64** to the surgical instrument **35** by the body of the patient **18** as the surgical instrument **35** advances into the tissue of the patient **18**. In some embodiments, the at least one force sensor can be a small conventional uniaxial load cell based on a conventional strain gauge mechanism. In some embodiments, the uniaxial load cell can be coupled to, for example, analog-to-digital filtering to supply a continuous digital data stream to the system **1**. Optionally, in some embodiments, the at least one force sensor can be configured to substantially continuously produce signals indicative of the force that is currently being applied to the surgical instrument **35**. In some embodiments, the surgical instrument **35** can be advanced into the tissue of the patient **18** by lowering the z-axis **70** while the position of the end-effectuator **30** and/or surgical instrument **35** along the x-axis **66** and y-axes **68** is adjusted such that alignment with the selected trajectory vector is substantially maintained. Furthermore, in some embodiments, the roll **62** and pitch **60** orientations can remain constant or self-adjust during movement of the x-(**66**), y-(**68**), and z-(**70**) axes such that the surgical instrument **35** remains oriented along the selected trajectory vector. In some embodiments, the position of the end-effectuator **30** along the z-axis **70** can be locked at a selected mid-range position (spaced a selected distance from the patient **18**) as the surgical instrument **35** advances into the tissue of the patient **18**. In some embodiments, the stiffness of the end-effectuator **30** and/or the surgical instrument **35** can be set at a selected level as further described herein. For example, in some embodiments, the stiffness of the Z-tube axis **64** position of the end-effectuator **30** and/or the surgical instrument **35** can be coupled to a conventional mechanical lock (not shown) configured to impart desired longitudinal stiffness characteristics to the end-effectuator **30** and/or surgical instrument **35**. In some embodiments, if the end-effectuator **30** and/or surgical instrument **35** lack sufficient longitudinal stiffness, then the counterforce applied by the tissue of the patient **18** during penetration of the surgical instrument **35** can oppose the direction of advancement of the surgical instrument **35** such that the surgical instrument **35** cannot advance along the selected trajectory vector. In other words, as the z-axis **70** advances downwards, the Z-tube axis **64** can be forced up and there can be no net advancement of the surgical instrument **35**. In some embodiments, the at least one force sensor can permit an agent (for example, a surgeon or other user, or equipment) to determine, (based on sudden increase in the level of applied force monitored by the force sensor at the end-effectuator **30** and/or the surgical instrument **35**), when the surgical instrument **35** has encountered a bone or other specific structure within the body of the patient **18**.

(145) In some alternative embodiments, the orientation angle of the end-effectuator **30** and/or surgical instrument **35** and the x-axis **66** and y-axis **68** can be configured to align the Z-tube axis **64** with the desired trajectory vector at a fully retracted Z-tube position, while a z-axis **70** position is set in which the distal tip of the surgical instrument **35** is poised to enter tissue. In this configuration, in some embodiments, the end-effectuator **30** can be positioned in a manner that the end-effectuator **30** can move, for example, exactly or substantially exactly down the trajectory vector if it were advanced only along guide tube **50**. In such scenario, in some embodiments, advancing the Z-tube axis **64** can cause the guide tube **50** to enter into tissue, and an agent (a surgeon or other user, equipment, etc.) can monitor change in force from the load sensor. Advancement can continue until a sudden increase in applied force is detected at the time the surgical instrument **35** contacts bone.

(146) In some embodiments, the robot **15** can be configured to deactivate the one or more motors **160** that advance the Z-tube axis **64** such that the end-effectuator **30** and/or the surgical instrument **35** can move freely in the Z-tube axis **64** direction while the position of the end-effectuator **30** and/or the surgical instrument **35** continues to be monitored. In some embodiments, the surgeon can then push the end-effectuator **30** down along the Z-tube axis **64**, (which coincides with the desired trajectory vector) by hand. In some embodiments, if the end-effectuator **30** position has been forced out of alignment with the trajectory vector, the position of the surgical instrument **35** can be corrected by adjustment along the x-(**66**) and/or y-(**68**) axes and/or in the roll **62** and/or pitch **60** directions. In some embodiments, when motor **160** associated with the Z-tube **50** movement of the surgical instrument **35** is deactivated, the agent (for example, a surgeon or other user, or equipment) can manually force the surgical instrument **35** to advance until a tactile sense of the surgical instrument **35** contacts bone, or another known region of the body).

(147) In some further embodiments, the robotic surgical system **1** can comprise a plurality of conventional tracking markers **720** configured to track the movement of the robot arm **23**, the end-effectuator **30**, and/or the surgical instrument **35** in three dimensions. It should be appreciated that three dimensional positional information from tracking markers **720** can be used in conjunction with the one dimensional linear positional information from absolute or relative conventional linear encoders on each axis of the robot **15** to maintain a high degree of accuracy. In some embodiments, the plurality of tracking markers **720** can be mounted (or otherwise secured) thereon an outer surface of the robot **15**, such as, for example and without limitation, on the base **25** of the robot **15**, or the robot arm **23**. In some embodiments, the plurality of tracking markers **720** can be configured to track the movement of the robot **15** arm, the end-effectuator **30**, and/or the surgical instrument **35**. In some embodiments, the computer **100** can utilize the tracking information to calculate the orientation and coordinates of

the distal tip 30a of the surgical instrument 35 based on encoder counts along the x-axis 66, y-axis 68, z-axis 70, the Z-tube axis 64, and the roll 62 and pitch 60 axes. Further, in some embodiments, the plurality of tracking markers 720 can be positioned on the base 25 of the robot 15 spaced from the surgical field 17 to reduce the likelihood of being obscured by the surgeon, surgical tools, or other parts of the robot 15. In some embodiments, at least one tracking marker 720 of the plurality of tracking markers 720 can be mounted or otherwise secured to the end-effector 30. In some embodiments, the positioning of one or more tracking markers 720 on the end-effector 30 can maximize the accuracy of the positional measurements by serving to check or verify the end-effector 30 position (calculated from the positional information from the markers on the base 25 of the robot 15 and the encoder counts of the x-(66), y-(68), roll 62, pitch 60, and Z-tube axes 64).

(148) In some further embodiments, at least one optical marker of the plurality of optical tracking markers 720 can be positioned on the robot 15 between the base 25 of the robot 15 and the end-effector 30 instead of, or in addition to, the markers 720 on the base 25 of the robot 15, (see FIG. 16). In some embodiments, the at least one tracking marker 720 can be mounted to a portion of the robot 15 that effects movement of the end-effector 30 and/or surgical instrument 35 along the x-axis to enable the tracking marker 720 to move along the x-axis 66 as the end-effector 30 and surgical instrument 35 move along the x-axis 66 (see FIG. 76). The placement of the tracking markers 720 in this way can reduce the likelihood of a surgeon blocking the tracking marker 720 from the cameras or detection device, or the tracking marker 720 becoming an obstruction to surgery. In certain embodiments, because of the high accuracy in calculating the orientation and position of the end-effector 30 based on the tracking marker 720 outputs and/or encoder counts from each axis, it can be possible to very accurately determine the position of the end-effector 30. For example, in some embodiments, without requiring knowledge of the counts of axis encoders for the z-axis 70, which is between the x-axis 66 and the base 25, knowing only the position of the markers 720 on the x-axis 66 and the counts of encoders on the y-(68), roll 62, pitch 60, and Z-tube axes 64 can enable computation of the position of the end-effector 30. In some embodiments, the placement of markers 720 on any intermediate axis of the robot 15 can permit the exact position of the end-effector 30 to be calculated based on location of such markers 720 and counts of encoders on axes (66, 62, 60, 64) between the markers 720 and the end-effector 30. In some embodiments, from the configuration of the robot 15 (see for example, FIG. 2), the order of axes from the base 25 to the end-effector 30 is z-(70) then x-(66) then y-(68) then roll 62 then pitch 60 then Z-tube 64. Therefore, for example, within embodiments in which tracking markers 720 are placed on the housing 27 of the robot 15 that moves with the roll 62 axis, the locations of such tracking markers 720 and the encoder counts of the pitch 60 and Z-tube axes 64 can be sufficient to calculate the end-effector 30 position.

(149) In some embodiments, when the surgical instrument 35 is advanced into the tissue of the patient 18 with the assistance of a guide tube 50, the surgical instrument 35 can comprise a stop mechanism 52 that is configured to prevent the surgical instrument 35 from advancing when it reaches a predetermined amount of protrusion (see for example, FIGS. 17A-B). In some embodiments, by knowing the lengths of the guide tube 50 and the surgical instrument 35, the distance between the respective ends of the surgical instrument 35, and the location where the stop mechanism 52 is attached, it is possible to determine the maximum distance past the end of the guide tube 50 that the surgical instrument 35 can protrude.

(150) In some embodiments, it can be desirable to monitor not just the maximum protrusion distance of the surgical instrument 35, but also the actual protrusion distance at any instant during the insertion process. Therefore, in some embodiments, the robot 15 can substantially continuously monitor the protrusion distance, and in some embodiments, the distance can be displayed on a display (such as display means 29). In some embodiments, protrusion distance can be substantially continuously monitored using a spring-loaded plunger 54 including a spring-loaded mechanism 55a and sensor pad 55b that has a coupled wiper 56 (see for example FIG. 17B). In some embodiments, the stop mechanism 52 on the surgical instrument 35 can be configured to contact the spring-loaded mechanism 55 well before it encounters the end of the guide tube 50. In some embodiments, when the wiper 56 moves across the position sensor pad 55b, its linear position is sampled, thereby permitting calculation of the distance by which the surgical instrument 35 protrudes past the end of the guide tube 50 substantially in real-time. In some embodiments, any conventional linear encoding mechanism can be used to monitor the plunger's depth of depression and transmit that information to the computer 100 as further described herein.

(151) Some embodiments include instruments that enable the stop on a drill bit 42 to be manually adjusted with reference to markings 44 on the drill bit 42. For example, FIGS. 17C-17E depict tools for manually adjusting a drill stop 46 with reference to drill bit markings 44 in accordance with one embodiment of the invention. As shown, in some embodiments, the drill bit 42 can include release mechanisms 48 on each end of the drill stop 46. In some embodiments, if the release 48 on one end of the drill stop 46 is pulled, it is possible to move the drill stop 46 up the shaft of the drill bit 42. In some embodiments, if the release 48 on the other end of the drill stop 46

is pulled, it is possible to move the drill stop **46** down the shaft (see the direction of movement in FIGS. **17D** and **17E**). In some embodiments, if neither release mechanism **48** is pulled, the drill stop **46** will not move in either direction, even if bumped.

(152) Some embodiments include the ability to lock and hold the drill bit **42** in a set position relative to the tube **50** in which it is housed. For example, in some embodiments, the drill bit **42** can be locked by locking the drill stop **46** relative to the tube **50** using a locking mechanism. FIGS. **17F-J** illustrates tools for locking and holding a drill bit **42** in a set position in accordance with one embodiment of the invention. In some embodiments, the locking mechanism **49** shown in FIG. **17H** can comprise two clam shells **49** (shown in FIG. **17F**). In some embodiments, a drill bit **42** can be locked into position by assembling the clam shells around the drill stop **46** (shown in FIG. **17G**). This feature allows the user to lock the drill bit **42** in a position such that the tip slightly protrudes past the end of the tube **50** (see FIGS. **17I** and **17J**). In this position, the user can force the tube **50** to penetrate through soft tissues to force the tube **50** to contact bone (for example during a percutaneous spine screw insertion).

(153) In some further embodiments, the end-effectuator **30** can be configured not block the tracking optical markers **720** or interfere with the surgeon. For example, in some embodiments, the end-effectuator **30** can comprise a clearance mechanism **33** including an actuator **33a** that permits this configuration, as depicted in FIGS. **18A** and **18B**. As shown, the guide tube **50** can be secured within a housing of the end-effectuator **30** with two shafts **32**. In some embodiments, the shafts **32** move relative to one other, due to a parallelogram effect of the clearance mechanism **33**, the position of the guide tube **50** can mimic the position of the end-effectuator **30** (see FIG. **18B**).

(154) In applications such as cervical or lumbar fusion surgery, it can be beneficial to apply distraction or compression across one or more levels of the spine (anteriorly or posteriorly) before locking hardware in place. In some embodiments, the end-effectuator **30** can comprise an attachment element **37** that is configured to apply such forces (see for example FIGS. **19A-B**). In some embodiments, the end-effectuator **30** attachment element **37** can be configured for coupling to the end-effectuator **30** at substantially the same location as the clearance mechanism **33**. In some embodiments, the end-effectuator **30** with attachment element **37** snaps into the same place as the end-effectuator **30** without the attachment element **37**. In some embodiments, during use of the end-effectuator **30** attachment element **37**, the relative movement of the two shafts **32** caused by angulation **30e** will not cause movement in the pitch **60** direction and will instead cause distraction (illustrated as moving from an attachment element **37** distance **37a** in FIG. **19A** to distance **37b** in FIG. **19B**). Further, although shaft **32** movement as shown in FIGS. **19A-B** would cause distraction, rotation of the actuator **33a** in the opposite direction to that represented by **30e** would cause compression (i.e. the distance **37b** in FIG. **19B** would move towards the distance **37a** in FIG. **19A**).

(155) In view of the embodiments described hereinbefore, some embodiments that can be implemented in accordance with the disclosed subject matter can be better appreciated with reference to the flowcharts in FIGS. **24-33**. For purposes of simplicity of explanation, the method disclosed by the embodiments described herein is presented and described as a series of steps; however, it is to be understood and appreciated that the claimed subject matter is not limited by the order of acts, as some acts may occur in different orders and/or concurrently with other acts from that shown and described herein. For example, the various methods or processes of some embodiments of the invention can alternatively be represented as a series of interrelated states or events, such as in a state diagram. Furthermore, not all illustrated acts may be required to implement a method in accordance with some embodiments of the invention. Further yet, two or more of the disclosed methods or processes can be implemented in combination with each other, to accomplish one or more features or advantages herein described.

(156) It should be further appreciated that the methods disclosed in the various embodiments described throughout the subject specification can be stored on an article of manufacture, or computer-readable medium, to facilitate transporting and transferring such methods to a computing device (e.g., a desktop computer, a mobile computer, a mobile telephone, a blade computer, a programmable logic controller, and the like) for execution, and thus implementation, by a processor of the computing device or for storage in a memory thereof.

(157) In some embodiments, the surgical robot **15** can adjust its position automatically continuously or substantially continuously in order to move the end-effectuator **30** to an intended (i.e. planned) position. For example, in some embodiments, the surgical robot **15** can adjust its position automatically continuously or substantially continuously based on the current position of the end-effectuator **30** and surgical target as provided by a current snapshot of tracking markers, LPS, or other tracking data. It should further be appreciated that certain position adjustment strategies can be inefficient. For example, an inefficient strategy for the robot **15** to find a target location can be an iterative algorithm to estimate the necessary direction of movement, move toward the target location, and then assess a mismatch between a current location and the target location (the mismatch referred to as an error), and estimate a new direction, repeating the cycle of estimate-movement-assessment until

the target location is reached within a satisfactory error. Conversely, the position adjustment strategies in accordance with some embodiments of the invention are substantively more efficient than iterative strategies. For example, in some embodiments, a surgical robot **15** can make movements and adjust its location by calibrating the relative directions of motions in each axis (permitting computation via execution of software or firmware with the computer **100**) at each frame of tracking data, of a unique set of necessary motor encoder counts that can cause each of the individual axes to move to the correct location. In some embodiments, the Cartesian design of the disclosed robot **15** can permit such a calibration to be made by establishing a coordinate system for the robot **15** and determining key axes of rotation.

(158) As described in greater detail below, in some embodiments, methods for calibrating the relative directions of the robot's **15** axes can utilize a sequence of carefully planned movements, each in a single axis. In some embodiments, during these moves, temporary tracking markers **720** are attached to the end-effector **30** to capture the motion of the end-effector **30**. It should be appreciated that the disclosed methods do not require the axes of the robot **15** to be exactly or substantially perpendicular, nor do they require the vector along which a particular axis moves (such as the x-axis **66**) to coincide with the vector about which rotation occurs (such as pitch **60**, which occurs primarily about the x-axis **66**). In certain embodiments, the disclosed methods include motion along a specific robot **15** axis that occurs in a straight line. In some embodiments, the disclosed methods for calibrating the relative directions of movement of the robot's **15** axes can utilize one or more frames of tracking data captured at the ends of individual moves made in x-(**66**), y-(**68**), roll (**62**), pitch (**60**), and Z-tube axes **64** from markers **720** temporarily attached to the end-effector's **30** guide tube **50**. In some embodiments, when moving individual axes, all other axes can be configured at the zero position (for example, the position where the encoder for the axis reads 0 counts). Additionally or alternatively, one or more frames of tracking data with all robot **15** axes at 0 counts (neutral position) may be necessary, and one or more frames of data with the temporary markers **720** rotated to a different position about the longitudinal axis of the guide tube **50** may be necessary. In some embodiments, the marker **720** positions from these moves can be used to establish a Cartesian coordinate system for the robot **15** in which the origin (0,0,0) is through the center of the end-effector **30** and is at the location along the end-effector **30** closest to where pitch **60** occurs. Additionally or alternatively, in some embodiments, this coordinate system can be rotated to an alignment in which y-axis **68** movement of the robot **15** can occur exactly or substantially along the coordinate system's y-axis **68**, while x-axis **66** movement of the robot **15** occurs substantially perpendicular to the y-axis **68**, but by construction of the coordinate system, without resulting in any change in the z-axis **70** coordinate. In certain embodiments, the steps for establishing the robot's **15** coordinate system based at least on the foregoing individual moves can comprise the following: First, from the initial and final positions of the manual rotation of tracking markers **720** about the long axis of the end-effector **30**, a finite helical axis of motion is calculated, which can be represented by a vector that is centered in and aligned with the end-effector **30**. It should be appreciated that methods for calculating a finite helical axis of motion from two positions of three or more markers are described in the literature, for example, by Spoor and Veldpaus (Spoor, C. W. and F. E. Veldpaus, "Rigid body motion calculated from spatial co-ordinates of markers," J Biomech 13(4): 391-393 (1980)). In some embodiments, rather than calculating the helical axis, the vector that is centered in and aligned with the end-effector **30** can be defined, or constructed, by interconnecting two points that are attached to two separate rigid bodies that can be temporarily affixed to the entry and exit of the guide tube **50** on the Z-tube axis **64**. In this instance, each of the two rigid bodies can include at least one tracking marker **720** (e.g., one tracking marker **720**, two tracking markers **720**, three tracking markers **720**, more than three tracking markers **720**, etc.), and a calibration can be performed that provides information indicative of the locations on the rigid bodies that are adjacent to the entry and exit of the guide tube **50** relative to the tracking markers.

(159) A second helical axis can be calculated from the pitch **60** movements, providing a vector substantially parallel to the x-axis of the robot **15** but also close to perpendicular with the first helical axis calculated. In some embodiments, the closest point on the first helical axis to the second helical axis (or vector aligned with the end-effector **30**) is calculated using simple geometry and used to define the origin of the robot's coordinate system (0,0,0). A third helical axis is calculated from the two positions of the roll **62** axis. In certain scenarios, it cannot be assumed that the vector about which roll occurs (third helical axis) and the vector along which the y-axis **68** moves are exactly or substantially parallel. Moreover, it cannot be assumed that the vector about which pitch **60** occurs and the vector along which x-axis **66** motion occurs are exactly or substantially parallel. Vectors for x-axis **66** and y-axis **68** motion can be determined from neutral and extended positions of x-axis **66** and y-axis **68** and stored separately. As described herein, in some embodiments, the coordinate system can be realigned to enable y-axis movement of the robot **15** to occur exactly or substantially in the y-axis **68** direction of the coordinate system, and x-axis **66** movement of the robot **15** without any change in the z-coordinate (**70**). In general, to perform such a transformation of coordinate systems, a series of rotations about a coordinate axis is performed and applied to

every point of interest in the current coordinate system. Each point is then considered to be represented in the new coordinate system. In some embodiments, to apply a rotation of a point represented by a 3×1 vector about a particular axis, the vector can be pre-multiplied by a 3×3 rotation matrix. The 3×3 rotation matrix for a rotation of R.sub.x degrees about the x-axis is:

$$(160) \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos R_x & -\sin R_x \\ 0 & \sin R_x & \cos R_x \end{bmatrix}$$

The 3×3 rotation matrix for a rotation of R.sub.y degrees about the y-axis is:

$$(161) \begin{bmatrix} \cos R_y & 0 & \sin R_y \\ 0 & 1 & 0 \\ -\sin R_y & 0 & \cos R_y \end{bmatrix}$$

The 3×3 rotation matrix for a rotation of R.sub.z degrees about the z-axis is:

$$(162) \begin{bmatrix} \cos R_z & -\sin R_z & 0 \\ \sin R_z & \cos R_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(163) In some embodiments, to transform coordinate systems, a series of three rotations can be performed. For example, such rotations can be applied to all vectors and points of interest in the current coordinate system, including the x-movement vector, y-movement vector and each of the helical axis, to align the y movement vector with the new coordinate system's y-axis, and to align the x movement vector as closely as possible to the new coordinate system's x-axis at $z=0$. It should be appreciated that more than one possible sequence of three rotations can be performed to achieve substantially the same goal. For example, in some embodiments, a sequence of three rotations can comprise (1) a rotation about x using an R.sub.x value appropriate to rotate the y-movement vector until its z coordinate equal 0, followed by (2) a rotation about z using an R.sub.z value appropriate to rotate the y-movement vector until its x coordinate equal 0, followed by (3) a rotation about y using an R.sub.y value appropriate to rotate the x-movement vector until its z coordinate equals 0. In some embodiments, to find the rotation angle appropriate to achieve a given rotation, the arctangent function can be utilized. For example, in some embodiments, the angle needed to rotate a point or vector (x_1, y_1, z_1) about the z axis to $y_1=0$ is— $\arctan(y_1/x_1)$.

(164) It should be appreciated that after transformation of the coordinate system, in some embodiments, although the new coordinate system is aligned such that the y-movement axis of the surgical robot **15** is exactly or substantially exactly aligned with the coordinate system's y-axis **68**, the roll **62** rotation movement of the robot **15** should not be assumed to occur exactly or substantially exactly about a vector aligned with the coordinate system's y-axis **68**. Similarly, in some embodiments, the pitch **60** movement of the surgical robot **15** should not be assumed to occur exactly or substantially exactly about a vector aligned with the coordinate system's x-axis. In some embodiments, in roll **62** and pitch **60** rotational movement there can be linear and orientational “offsets” from the helical axis of motion to the nearest coordinate axis. In some embodiments, from the helical axes determined above using tracked markers, such offsets can be calculated and retained (e.g., stored in a computing device's memory) so that for any rotation occurring during operation, the offsets can be applied, rotation can be performed, and then negative offsets can be applied so that positional change occurring with rotation motion accounts for the true center of rotation.

(165) In some embodiments, during tracking, the desired trajectory can be first calculated in the medical image coordinate system, then transformed to the robot **15** coordinate system based at least on known relative locations of active markers. For example, in some embodiments, conventional light-emitting markers and/or conventional reflective markers associated with an optical tracking system **3417** can be used (see for example active markers **720** in FIG. **20A**). In other embodiments, conventional electromagnetic sensors associated with an electromagnetic tracking system can be used. In some other embodiments, radio-opaque markers (for example markers **730** shown in FIG. **20A**) can be used with a CT imaging system. In some embodiments, radio-opaque markers **730** (spheres formed, at least in part from metal or other dense material), can be used to provide a marker **730** that can at least partially absorb x-rays to produce a highly contrasted image of the sphere in a CT scan image.

(166) In some embodiments, the necessary counts for the end-effectuator **30** to reach the desired position in the robot's **15** coordinate system can be calculated based on the following example process. First the necessary counts to reach the desired angular orientation can be calculated. In some embodiments, a series of three rotations can be applied to shift the coordinate system temporarily to a new coordinate system in which the y-axis **68** coincides or substantially coincides with the helical axis of motion for roll **62**, and the x-axis **66** is largely aligned with the helical axis of motion for pitch **60** and by definition, and the helical axis of motion for pitch **60** has constant $z=0$. Then, the number of counts necessary to achieve the desired pitch **60** can be determined, keeping track of how this pitch **60** can affect roll **62**. In one implementation, to find the necessary counts to achieve the desired pitch, the change in pitch angle **60** can be multiplied by the previously calibrated motor counts per degree for pitch. The

change in roll **62** caused by this change in pitch **60** can be calculated from the orientation of the helical axis and the rotation angle (pitch) about the helical axis. Then, the necessary roll **62** to get to the desired roll **62** to reach the planned trajectory alignment can be calculated, with the benefit that applying roll **62** does not, by definition of the coordinate system, result in any further change in pitch. The coordinate system is then shifted back to the previously described robot **15** coordinate system by the inverse of the three rotations applied above. Then the necessary counts to reach the desired x-axis **66** position can be calculated, also keeping track of how this x-axis **66** position change will affect y-axis **68** position. Then the necessary y-axis **68** counts to reach the desired y-axis position can be readily calculated with the benefit that changing the y-axis **68** coordinate can have no effect on any other axis since the y-axis motion vector is by definition aligned with the robot's y-axis **68**. In a scenario in which the Z-tube **50** position is being actively controlled, the orientation of the Z-tube **50** movement vector is adjusted when adjusting roll **62** and pitch **60** and the counts necessary to move it to the desired position along the trajectory vector is calculated from the offset. In some embodiments, after the necessary counts to achieve the desired positions in all axes are calculated as described, these counts can be sent as computer-accessible instructions (e.g., computer-readable and/or computer-executable instructions) to respective controllers for each axis in order to move the axes to the computed positions.

(167) FIG. **24** is a flowchart of a method **2400** for positioning and advancing through soft tissue in accordance with one or more aspects according to one embodiment of the invention. As shown, in some embodiments, at block **2410**, a medical image is accessed (e.g., received, retrieved, or otherwise acquired). As described herein, the medical image can be a 3D anatomical image scan including, but not limited to a CT scan, a magnetic resonance imaging scan (hereinafter referred to as an "MRI scan"), an X-ray image, or other anatomical scan. It should be appreciated that any 3D anatomical scan may be utilized with the surgical robot **15** and is within the scope of the present invention. In some embodiments, at block **2420**, a targeting fixture **690** is calibrated to the medical image. In some embodiments, the calibration can be semi-automated or automated. In some embodiments, at block **2430**, data indicative of an intended trajectory associated with the medical image is received. In some embodiments, at block **2440**, a robot **15** is substantially maintained on the intended trajectory. In some embodiments, a control platform (for example, platform **3400** shown in FIG. **34**) can adjust movement of the robot **15** in order to substantially maintain the intended trajectory.

(168) FIGS. **25-26** are flowcharts of methods for calibrating a targeting fixture **690** to a medical image in accordance with one or more embodiments of the invention. As shown in FIG. **25**, in some embodiments, the method **2500** can embody a semi-automated calibration method and can be implemented (e.g., executed) as part of block **2420** in certain scenarios. In some embodiments, at block **2510**, data indicative of a medical image having a representation of a plurality of radio-opaque markers (for example radio-opaque markers **730**) is received. In one embodiment, as described herein, such plurality can contain four radio-opaque markers **730**. In some embodiments, at block **2520**, a geometrical center for each radio-opaque marker **730** is determined in a coordinate system associated with the medical image. In some embodiments, image thresholding can be utilized to define one or more edges of each radio-opaque marker **730** and a geometrical center thereof. Thresholding refers to an image processing technique in which pixel intensity within a 2D region can be monitored. For example, the x, y positions (for instance expressed in mm) of pixels of an intensity that reach a predetermined value can be retrieved. Stated similarly, the threshold refers to the transition pixel intensity from light to dark. In some embodiments, on 2D slices of the medical image, the radio-opaque marker **730** can appear light and the adjacent space (such as tissue or air) can appear dark. In some embodiments, displaying pixels that satisfy a thresholding criterion at an intensity encountered at the edge of a radio-opaque marker can yield a largely circular trace outlining the marker on the medical image. Since in some embodiments, markers **730** can be spherical, a method for finding the center of the marker **730** in a 2D view can include firstly restricting the 2D view to a sampling region with the high-intensity image of the sphere toward the center of the region and pixels of lower intensity toward the outer edges of the region. Secondly, the method can include finding the mean x threshold position (e.g., the maximum x coordinate of pixels satisfying the threshold criterion plus minimum x coordinate of pixels satisfying the threshold criterion divided by two), and finding the mean y threshold position using a similar method. In some embodiments, the center of the sphere can be found by determining 2D centers of slices through the same marker **730** in two orthogonal views. For example, in some embodiments, the method can include finding mean x and mean y from an xy slice, then finding mean x and mean z from an xz slice to get a mean x, y, and z axis coordinate representing the center of the marker **730**. Further, upon or after the mean x, mean y, and mean z are found, new xy and xz slices can be evaluated again and the maximum and minimum x, y, and z threshold values can be again determined to evaluate the dimensions of the thresholded object in each view. It can be appreciated from this method that in some embodiments, a non-spherical object of high intensity, such as a small process of cortical bone extending away from the side of the spine, may fail to satisfy (1) a condition where there is high intensity near the middle of the region, but low intensity all around, since the process may extend out

of the region in one or more directions; or (2) a condition where the dimensions in x, y, and z of the centered object do not match each other (e.g., non-spherical case).

(169) As shown in FIG. 25, in some embodiments, at block **2530**, it is ascertained if one centered sphere is determined for each radio-opaque marker **730** for the fixture being calibrated. In some embodiments, when at least one such sphere is not determined, or identified, the threshold setting is adjusted and flow is directed to block **2510**. In some embodiments, at block **2540**, each centered sphere is mapped to each radio-opaque marker **730** of the plurality of radio-opaque markers **730**. As shown, in some embodiments, block **2540** can represent a mapping action which, in some embodiments, can comprise implementing a sorting process to establish a specific centered sphere is associated with a specific one of the plurality of radio-opaque markers **730**. In some embodiments, a plurality of radio-opaque markers **730** contains four radio-opaque markers **730** (represented, for example, as OP1, OP2, OP3, and OP4). In some embodiments, the sorting process can map each one of four centered markers **730** to one of OP1, OP2, OP3, or OP4. In some embodiments, the sorting process can distinguish a specific marker **730** by measuring inter-marker distances from mean positions of the four unidentified markers **730**, and comparing such distances to extant inter-marker distances (for example, those that are pre-measured and retained in memory, such as mass storage device **3404**) for each marker **730** on a marker fixture. In some embodiments, the opaque markers **730** on the fixture **690** can be placed asymmetrically, each marker **730** can be identified from a unique set of inter-marker distances corresponding to such marker **730**. For example, in some embodiments where the sum of inter-marker distances of one unknown marker **730** relative to the other three markers **730** measured from the medical image is D, a single physical marker **730** (one of OP1, OP2, OP3, or OP4) can have a matching inter-marker distance sum within a specified tolerance (such as ± 1 mm) of D. In some embodiments, at block **2550**, coordinates of each centered sphere can be retained (for example in memory of a computer platform **3400**). As described herein, in some embodiments, such coordinates can be utilized in a process for tracking movement of a robot **15**.

(170) Some embodiments include method **2600** (shown as a flowchart in FIG. 26) that can embody an automated calibration method and can be implemented (e.g., executed) as part of block **2420** in certain scenarios. In some embodiments, at block **2605**, for a Z position, an x-y grid of test area squares is created. In some embodiments, each test area square can be larger than the diameter of a sphere (a radio-opaque marker **730**) associated with a targeting fixture **690** comprised of material that, when imaged, appears as opaque. In some embodiments, each test area square can be at least partially overlapping with at least one adjacent test area square. In one embodiment of the invention, a nearly half the surface of a test area square can overlap with the surface of an adjacent test area square. In some embodiments, at block **2610**, calibration is initiated at place $Z=0$, x-y grid row 0, x-y grid column 0. In some embodiments, at block **2615**, borders of a medical image within a first test area square are determined. It should be appreciated that in some embodiments, a sphere can be rendered as a circular area, but a section of bone represented in the medical image can be asymmetrical. In some embodiments, a thresholding process in accordance with one or more aspects described herein can be implemented to exclude one or more invalid markers **730** by assessing if the x, y, and z axes boundaries of the object are of substantially equivalent dimensions, consistent with the shape being spherical.

(171) In some embodiments, at block **2620**, it is determined if a maximum (max) border coordinate is less than the maximum coordinate of the test area, and a minimum (min) border coordinate is greater than the minimum coordinate of the test area, and vertical span of features rendered in the image are equal or substantially equal to horizontal span of such features. As shown in FIG. 26, in some embodiments, in the negative case, flow is directed to block **2645**, at which the first test area is moved to next grid location and next Z plane. Conversely, in case the three foregoing conditions are fulfilled, flow is directed to block **2625**, at which X coordinate and Y coordinate are centered at the center of the current test area. In some embodiments, at block **2630**, Z coordinate is probed by creating a second test area square spanning upwards and downwards in XZ plane and/or YZ plane to determine one or more borders of an object. In some embodiments, at block **2635**, it is determined if borders of the object observed in XZ plane are of substantially equivalent relative spacing (vertically and horizontally) to borders in the x-y plane, consistent with the shape of the object being spherical. In some embodiments, when such borders are of different spacing, flow is directed to block **2645**. Conversely, when spacing of such borders is substantially equivalent between views, a sphere having a center at X coordinate, Y coordinate, and Z coordinate is identified at block **2640** and flow is directed to block **2645**.

(172) In some embodiments, at block **2650**, it is determined if last row and column in x-y grid are reached and last Z plane is reached as a result of updating the first test area at block **2645**. In some embodiments, in the negative case, flow is directed to block **2615**, in which the first area is the updated instance of a prior first area, with the flow reiterating one or more of blocks **2620** through **2645**. Conversely, in the affirmative case, flow is directed to block **2655** at which invalid marker(s) **730** can be excluded. In some embodiments, a paring process can be implemented to exclude one or more invalid markers **730**. For this paring process, in some embodiments, the

known spacings between the N radio-opaque markers **730** (with N a natural number) on the targeting fixture **690** and each other radio-opaque marker **730** on the targeting fixture **690** can be compared to the markers **730** that have been found on the medical image. In a scenario in which more than N number of markers **730** can be found on the medical image, any sphere found on the medical image that does not have spacings relative to N-1 other markers **730** that are within an acceptable tolerance of known spacings retained, for example, on a list can be considered to be invalid. For example, if a targeting fixture **690** has four radio-opaque markers **730**, there are six known spacings, with each marker **730** having a quantifiable spacing relative to three other markers **730**: the inter-marker spacings for markers 1-2, 1-3, 1-4, 2-3, 2-4, and 3-4. On the 3D medical image of the targeting fixture **690**, in some embodiments, if five potential markers **730** are found on the medical image, their inter-marker spacings can be calculated. In this scenario, there are 10 inter-marker spacings: 1-2, 1-3, 1-4, 1-5, 2-3, 2-4, 2-5, 3-4, 3-5, and 4-5, with each sphere having a quantifiable spacing relative to four other markers **730**. Considering each of the five potential markers **730** individually, if any one of such five markers **730** does not have three of its four inter-marker spacings within a very small distance of the spacings on the list of six previously quantified known spacings, it is considered invalid.

(173) In some embodiments, at block **2660**, each centered radio-opaque marker **730**, identified at block **2640**, can be mapped to each radio-opaque marker **730** of a plurality of radio-opaque markers **730**. In some embodiments, a sorting process in accordance with one or more aspects described herein can be implemented to map such markers **730** to radio opaque markers **730**. In some embodiments, at block **2665**, coordinates of each centered sphere can be retained (e.g., in memory of a computer platform **3400**). As described herein, in some embodiments, such coordinates can be utilized in a process for tracking movement of a robot **15**. In some embodiments, during tracking, the established (e.g., calibrated) spatial relationship between active markers **720** and radio-opaque markers **730** can be utilized to transform the coordinate system from the coordinate system of the medical image to the coordinate system of the tracking system **3417**, or vice versa. Some embodiments include a process for transforming coordinates from the medical image's coordinate system to the tracking system's coordinate system can include a fixture **690** comprising four radio-opaque markers OP1, OP2, OP3, and OP4 (for example radio-opaque markers **730**) in a rigidly fixed position relative to four active markers AM1, AM2, AM3, AM4 (for example, active markers **720**). In some embodiments, at the time the calibration of the fixture **690** occurred, this positional relationship can be retained in a computer memory (e.g., system memory **3412**) for later access on real-time or substantially on real-time in a set of four arbitrary reference Cartesian coordinate systems that can be readily reachable through transformations at any later frame of data. In some embodiments, each reference coordinate system can utilize an unambiguous positioning of three of the active markers **720**. Some embodiments can include a reference coordinate system for AM1, AM2, and AM3 can be coordinate system in which AM1 can be positioned at the origin (e.g., the three-dimensional vector (0,0,0)); AM2 can be positioned on the x-axis (e.g., x-coordinate $AM2x > 0$, y-coordinate $AM2y = 0$, and z-coordinate $AM2z = 0$); and AM3 can be positioned on the x-y plane (e.g., x-coordinate $AM3x$ unrestricted, y-coordinate $AM3y > 0$, and z-coordinated $AM3z = 0$). Some embodiments include a method to generate a transformation to such coordinate system can comprise (1) translation of AM1, AM2, AM3, OP1, OP2, OP3, and OP4 in a manner that AM1 vector position is (0,0,0); (2) rotation about the x-axis by an angle suitable to position AM2 at $z=0$ (e.g., rotation applied to AM2, AM3 and OP1-OP4); (3) rotation about the z-axis by an angle suitable to position AM2 at $y=0$ and $x > 0$ (e.g., rotation applied to AM2, AM3 and OP1-OP4); (4) rotation about the x-axis by an angle suitable to position AM3 at $z=0$ and $y > 0$ (e.g., rotation applied to AM3 and OP1-OP4). It should be appreciated that, in some embodiments, it is unnecessary to retain these transformations in computer memory, for example; rather, the information retained for later access can be the coordinates of AM1-AM3 and OP1-OP4 in such reference coordinate system. In some embodiments, another such reference coordinate system can transform OP1-OP4 by utilizing AM2, AM3, and AM4. In some embodiments, another such reference coordinate system can transform OP1-OP4 by utilizing AM1, AM3, and AM4. In some further embodiments, another such reference coordinate system can transform OP1-OP4 by utilizing AM1, AM2, and AM4.

(174) In some embodiments, at the time of tracking, during any given frame of data, the coordinates of the active markers AM1-AM4 can be provided by the tracking system **3417**. In some embodiments, by utilizing markers AM1, AM2, and AM3, transformations suitable to reach the conditions of the reference coordinate system can be applied. In some embodiments, such transformations can position AM1, AM2, and AM3 on the x-y plane in a position in proximity to the position that was earlier stored in computer memory for this reference coordinate system. In some embodiments, for example, to achieve a best fit of the triad of active markers **720** on their stored location, a least squares algorithm can be utilized to apply an offset and rotation to the triad of markers **720**. In one implementation, the least squares algorithm can be implemented as described by Sneath (Sneath P. H. A., Trend-surface analysis of transformation grids, J. Zoology 151, 65-122 (1967)). In some embodiments, transformations suitable to reach the reference coordinate system, including the least squares adjustment, can be

in memory (e.g., system memory **3412** and/or mass storage device **3404**). In some embodiments, the retained coordinates of OP1-OP4 in such reference coordinate system can be retrieved and the inverse of the retained transformations to reach the reference coordinate system can be applied to such coordinates. It should be appreciated that the new coordinates of OP1-OP4 (the coordinates resulting from application of the inverse of the transformations) are in the coordinate system of the tracking system **3417**. Similarly, in some embodiments, by utilizing the remaining three triads of active markers **720**, the coordinates of OP1-OP4 can be retrieved.

(175) In some embodiments, the four sets of OP1-OP4 coordinates in the tracking system's coordinate system that can be calculated from different triads of active markers **720** are contemplated to have coordinates that are approximately equivalent. In some embodiments, when coordinates are not equivalent, the data set can be analyzed to determine which of the active markers **720** provides non-suitable (or poor) data by assessing how accurately each triad of active markers **720** at the current frame overlays onto the retained positions of active markers **720**. In some other embodiments, when the coordinates are nearly equivalent, a mean value obtained from the four sets can be utilized for each radio-opaque marker **730**. In some embodiments, to transform coordinates of other data (such as trajectories from the medical image coordinate system) to the tracking system's coordinate system, the same transformations can be applied to the data. For example, in some embodiments, the tip and tail of a trajectory vector can be transformed to the four reference coordinate systems and then retrieved with triads of active markers **720** at any frame of data and transformed to the tracking system's coordinate system.

(176) FIG. **27** is a flowchart of a method **2700** for automatically maintaining a surgical robot **15** substantially on a trajectory in accordance some embodiments of the invention. In some embodiments, at block **2710**, data indicative of position of one or more of Z-frame **72** or Z-tube **50** are received. In some embodiments, at block **2720**, data indicative of each robot **15** joint in the surgical robot **15**, such as encoder counts from each axis motor **160**, are accessed. In some embodiments, at block **2730**, a current robot **15** position on a planned trajectory is accessed (the position being represented in a camera coordinate system or the coordinate system of other tracking device). In one embodiment, the planned trajectory can be generated by an operator. For example, the operator (e.g., a surgeon) can scroll and rotate through the image slices until the desired anatomy can be viewed on three windows representing three orthogonal planes (typically sagittal, coronal, and axial slices). The operator can then draw a line at the desired slope and location on one window; the line simultaneously is calculated and appears on the other two windows, constrained by the views of the screens and orientation on the window on which it was drawn.

(177) In another embodiment, a line (e.g., referred to as line *t*) that is fixed on the image both in angle and position represents the desired trajectory; the surgeon has to rotate and scroll the images to align this trajectory to the desired location and orientation on the anatomy. At least one advantage of such embodiment is that it can provide a more complete, holistic picture of the anatomy in relationship to the desired trajectory that may not require the operator to erase and start over or nudge the line after it is drawn, and this process was therefore adopted. In some embodiments, a planned trajectory can be retained in a memory of a computing device (for example, computing device **3401**) that controls the surgical robot **15** or is coupled thereto for use during a specific procedure. In some embodiments, each planned trajectory can be associated with a descriptor that can be retained in memory with the planned trajectory. As an example, the descriptor can be the level and side of the spine where screw insertion is planned.

(178) In another embodiment, the line *t* that is (fixed on the image both in angle and position representing the desired trajectory) is dictated by the current position of the robot's end effector **30**, or by an extrapolation of the end effector guide tube **50** if an instrument **35** were to extend from it along the same vector. In some embodiments, as the robot **15** is driven manually out over the patient **18** by activating motors **160** controlling individual or combined axes **64**, **66**, **68**, **70**, the position of this extrapolated line (robot's end effector **30**) is updated on the medical image, based on markers **720** attached to the robot, conventional encoders showing current position of an axis, or a combination of these registers. In some embodiments, when the desired trajectory is reached, that vector's position in the medical image coordinate system is stored into the computer memory (for example in memory of a computer platform **3400**) so that later, when recalled, the robot **15** will move automatically in the horizontal plane to intersect with this vector. In some embodiments, instead of manually driving the robot **15** by activating motors **160**, the robot's axes can be put in a passive state. In some embodiments, in the passive state, the markers **720** continue to collect data on the robot arm **23** position and encoders on each axis **64**, **66**, **68**, **70** continue to provide information regarding the position of the axis; therefore the position of an extrapolated line can be updated on the medical image as the passive robot **15** is dragged into any orientation and position in the horizontal plane. In some embodiments, when a desired trajectory is reached, the position can be stored into the computer memory. Some embodiments include conventional software control or a conventional switch activation capable of placing the robot **15** into an active state to immediately rigidly hold the position or trajectory, and to begin compensating for movement of the patient **18**.

(179) In some further embodiments, the computing device that implements the method **2700** or that is coupled to

the surgical robot **15** can render one or more planned trajectories. Such information can permit confirming that the trajectories planned are within the range of the robot's **15** reach by calculating the necessary motor **160** encoder counts to reach each desired trajectory, and assessing if the counts are within the range of possible counts of each axis.

(180) In some embodiments, information including whether each trajectory is in range, and how close each trajectory is to being out of range can be provided to an agent (such as a surgeon or other user, or equipment). For example, in some embodiments, a display means **29** (such as a display device **3411**) can render (i.e. display) the limits of axis counts or linear or angular positions of one or more axes and the position on each axis where each targeted trajectory is currently located.

(181) In another embodiment, the display device **3411** (for example, a display **150**) can render a view of the horizontal work field as a rectangle with the robot's x-axis **66** movement and y-axis **68** movement ranges defining the horizontal and vertical dimensions of the rectangle, respectively. In some embodiments, marks (for example, circles) on the rectangle can represent the position of each planned trajectory at the current physical location of the robot **15** relative to the patient **18**. In another embodiment, a 3D Cartesian volume can represent the x-axis **66** movement, y-axis **68** movement and z-axis **70** movement ranges of the robot **15**. In some embodiments, line segments or cylinders rendered in the volume can represent the position of each planned trajectory at the current location of the robot **15** relative to the patient **18**. Repositioning of the robot **15** or a patient **18** is performed at this time to a location that is within range of the desired trajectories. In other embodiments, the surgeon can adjust the Z Frame **72** position, which can affect the x-axis **66** range and the y-axis **68** range of trajectories that the robot **15** is capable of reaching (for example, converging trajectories require less x-axis **66** or y-axis reach the lower the robot **15** is in the z-axis **70**). During this time, simultaneously, a screen shows whether tracking markers on the patient **18** and robot **15** are in view of the detection device of the tracking system (for example, optical tracking system **3417** shown in FIG. **34** and cameras **8200** in FIG. **81**). Repositioning of the cameras **8200**, if necessary, is also performed at this time for good visibility or optimal detection of tracking sensors.

(182) In some embodiments, at block **2740**, orientation of an end-effectuator **30** in a robot **15** coordinate system is calculated. In some embodiments, at block **2750**, position of the end-effectuator **30** in the robot **15** coordinate system is calculated. In some embodiments, at block **2760**, a line *t* defining the planned trajectory in the robot **15** coordinate system is determined. In some embodiments, at block **2770**, robot **15** position is locked on the planned trajectory at a current Z level. In some embodiments, at block **2780**, information indicative of quality of the trajectory lock can be supplied. In some embodiments, actual coordinate(s) of the surgical robot **15** can be rendered in conjunction with respective coordinate(s) of the planned trajectory. In some embodiments, aural indicia can be provided based on such quality. For instance, in some embodiments, a high-frequency and/or high-amplitude noise can embody aural indicia suitable to represent a low-quality lock. In some alternative embodiments, a brief melody may be repeatedly played, such as the sound associated with successful recognition of a USB memory device by a computer, to indicate successful lock on the planned trajectory. In other embodiments, a buzz or other warning noise may be played if the robot **15** is unable to reach its target due to the axis being mechanically overpowered, or if the tracking markers **720** are undetectable by cameras **8200** or other marker position sensors.

(183) In some embodiments, at block **2790**, it is determined if a surgical procedure is finished and, in the affirmative case, the flow terminates. In other embodiments, the flow is directed to block **2710**. In some embodiments, the method **2700** can be implemented (i.e., executed) as part of block **2440** in certain scenarios. It should be appreciated that in some embodiments, the method **2700** also can be implemented for any robot **15** having at least one feature that enable movement of the robot **15**.

(184) FIG. **28A** is a flowchart of a method **2800a** for calculating position and/or orientation of an end-effectuator **30** in a robot **15** according to one at least one embodiment of the invention. In some embodiments, the position and/or orientation can be calculated based at least on monitored position of a tracking array **690** mounted on the robot's x-axis **66** and monitored counts of encoders on the y-axis **68**, roll **62**, pitch **60**, and Z-tube axis **64** actuators. In some embodiments, position and/or orientation are calculated in a robot **15** coordinate system. In some embodiments, the method **2800a** can embody one or more of blocks **2740** or **2750**. In some embodiments, at block **2805a**, the current position (i.e., 3D position) of x-axis **66** mounted robot **15** tracking markers is accessed. In some embodiments, at block **2810a**, the current position of the tracking array **690** mounted to the robot **15** is transformed to neutral position. This is a position that was previously stored and represents the position of the tracking array **690** when the robot **15** was at zero counts on each axis between the tracker **690** and the robot **15** base (x-axis **66** and z-axis **70** in this configuration). In some embodiments, the set of transformations (T1) to transform from the current position to the neutral position can be retained in computer memory (for example, the system memory **3412** and/or mass storage device **3404**). In some embodiments, a tip and tail of a line segment representing the vector in line with the end-effectuator **30** can be computed based at least on the process described

herein. In some embodiments, this process can establish a robot **15** coordinate system and calibrate the relative orientations of the axes of movement of the robot **15** where tracking markers **720** can be attached temporarily to the end-effector **30**. In some embodiments, the vector's position in space can be determined by finding the finite helical axis of motion of markers **720** manually rotated to two positions around the guide tube **50**. In some embodiments, the vector's position in space can be determined by connecting a point located at the entry of the guide tube **50** (identified by a temporarily mounted rigid body **690** with tracking markers **720**) to a point located at the exit of the guide tube **50** (identified by a second temporarily mounted rigid body **690** with tracking markers **720**).

(185) In some embodiments, the tip of the line segment can be obtained as the point along the vector that is closest to the vector representing the helical axis of motion during pitch. In some embodiments, the tail of the line segment can be set an arbitrary distance (for example about 100 mm) up the vector aligned with the guide tube **50** and/or first helical axis. In some embodiments, the Cartesian coordinates of such tip and tail positions can be transformed to a coordinate system described herein in which the y-axis **68** movement can coincide with the y-axis **68** of the coordinate system, and the x-axis **66** can be aligned such that x-axis **66** movement can cause the greatest change in direction in the x-axis **66**, moderate change in the y-axis **68**, and no change in the z-axis **70**. In some embodiments, these coordinates can be retained in a computer memory (for example system memory **3412**) for later retrieval. In some embodiments, at block **2815a**, tip and tail coordinates for neutral are accessed (i.e., retrieved). In some embodiments, at block **2820a**, tip and tail are translated along Z-tube **50** neutral unit vector by monitored Z-tube **50** counts. In some embodiments, at block **2825a**, an instantaneous axis of rotation ("IAR") is accessed. The IAR is the same as the helical axis of motion ignoring the element of translation along the helical axis for pitch **60** for neutral. As described earlier, in some embodiments, the vectors for this IAR were previously stored in computer memory at the time the coordinate system of the robot **15** was calibrated. In some embodiments, at block **2830a**, tip coordinate, tail coordinate, and IAR vector direction and location coordinates are transformed (for example, iteratively transformed) to a new coordinate system in which IAR is aligned with X axis. In some embodiments, data indicative of such transformations (T2) can be stored. In some embodiments, at block **2835a**, tip coordinate and tail coordinate are rotated about X axis by pitch **60** angle. In some embodiments, at block **2840a**, tip coordinate and tail coordinate are transformed back by inverse of T2 to the previous coordinate system. In some embodiments, at block **2845a**, previously stored vectors that represent the IAR for roll **62** are accessed. In some embodiments, at block **2850a**, tip coordinate, tail coordinate, IAR coordinate are transformed (for example, iteratively transformed) to a new coordinate system in which IAR is aligned with y-axis **68**. In some embodiments, data indicative of such transformation(s) (T3) can be retained in memory. In some embodiments, at block **2855a**, tip coordinate and tail coordinate are rotated about y-axis **68** by roll **62** angle. In some embodiments, at block **2860a**, tip coordinate and tail coordinate are transformed back by inverse of T3 to the previous coordinate system. In some embodiments, at block **2865a**, tip coordinate and tail coordinate are translated along a y-axis **68** unit vector (e.g., a vector aligned in this coordinate system with the y-axis **68**) by monitored counts. In some embodiments, at block **2870a**, tip coordinate and tail coordinate are transformed back by inverse of T1 to the current coordinate system monitored by the tracking system **3417**.

(186) FIG. **28B** is a flowchart of a method **2800b** for calculating position and/or orientation of an end-effector **30** in a robot **15** in accordance with one embodiment of the invention. In some embodiments, the position and/or the orientation can be calculated based at least on monitored position of a tracking array **690** mounted on the robot's **15** roll **62** axis and monitored counts of encoders on the pitch **60** and Z-tube **50** actuators. In some embodiments, position and/or orientation can be calculated in a robot **15** coordinate system. In accordance with some embodiments of the invention, the method **2800B** can embody one or more of blocks **2740** or **2750**. In some embodiments, at block **2805b**, current position of an array of one or more robot **15** tracking markers **720** is accessed. In some embodiments, the current position is a 3D position and the array of robot **15** tracking markers **720** can be mounted to the roll **62** axis of the robot **15**. In some embodiments, at block **2810b**, the current position of the array of robot **15** tracking markers **720** mounted to the robot **15** is transformed to neutral position. In some embodiments, the neutral position can be a position that was previously stored and can represent the position of the robot **15** tracking array **690** when the robot **15** had zero counts on each axis between the tracker **690** and the robot base **25** (e.g., z-axis **70**, x-axis **68**, y-axis **66**, and roll **62** axis in this configuration). In some embodiments, data indicative of a set of transformations (T1) to go from the current position to the neutral position can be stored in a computer memory (for example, mass storage device **3404** or system memory **3412**).

(187) In some embodiments, in order to establish a robot **15** coordinate system and calibrate the relative orientations of the axes of movement of the robot **15**, a tip and tail of a line segment representing the vector in line with the end-effector **30** with temporarily attached tracking markers **720** is located. In some embodiments, the vector's position in space can be determined by finding the finite helical axis of motion of markers manually rotated to two positions around the guide tube **50**. In other embodiments, the vector's position in space can be

determined by connecting a point located at the entry of the guide tube **50** (identified by a temporarily mounted rigid body **690** with tracking markers **720**) to a point located at the exit of the guide tube **50** (identified by a second temporarily mounted rigid body **690** with tracking markers **720**).

(188) In some embodiments, the tip of the line segment can be found as the point along the vector that is closest to the vector representing the helical axis of motion during pitch. In some embodiments, the tail of the line segment can be set an arbitrary distance (for example, nearly 100 mm) up the vector aligned with the guide tube/first helical axis. In some embodiments, the Cartesian coordinates of these tip and tail positions can be transformed to a coordinate system described herein in which the y-axis **68** movement substantially coincides with the y-axis **68** of the coordinate system, and the x-axis **66** movement is aligned in a manner that, in some embodiments, x-axis **66** movement causes the greatest change in direction in the x-axis **66**, slight change in y-axis **68**, and no change in the z-axis **70**. It should be appreciated that such coordinates can be retained in memory (for example system memory **3412**) for later retrieval. In some embodiments, at block **2815b**, tip and tail coordinates for the neutral position are accessed (i.e., retrieved or otherwise obtained). In some embodiments, at block **2820b**, tip and tail are translated along Z-tube **50** neutral unit vector by monitored Z-tube **50** counts. In some embodiments, at block **2825b**, IAR is accessed. In one implementation, the vectors for this IAR may be available in a computer memory, for example, such vectors may be retained in the computer memory at the time the coordinate system of the robot **15** is calibrated in accordance with one or more embodiments described herein. In some embodiments, at block **2830b**, tip coordinate, tail coordinate, and IAR vector direction and location coordinates are transformed to a new coordinate system in which IAR is aligned with x-axis **66**. In some embodiments, data indicative of the applied transformations (T2) can be retained in a computer memory. In some embodiments, at block **2835b**, tip coordinate and tail coordinate are rotated about x-axis **66** by pitch **60** angle. In some embodiments, at block **2840b**, tip coordinate and tail coordinate are transformed back by applying the inverse of T2 to the previous coordinate system. In some embodiments, at block **2870b**, tip coordinate and tail coordinate are transformed back by applying the inverse of T1 to the current coordinate system monitored by the tracking system **3417**.

(189) FIG. **29** is a flowchart of a method **2900** for determining a line indicative of a trajectory in a robot **15** coordinate system in accordance with one embodiment of the invention. In some embodiments, the trajectory can be a planned trajectory associated with a surgical procedure. In some embodiments, at block **2910**, for a set of current active marker **720** positions on a targeting fixture **690**, respective opaque marker **730** positions are accessed from a rigid body source (such as a fixture **690**). In some embodiments, at block **2920**, an opaque marker **730** position is transformed from a representation in an image coordinate system to a representation in a current active marker **720** coordinate system. In some embodiments, at block **2930**, a planned trajectory is transformed from a representation in the image coordinate system to a representation in the current active marker **720** coordinate system. In some embodiments, at block **2940**, a set of current marker **720** positions on a robot **15** is transformed to a representation in a robot **15** coordinate system. In some embodiments, at block **2950**, the planned trajectory is transformed from a representation in the current active marker **720** coordinate system to the robot **15** coordinate system.

(190) FIG. **30** is a flowchart of a method **3000** for adjusting a robot **15** position to lock on a trajectory in accordance with one embodiment of the invention. As illustrated, in some embodiments, the trajectory can be locked at a current Z plane, or level above the surgical field **17**. In some embodiments, at block **3005**, it is determined if roll **62** of an end-effector **30** matches roll of the trajectory (represented by a line t (or t)). In the negative case, in some embodiments, an instruction to move a roll **62** axis is transmitted at block **3010** and flow is directed to block **3015**. Conversely, in the affirmative case, in some embodiments, flow is directed to block **3015** where it is determined if the pitch **60** of the end-effector **30** has matched the pitch of the trajectory. In the negative case, an instruction to move a pitch **60** axis is transmitted at block **3020** and flow is directed to block **3025**. In the affirmative case, in some embodiments, flow is directed to block **3025** where it is determined if x-axis **66** coordinates of points on the vector of the end-effector **30** intercept the x-axis **66** coordinates of the desired trajectory vector. In the negative case, in some embodiments, an instruction to move the x-axis **66** can be transmitted and flow is directed to **3035**. In the affirmative case, in some embodiments, flow is directed to block **3035** where it is determined if y-axis **68** coordinates of points on the vector of the end-effector **30** intercept the y-axis **68** coordinates of the desired trajectory vector. In the negative case, in some embodiments, an instruction to move the y-axis **68** can be transmitted at block **3040** and flow is directed to block **3045**. In the affirmative case, in some embodiments, flow is directed to block **3045** in which it is determined if a Z-tube **50** is being adjusted. In some embodiments, an end-user can configure information (i.e., data or metadata) indicative of the Z-tube **50** being adjusted to control it to a desired position, for example. In the negative case, in some embodiments, flow is terminated. In the affirmative case, in some embodiments, flow is directed to block **3050** where it is determined if the Z-tube **50** is positioned at a predetermined distance from anatomy. In the affirmative case, in some embodiments, flow terminates and the Z-tube **50** is located at a desired position with respect to a target location in

the anatomy (bone, biopsy site, etc.). In the negative case, in some embodiments, an instruction to move the Z-tube axis **64** is transmitted at block **3055** and the flow is directed to block **3050**. It should be noted that the subject method **3000** in some embodiments, but not all embodiments, may require that movement in each of the indicated axes (x-axis **66**, y-axis **68**, Z-tube axis **64**, roll **62**, and pitch **60**) occurs without affecting the other axes earlier in the method flow. For example, in some embodiments, the y-axis **68** movement at block **3040** should not cause change in the position of x-axis **66** coordinate, which was already checked at block **3025**. In some embodiments, the method **3000** can be implemented iteratively in order to reach a desired final position in instances where the axes do not move completely independently. In certain embodiments, the method **3000** can account for all axis positions nearly simultaneously, and can determine the exact amount of movement necessary in each axis, and thus it can be more efficient.

(191) FIG. **31** is a flowchart of a method **3100** for positioning an end-effectuator **30** in space in accordance with one embodiment of the invention. In some embodiments, the positioning can comprise positioning the end-effectuator **30** in a first plane (for example, the x-y plane or horizontal plane) and moving the end-effectuator **30** along a direction substantially normal to the first plane. FIGS. **32-33** are flowcharts of methods for driving an end-effectuator **30** to a procedure location in accordance with one embodiment of the invention. As an example, in some embodiments, the procedure location can be a position at the surface of a bone into which a conventional screw or other piece of hardware is to be inserted. In some embodiments, the end-effectuator **30** can be fitted with a guide tube **50** or conventional dilator. In some embodiments, in scenarios in which a Z-tube **50** of a surgical robot **15** comprising the end-effectuator **30** is to be locked and a Z-frame **72** is to be advanced, the method **3200** can be implemented (i.e., executed). In applications where conventional screws are to be driven into bone, the surgeon may want to move the end-effectuator tip **30**, fitted with a guide tube **50** or a conventional dilator, all the way down to the bone (see for example FIG. **62** described below). It should be appreciated that in some embodiments, since the first lateral movement occurs above the level where the patient **18** is lying, the methods depicted in FIGS. **31-33** and **62** can mitigate the likelihood that the robot **15** randomly collides with a patient **18**. In some embodiments, the method can also utilize the robot's Cartesian architecture, and the ease with which a coordinated movement down the infinite trajectory vector can be made. That is, in some embodiments, to move down this vector, the roll **62** and pitch **60** axes need no adjustment, while the x-axis **66**, y-axis **68**, and Z-frame **72** axes are moved at a fixed rate. In certain embodiments, for an articular robot **15** to make such a move, the multiple angular axes would have to be synchronized nonlinearly, with all axes simultaneously moved at varying rates.

(192) FIG. **34** illustrates a block diagram of a computer platform **3400** having a computing device **3401** that enables various features of the invention, and performance of the various methods disclosed herein in accordance with some embodiments of the invention. In some embodiments, the computing device **3401** can control operation of a surgical robot **15** and an optical tracking system **3417** in accordance with aspects described herein. In some embodiments, control can comprise calibration of relative systems of coordinates, generation of planned trajectories, monitoring of position of various units of the surgical robots **15** and/or units functionally coupled thereto, and implementation of safety protocols, and the like. For example, in some embodiments, computing device **3401** can embody a programmable controller that can control operation of a surgical robot **15** as described herein. It should be appreciated that in accordance with some embodiments of the invention, the operating environment **3400** is only an example of an operating environment and is not intended to suggest any limitation as to the scope of use or functionality of operating environment architecture. In some embodiments of the invention, the operating environment **3400** should not be interpreted as having any dependency or requirement relating to any one functional element or combination of functional elements (e.g., units, components, adapters, or the like).

(193) The various embodiments of the invention can be operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that can be suitable for use with the systems and methods of the invention comprise personal computers, server computers, laptop devices or handheld devices, and multiprocessor systems. Additional examples comprise mobile devices, programmable consumer electronics, network PCs, minicomputers, mainframe computers, distributed computing environments that comprise any of the above systems or devices, and the like.

(194) In some embodiments, the processing effected in the disclosed systems and methods can be performed by software components. In some embodiments, the disclosed systems and methods can be described in the general context of computer-executable instructions, such as program modules, being executed by one or more computers, such as computing device **3401**, or other computing devices. Generally, program modules comprise computer code, routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types. The disclosed methods also can be practiced in grid-based and distributed computing environments where tasks are performed by remote processing devices that are linked through a

communications network. In a distributed computing environment, program modules can be located in both local and remote computer storage media including memory storage devices.

(195) Further, one skilled in the art will appreciate that the systems and methods disclosed herein can be implemented via a general-purpose computing device in the form of the computing device **3401**. In some embodiments, the components of the computing device **3401** can comprise, but are not limited to, one or more processors **3403**, or processing units **3403**, a system memory **3412**, and a system bus **3413** that couples various system components including the processor **3403** to the system memory **3412**. In some embodiments, in the case of multiple processing units **3403**, the system can utilize parallel computing.

(196) In general, a processor **3403** or a processing unit **3403** refers to any computing processing unit or processing device comprising, but not limited to, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally or alternatively, a processor **3403** or processing unit **3403** can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Processors or processing units referred to herein can exploit nano-scale architectures such as, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of the computing devices that can implement the various aspects of the subject invention. In some embodiments, processor **3403** or processing unit **3403** also can be implemented as a combination of computing processing units.

(197) The system bus **3413** represents one or more of several possible types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, such architectures can comprise an Industry Standard Architecture (ISA) bus, a Micro Channel Architecture (MCA) bus, an Enhanced ISA (EISA) bus, a Video Electronics Standards Association (VESA) local bus, an Accelerated Graphics Port (AGP) bus, and a Peripheral Component Interconnects (PCI), a PCI-Express bus, a Personal Computer Memory Card Industry Association (PCMCIA), Universal Serial Bus (USB) and the like. The bus **3413**, and all buses specified in this specification and annexed drawings also can be implemented over a wired or wireless network connection and each of the subsystems, including the processor **3403**, a mass storage device **3404**, an operating system **3405**, robotic guidance software **3406**, robotic guidance data storage **3407**, a network adapter **3408**, system memory **3412**, an input/output interface **3410**, a display adapter **3409**, a display device **3411**, and a human machine interface **3402**, can be contained within one or more remote computing devices **3414a,b** at physically separate locations, functionally coupled (e.g., communicatively coupled) through buses of this form, in effect implementing a fully distributed system.

(198) In some embodiments, robotic guidance software **3406** can configure the computing device **3401**, or a processor thereof, to perform the automated control of position of the local robot **3416** (for example, surgical robot **15**) in accordance with aspects of the invention. Such control can be enabled, at least in part, by a tracking system **3417**. In some embodiments, when the computing device **3401** embodies the computer **100** functionally coupled to surgical robot **15**, robotic guidance software **3406** can configure such computer **100** to perform the functionality described in the subject invention. In some embodiments, robotic guidance software **3406** can be retained in a memory as a group of computer-accessible instructions (for instance, computer-readable instructions, computer-executable instructions, or computer-readable computer-executable instructions). In some embodiments, the group of computer-accessible instructions can encode the methods of the invention (such as the methods illustrated in FIGS. **24-33** in accordance with some embodiments of the invention). In some embodiments, the group of computer-accessible instructions can encode various formalisms (e.g., image segmentation) for computer vision tracking. Some embodiments include robotic guidance software **3406** that can include a compiled instance of such computer-accessible instructions, a linked instance of such computer-accessible instructions, a compiled and linked instance of such computer-executable instructions, or an otherwise executable instance of the group of computer-accessible instructions.

(199) Some embodiments include robotic guidance data storage **3407** that can comprise various types of data that can permit implementation (e.g., compilation, linking, execution, and combinations thereof) of the robotic guidance software **3406**. In some embodiments, robotic guidance data storage **3407** can comprise data associated with intraoperative imaging, automated adjustment of position of the local robot **3416** and/or remote robot **3422**, or the like. In some embodiments, the data retained in the robotic guidance data storage **3407** can be formatted according to any image data in industry standard format. As illustrated, in some embodiments, a remote tracking system **3424** can enable, at least in part, control of the remote robot **3422**. In some embodiments, the information

can comprise tracking information, trajectory information, surgical procedure information, safety protocols, and so forth.

(200) In some embodiments of the invention, the computing device **3401** typically comprises a variety of computer readable media. The readable media can be any available media that is accessible by the computer **3401** and comprises, for example and not meant to be limiting, both volatile and non-volatile media, removable and non-removable media. In some embodiments, the system memory **3412** comprises computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as read only memory (ROM). In some embodiments, the system memory **3412** typically contains data (such as a group of tokens employed for code buffers) and/or program modules such as operating system **3405** and robotic guidance software **3406** that are immediately accessible to, and/or are presently operated-on by the processing unit **3403**. In some embodiments, operating system **3405** can comprise operating systems such as Windows operating system, Unix, Linux, Symbian, Android, Apple iOS operating system, Chromium, and substantially any operating system for wireless computing devices or tethered computing devices. Apple® is a trademark of Apple Computer, Inc., registered in the United States and other countries. iOS® is a registered trademark of Cisco and used under license by Apple Inc. Microsoft® and Windows® are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries. Android® and Chrome® operating system are a registered trademarks of Google Inc. Symbian® is a registered trademark of Symbian Ltd. Linux® is a registered trademark of Linus Torvalds. UNIX® is a registered trademark of The Open Group.

(201) In some embodiments, computing device **3401** can comprise other removable/non-removable, volatile/non-volatile computer storage media. As illustrated, in some embodiments, computing device **3401** comprises a mass storage device **3404** which can provide non-volatile storage of computer code (e.g., computer-executable instructions), computer-readable instructions, data structures, program modules, and other data for the computing device **3401**. For instance, in some embodiments, a mass storage device **3404** can be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

(202) In some embodiments, optionally, any number of program modules can be stored on the mass storage device **3404**, including by way of example, an operating system **3405**, and tracking software **3406**. In some embodiments, each of the operating system **3405** and tracking software **3406** (or some combination thereof) can comprise elements of the programming and the tracking software **3406**. In some embodiments, data and code (for example, computer-executable instructions, patient-specific trajectories, and patient **18** anatomical data) can be retained as part of tracking software **3406** and stored on the mass storage device **3404**. In some embodiments, tracking software **3406**, and related data and code, can be stored in any of one or more databases known in the art. Examples of such databases comprise, DB2®, Microsoft® Access, Microsoft® SQL Server, Oracle®, MySQL, PostgreSQL, and the like. Further examples include membase databases and flat file databases. The databases can be centralized or distributed across multiple systems.

(203) DB2® is a registered trademark of IBM in the United States.

(204) Microsoft®, Microsoft® Access®, and Microsoft® SQL Server™ are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

(205) Oracle® is a registered trademark of Oracle Corporation and/or its affiliates.

(206) MySQL® is a registered trademark of MySQL AB in the United States, the European Union and other countries.

(207) PostgreSQL® and the PostgreSQL® logo are trademarks or registered trademarks of The PostgreSQL Global Development Group, in the U.S. and other countries.

(208) In some embodiments, an agent (for example, a surgeon or other user, or equipment) can enter commands and information into the computing device **3401** via an input device (not shown). Examples of such input devices can comprise, but are not limited to, a camera (or other detection device for non-optical tracking markers), a keyboard, a pointing device (for example, a mouse), a microphone, a joystick, a scanner (for example, a barcode scanner), a reader device such as a radiofrequency identification (RFID) readers or magnetic stripe readers, gesture-based input devices such as tactile input devices (for example, touch screens, gloves and other body coverings or wearable devices), speech recognition devices, or natural interfaces, and the like. In some embodiments, these and other input devices can be connected to the processing unit **3403** via a human machine interface **3402** that is coupled to the system bus **3413**. In some other embodiments, they can be connected by other interface and bus structures, such as a parallel port, game port, an IEEE 1394 port (also known as a firewire port), a serial port, or a universal serial bus (USB).

(209) In some further embodiments, a display device **3411** can also be functionally coupled to the system bus **3413** via an interface, such as a display adapter **3409**. In some embodiments, the computer **3401** can have more

than one display adapter **3409** and the computer **3401** can have more than one display device **3411**. For example, in some embodiments, a display device **3411** can be a monitor, a liquid crystal display, or a projector. Further, in addition to the display device **3411**, some embodiments can include other output peripheral devices that can comprise components such as speakers (not shown) and a printer (not shown) capable of being connected to the computer **3401** via input/output Interface **3410**. In some embodiments, the input/output interface **3410** can be a pointing device, either tethered to, or wirelessly coupled to the computing device **3410**. In some embodiments, any step and/or result of the methods can be output in any form to an output device. In some embodiments, the output can be any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like.

(210) In certain embodiments, one or more cameras (for example, camera **8200** shown in FIG. **81**) can be contained or functionally coupled to the tracking system **3417**, which is functionally coupled to the system bus **3413** via an input/output interface of the one or more input/output interfaces **3410**. Such functional coupling can permit the one or more camera(s) to be coupled to other functional elements of the computing device **3401**. In one embodiment, the input/output interface, at least a portion of the system bus **3413**, and the system memory **3412** can embody a frame grabber unit that can permit receiving imaging data acquired by at least one of the one or more cameras. In some embodiments, the frame grabber can be an analog frame grabber, a digital frame grabber, or a combination thereof. In some embodiments, where the frame grabber is an analog frame grabber, the processor **3403** can provide analog-to-digital conversion functionality and decoder functionality to enable the frame grabber to operate with medical imaging data. Further, in some embodiments, the input/output interface can include circuitry to collect the analog signal received from at least one camera of the one or more cameras. In some embodiments, in response to execution by processor **3403**, tracking software **3406** can operate the frame grabber to receive imaging data in accordance with various aspects described herein.

(211) Some embodiments include a computing device **3401** that can operate in a networked environment (for example, an industrial environment) using logical connections to one or more remote computing devices **3414a,b**, a remote robot **3422**, and a tracking system **3424**. By way of example, in some embodiments, a remote computing device can be a personal computer, portable computer, a mobile telephone, a server, a router, a network computer, a peer device or other common network node, and so on. In particular, in some embodiments, an agent (for example, a surgeon or other user, or equipment) can point to other tracked structures, including anatomy of a patient **18**, using a remote computing device **3414** such as a hand-held probe that is capable of being tracked and sterilized. In some embodiments, logical connections between the computer **3401** and a remote computing device **3414a,b** can be made via a local area network (LAN) and a general wide area network (WAN). In some embodiments, the network connections can be implemented through a network adapter **3408**. In some embodiments, the network adapter **3408** can be implemented in both wired and wireless environments. Some embodiments include networking environments that can be conventional and commonplace in offices, enterprise-wide computer networks, intranets. In some embodiments, the networking environments generally can be embodied in wire-line networks or wireless networks (for example, cellular networks, such as third generation (“3G”) and fourth generation (“4G”) cellular networks, facility-based networks (for example, femtocell, picocell, wifi networks). In some embodiments, a group of one or more networks **3415** can provide such networking environments. In some embodiments of the invention, the one or more network(s) can comprise a LAN deployed in an industrial environment comprising the system **1** described herein.

(212) As an illustration, in some embodiments, application programs and other executable program components such as the operating system **3405** are illustrated herein as discrete blocks, although it is recognized that such programs and components reside at various times in different storage components of the computing device **3401**, and are executed by the data processor(s) of the computer **100**. Some embodiments include an implementation of tracking software **3406** that can be stored on or transmitted across some form of computer readable media. Any of the disclosed methods can be performed by computer readable instructions embodied on computer readable media. Computer readable media can be any available media that can be accessed by a computer. By way of example and not meant to be limiting, computer-readable media can comprise “computer storage media,” or “computer-readable storage media,” and “communications media.” “Computer storage media” comprise volatile and non-volatile, removable and non-removable media implemented in any methods or technology for storage of information such as computer readable instructions, data structures, program modules, or other data. In some embodiments of the invention, computer storage media comprises, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by a computer.

(213) As described herein, some embodiments include the computing device **3401** that can control operation of local robots **3416** and/or remote robots **3422**. Within embodiments in which the local robot **3416** or the remote

robot **3422** are surgical robots **15**, the computing device **3401** can execute robotic guidance software **3407** to control such robots **3416**, **3422**, **15**. In some embodiments, the robotic guidance software **3407**, in response to execution, can utilize trajectories (such as, tip and tail coordinates) that can be planned and/or configured remotely or locally. In an additional or alternative aspect, in response to execution, the robotic guidance software **3407** can implement one or more of the methods described herein in a local robot's computer or a remote robot's computer to cause movement of the remote robot **15** or the local robot **15** according to one or more trajectories. (214) In some embodiments, the computing device **3401** can enable pre-operative planning of the surgical procedure.

(215) In some embodiments, the computing device **3401** can permit spatial positioning and orientation of a surgical tool (for example, instrument **35**) during intraoperative procedures. In some further embodiments, the computing device **3401** can enable open procedures. In some other embodiments, the computing device **3401** can enable percutaneous procedures.

(216) In certain embodiments, the computing device **3401** and the robotic guidance software **3407** can embody a 3D tracking system **3417** to simultaneously monitor the positions of the device and the anatomy of the patient **18**. In some embodiments, the 3D tracking system **3417** can be configured to cast the patient's anatomy and the end-effector **30** in a common coordinate system.

(217) In some embodiments, the computing device **3401** can access (i.e., load) image data from a conventional static storage device. In some embodiments, the computing device **3401** can permit a 3D volumetric representation of patient **18** anatomy to be loaded into memory (for example, system memory **3412**) and displayed (for example, via display device **3411**).

(218) In some embodiments, the computing device **3401**, in response to execution of the robotic guidance software **3407** can enable navigation through the 3D volume representation of a patient's anatomy.

(219) In some embodiments, the computing device **3401** can operate with a conventional power source required to offer the device for sale in the specified country. A conventional power cable that supplies power can be a sufficient length to access conventional hospital power outlets. In some embodiments, in the event of a power loss, the computing device **3401** can hold the current end-effector **30** in a position unless an agent (for example, a surgeon or other user, or equipment) manually moves the end-effector **30**.

(220) In some embodiments, the computing device **3401** can monitor system physical condition data. In some embodiments, the computing device **3401** can report to an operator (for example, a surgeon) each of the physical condition data and indicate an out-of-range value.

(221) In some embodiments, the computing device **3401** can enable entry and storage of manufacturing calibration values for end-effector **30** positioning using, for example, the input/output interface **3410**.

(222) In some embodiments, the computing device **3401** can enable access to manufacturing calibration values by an agent (for example, a surgeon or other user, or equipment) authenticated to an appropriate access level. In some embodiments, the data can be retained in robotic guidance data storage **3407**, or can be accessed via network(s) **3415** when the data is retained in a remote computing device **3414a**.

(223) In some embodiments, the computing device **3401** can render (using for example display device **3411**) a technical screen with a subset of the end-effector **30** positioning calibration and system health data. The information is only accessible to an agent (for example, a surgeon or other user, or equipment) authenticated to an appropriate level.

(224) In some embodiments, the computing device **3401** can enable field calibration of end-effector **30** positioning only by an agent (for example, a surgeon or other user, or equipment) authenticated to an appropriate access level.

(225) In some embodiments, the computing device **3401** can convey the status of local robot **3416**, remote robot **3422**, and/or other device being locked in position using a visual or aural alert.

(226) In some further embodiments, the computing device **3401** can include an emergency stop control that upon activation, disables power to the device's motors **160** but not to the processor **3403**. In some embodiments, the emergency stop control can be accessible by the operator of computing device **3401**. In some embodiments, the computing device **3401** can monitor the emergency stop status and indicate to the operator that the emergency stop has been activated.

(227) In some other embodiments, the computing device **3401** can be operated in a mode that permits manual positioning of the end-effector **30**.

(228) In some embodiments, the computing device **3401** can boot directly to an application representing the robotic guidance software **3406**. In some embodiments, computing device **3401** can perform a system check prior to each use. In scenarios in which the system check fails, the computing device **3401** can notify an operator.

(229) In some embodiments, the computing device **3401** can generate an indicator for reporting system status.

(230) Some embodiments include the computing device **3401** that can minimize or can mitigate delays in

processing, and in the event of a delay in processing, notify an agent (for example, a surgeon or other user, or equipment). For example, in some embodiments, a delay may occur while a system scan is being performed to assess system status, and consequently the computing device **3401** can schedule (for example, generate a process queue) system scans to occur at low usage times. In some embodiments, a system clock of the computing device **3401** can be read before and after key processes to assess the length of time required to complete computation of a process. In some embodiments, the actual time to complete the process can be compared to the expected time. In some embodiments, if a discrepancy is found to be beyond an acceptable tolerance, the agent can be notified, and/or concurrently running non-essential computational tasks can be terminated. In one embodiment, a conventional system clock (not shown) can be part of processor **3403**.

(231) In some embodiments, the computing device **3401** can generate a display that follows a standardized workflow.

(232) In some embodiments, the computing device **3401** can render or ensure that text is rendered in a font of sufficient size and contrast to be readable from an appropriate distance.

(233) In some embodiments, the computing device **3401** can enable an operator to locate the intended position of a surgical implant or tool.

(234) In some further embodiments, the computing device **3401** can determine the relative position of the end-effectuator **30** to the anatomy of the patient **18**. For example, to at least such end, the computing device **3401** can collect data the optical tracking system **3417**, and can analyze the data to generate data indicative of such relative position.

(235) In some embodiments, the computing device **3401** can indicate the end-effectuator **30** position and orientation.

(236) In some embodiments, the computing device **3401** can enable continuous control of end-effectuator **30** position relative to the anatomy of a patient **18**.

(237) In some embodiments, the computing device **3401** can enable an agent (for example, a surgeon or other user, or equipment) to mark the intended position of a surgical implant or tool (for example, instrument **35**).

(238) In some embodiments, the computing device **3401** can allow the position and orientation of a conventional hand-held probe (or an instrument **35**) to be displayed overlaid on images of the patient's anatomy.

(239) In some embodiments, the computing device **3401** can enable an agent (for example, a surgeon or other user, or equipment) to position conventional surgical screws. In some embodiments, the computing device **3401** can enable selection of the length and diameter of surgical screws by the agent. In yet another aspect, the computing device can ensure that the relative position, size and scale of screws are maintained on the display **3411** when in graphical representation. In some embodiments, the computing device **3401** can verify screw path plans against an operation envelope and reject screw path plans outside this envelope. In still another aspect, the computing device **3401** can enable hiding of a graphical screw representation.

(240) In some embodiments, the computing device **3401** can enable a function that allows the current view to be stored. In some embodiments, the computing device **3401** can enable a view reset function that sets the current view back to a previously stored view.

(241) In some embodiments, the computing device **3401** can enable an authentication based tiered access system.

(242) In some embodiments, the computing device **3401** can log and store system activity. In some embodiments, the computing device **3401** can enable access to the system activity log to an agent authorized to an appropriate level.

(243) In some embodiments, the computing device **3401** can enable entry and storage of patient **18** data.

(244) In some embodiments, the computing device **3401** can enable the appropriate disposition of patient **18** data and/or procedure data. For example, in a scenario in which such data are being collected for research, the computing device **3401** can implement de-identification of the data in order to meet patient **18** privacy requirements. In some embodiments, the de-identification can be implemented in response to execution of computer-executable instruction(s) retained in memory **3412** or any other memory accessible to the computing device **3401**. In some embodiments, the de-identification can be performed automatically before the patient **18** data and/or procedure data are sent to a repository or any other data storage (including mass storage device **3404**, for example). In some embodiments, indicia (e.g., a dialog box) can be rendered (for example, at display device **3411**) to prompt an agent (e.g., machine or human) to permanently delete patient **18** data and/or procedure data at the end of a procedure.

(245) FIG. **11** shows a flow chart diagram **1100** for general operation of the robot **15** according to some embodiments is shown. In some embodiments, at step **210**, the local positioning system (herein referred to as "LPS") establishes a spatial coordinate measuring system for the room **10** where the invasive procedure is to occur; in other words, the LPS is calibrated. In some embodiments, in order to calibrate the LPS, a conventional mechanical fixture that includes a plurality of attached calibrating transmitters **120** is placed within the room **10**

where positioning sensors **12** are located. In some embodiments of the invention, at least three calibrating transmitters **120** are required, but any number of calibrating transmitters **120** above three is within the scope of the invention. Also, in some embodiments, at least three positioning sensors **12** are required, but any number of positioning sensors **12** above three is also within the scope of the invention, and the accuracy of the system is increased with the addition of more positioning sensors.

(246) In some embodiments, the distance between each of the calibrating transmitters **120** relative to each other is measured prior to calibration step **210**. Each calibrating transmitter **120** transmits RF signals on a different frequency so that the positioning sensors **12** can determine which transmitter **120** emitted a particular RF signal. In some embodiments, the signal of each of these transmitters **120** is received by positioning sensors **12**. In some embodiments, since the distance between each of the calibrating transmitters **120** is known, and the sensors **12** can identify the signals from each of the calibrating transmitters **120** based on the known frequency, using time of flight calculation, the positioning sensors **12** are able to calculate the spatial distance of each of the positioning sensors **12** relative to each other. The system **1** is now calibrated. As a result, in some embodiments, the positioning sensors **12** can now determine the spatial position of any new RF transmitter **120** introduced into the room **10** relative to the positioning sensors **12**.

(247) In some embodiments, a step **220a** in which a 3D anatomical image scan, such as a CT scan, is taken of the anatomical target. Any 3D anatomical image scan may be used with the surgical robot **15** and is within the scope of the present invention.

(248) In some embodiments, at step **230**, the positions of the RF transmitters **120** tracking the anatomical target are read by positioning sensors **110**. These transmitters **120** identify the initial position of the anatomical target and any changes in position during the procedure.

(249) In some embodiments, if any RF transmitters **120** must transmit through a medium that changes the RF signal characteristics, then the system will compensate for these changes when determining the transmitter's **120** position.

(250) In some embodiments, at step **240**, the positions of the transmitters **120** on the anatomy are calibrated relative to the LPS coordinate system. In other words, the LPS provides a reference system, and the location of the anatomical target is calculated relative to the LPS coordinates. In some embodiments, to calibrate the anatomy relative to the LPS, the positions of transmitters **120** affixed to the anatomical target are recorded at the same time as positions of temporary transmitters **120** placed on precisely known anatomical landmarks also identified on the anatomical image. This calculation is performed by a computer **100**.

(251) In some embodiments, at step **250**, the positions of the RF transmitters **120** that track the anatomical target are read. Since the locations of the transmitters **120** on the anatomical target have already been calibrated, the system can easily determine if there has been any change in position of the anatomical target.

(252) Some embodiments include a step **260**, where the positions of the transmitters **120** on the surgical instrument **35** are read. The transmitters **120** may be located on the surgical instrument **35** itself, and/or there may be transmitters **120** attached to various points of the surgical robot **15**.

(253) In some embodiments of the invention, the surgical robot **15** can also include a plurality of attached conventional position encoders that help determine the position of the surgical instrument **35**. In some embodiments, the position encoders can be devices used to generate an electronic signal that indicates a position or movement relative to a reference position. In some other embodiments, a position signal can be generated using conventional magnetic sensors, conventional capacitive sensors, and conventional optical sensors.

(254) In some embodiments, position data read from the position encoders may be used to determine the position of the surgical instrument **35** used in the procedure. In some embodiments, the data may be redundant of position data calculated from RF transmitters **120** located on the surgical instrument **35**. Therefore, in some embodiments, position data from the position encoders may be used to double-check the position being read from the LPS.

(255) In some embodiments, at step **270**, the coordinates of the positions of the transmitters **120** on the surgical instrument **35**, and/or the positions read from the position encoders, is calibrated relative to the anatomical coordinate system. In other words, in some embodiments, the position data of the surgical instrument **35** is synchronized into the same coordinate system as the patient's anatomy. In some embodiments, this calculation is performed automatically by the computer **100** since the positions of the transmitters **120** on the anatomical target and the positions of the transmitters **120** on the surgical instrument **35** are in the same coordinate system, and the positions of the transmitters **120** on the anatomical target are already calibrated relative to the anatomy.

(256) In some embodiments, at step **280**, the computer **100** superimposes a representation of the location calculated in step **270** of the surgical device on the 3D anatomical image of the patient **18** taken in step **220**. In some embodiments, the superimposed image can be displayed to an agent.

(257) In some embodiments, at step **290**, the computer **100** sends the appropriate signals to the motors **160** to drive the surgical robot **15**. In some embodiments, if the agent preprogrammed a trajectory, then the robot **15** is

driven so that the surgical instrument **35** follows the preprogrammed trajectory if there is no further input from the agent. In some embodiments, if there is agent input, then the computer **100** drives the robot **15** in response to the agent input.

(258) In some embodiments, at step **295**, the computer **100** determines whether the anatomy needs to be recalibrated. In some embodiments, the agent may choose to recalibrate the anatomy, in which case the computer **100** responds to agent input. Alternatively, in some embodiments, the computer **100** may be programmed to recalibrate the anatomy in response to certain events. For instance, in some embodiments, the computer **100** may be programmed to recalibrate the anatomy if the RF transmitters **120** on the anatomical target indicate that the location of the anatomical target has shifted relative to the RF transmitters **120** (i.e. this spatial relationship should be fixed). In some embodiments, an indicator that the anatomical target location has shifted relative to the transmitters **120** is if the computer **100** calculates that the surgical instrument **35** appears to be inside bone when no drilling or penetration is actually occurring.

(259) In some embodiments, if the anatomy needs to be calibrated, then the process beginning at step **230** is repeated. In some embodiments, if the anatomy does not need to be recalibrated, then the process beginning at step **250** is repeated.

(260) In some embodiments, at any time during the procedure, certain fault conditions may cause the computer **100** to interrupt the program and respond accordingly. For instance, in some embodiments, if the signal from the RF transmitters **120** cannot be read, then the computer **100** may be programmed to stop the movement of the robot **15**, or remove the surgical instrument **35** from the patient **18**. Another example of a fault condition is if the robot **15** encounters a resistance above a preprogrammed tolerance level.

(261) FIG. **12** shows a flow chart diagram **1200** for a closed screw/needle insertion procedure according to an embodiment of the invention is shown. In a closed pedicle screw insertion procedure, in some embodiments, the robot **15** holds a guide tube **50** adjacent to the patient **18** in the correct angular orientation at the point where a conventional pedicle screw is to be inserted through the tissue and into the bone of the patient **18**.

(262) In some embodiments, the distance between each of the calibrating transmitters **120** relative to each other is measured prior to calibration step **300**. In some embodiments, each calibrating transmitter **120** transmits RF signals on a different frequency so the positioning sensors **12** can determine which transmitter **120** emitted a particular RF signal. In some embodiments, the signal of each of these transmitters **120** is received by positioning sensors **12**. Since the distance between each of the calibrating transmitters **120** is known, and the sensors **12** can identify the signals from each of the calibrating transmitters **120** based on the known frequency, using time of flight calculation, in some embodiments, the positioning sensors **12** are able to calculate the spatial distance of each of the positioning sensors **12** relative to each other. The system **1** is now calibrated. As a result, in some embodiments, the positioning sensors **12** can now determine the spatial position of any new RF transmitter **120** introduced into the room **10** relative to the positioning sensors **12**.

(263) In some embodiments, at step **310**, a 3D anatomical image scan, such as a CT scan, is taken of the anatomical target. Any 3D anatomical image scan may be used with the surgical robot **15** and is within the scope of the present invention.

(264) In some embodiments, at step **320**, the operator selects a desired trajectory and insertion point of the surgical instrument **35** on the anatomical image captured at step **310**. In some embodiments, the desired trajectory and insertion point is programmed into the computer **100** so that the robot **15** can drive a guide tube **50** automatically to follow the trajectory.

(265) In some embodiments, at step **330**, the positions of the RF transmitters **120** tracking the anatomical target are read by positioning sensors **110**. In some embodiments, these transmitters **120** identify the initial position of the anatomical target and any changes in position during the procedure.

(266) In some embodiments, if any RF transmitters **120** must transmit through a medium that changes the RF signal characteristics, the system will compensate for these changes when determining the transmitter's **120** position.

(267) In some embodiments, at step **340**, the positions of the transmitters **120** on the anatomy are calibrated relative to the LPS coordinate system. In other words, the LPS provides a reference system, and the location of the anatomical target is calculated relative to the LPS coordinates. In some embodiments, to calibrate the anatomy relative to the LPS, the positions of transmitters **120** affixed to the anatomical target are recorded at the same time as positions of temporary transmitters **120** on precisely known anatomical landmarks also identified on the anatomical image. This calculation is performed by a computer.

(268) In some embodiments, at step **350**, the positions of the RF transmitters **120** that track the anatomical target are read. Since the locations of the transmitters **120** on the anatomical target have already been calibrated, in some embodiments, the system can easily determine if there has been any change in position of the anatomical target.

(269) In some embodiments, at step **360**, the positions of the transmitters **120** on the surgical instrument **35** are

read. In some embodiments, the transmitters **120** may be located on the surgical instrument **35**, and/or attached to various points of the surgical robot **15**.

(270) In some embodiments, at step **370**, the coordinates of the positions of the transmitters **120** on the surgical instrument **35**, and/or the positions read from the position encoders, are calibrated relative to the anatomical coordinate system. In other words, the position data of the surgical instrument **35** is synchronized into the same coordinate system as the anatomy. This calculation is performed automatically by the computer **100** since the positions of the transmitters **120** on the anatomical target and the positions of the transmitters **120** on the surgical instrument **35** are in the same coordinate system and the positions of the transmitters **120** on the anatomical target are already calibrated relative to the anatomy.

(271) In some embodiments, at step **380**, the computer **100** superimposes a representation of the location calculated in step **370** of the surgical device on the 3D anatomical image of the patient **18** taken in step **310**. The superimposed image can be displayed to the user.

(272) In some embodiments, at step **390**, the computer **100** determines whether the guide tube **50** is in the correct orientation and position to follow the trajectory planned at step **320**. If it is not, then step **393** is reached. If it is in the correct orientation and position to follow the trajectory, then step **395** is reached.

(273) In some embodiments, at step **393**, the computer **100** determines what adjustments it needs to make in order to make the guide tube **50** follow the preplanned trajectory. The computer **100** sends the appropriate signals to drive the motors **160** in order to correct the movement of the guide tube.

(274) In some embodiments, at step **395**, the computer **100** determines whether the procedure has been completed. If the procedure has not been completed, then the process beginning at step **350** is repeated.

(275) In some embodiments, at any time during the procedure, certain fault conditions may cause the computer **100** to interrupt the program and respond accordingly. For instance, if the signal from the RF transmitters **120** cannot be read, then the computer **100** may be programmed to stop the movement of the robot **15** or lift the guide tube **50** away from the patient **18**. Another example of a fault condition is if the robot **15** encounters a resistance above a preprogrammed tolerance level. Another example of a fault condition is if the RF transmitters **120** on the anatomical target shift so that actual and calculated positions of the anatomy no longer match. One indicator that the anatomical target location has shifted relative to the transmitters **120** is if the computer **100** calculates that the surgical instrument **35** appears to be inside bone when no drilling or penetration is actually occurring.

(276) In some embodiments, the proper response to each condition may be programmed into the system, or a specific response may be user-initiated. For example, the computer **100** may determine that in response to an anatomy shift, the anatomy would have to be recalibrated, and the process beginning at step **330** should be repeated. Alternatively, a fault condition may require the flowchart to repeat from step **300**. Another alternative is the user may decide that recalibration from step **330** is desired, and initiate that step himself.

(277) Referring now to FIG. **13**, a flow chart diagram **1300** for a safe zone surgical procedure performed using the system described herein is shown in accordance with some embodiments of the invention. In a safe zone surgical procedure, there is a defined safe zone around the surgical area within which the surgical device must stay. The physician manually controls the surgical device that is attached to the end-effector **30** of the surgical robot **15**. If the physician moves the surgical device outside of the safe zone, then the surgical robot **15** stiffens the arm **23** so that the physician cannot move the instrument **35** in any direction that would move the surgical instrument **35** outside the safe zone.

(278) In some embodiments, the distance between each of the calibrating transmitters **120** relative to each other is measured prior to calibration step **400**. Each calibrating transmitter **120** transmits RF signals on a different frequency so the positioning sensors **12** can determine which transmitter **120** emitted a particular RF signal. The signal of each of these transmitters **120** is received by positioning sensors **12**. Since the distance between each of the calibrating transmitters **120** is known, and the sensors **12** can identify the signals from each of the calibrating transmitters **120** based on the known frequency, the positioning sensors **12** are able to calculate, using time of flight calculation, the spatial distance of each of the positioning sensors **12** relative to each other. The system **1** is now calibrated. As a result, the positioning sensors **12** can now determine the spatial position of any new RF transmitter **120** introduced into the room **10** relative to the positioning sensors **12**.

(279) In some embodiments, at step **410**, a 3D anatomical image scan, such as a CT scan, is taken of the anatomical target. Any 3D anatomical image scan may be used with the surgical robot **15** and is within the scope of the present invention.

(280) In some embodiments, at step **420**, the operator inputs a desired safe zone on the anatomical image taken in step **410**. In an embodiment of the invention, the operator uses an input to the computer **100** to draw a safe zone on a CT scan taken of the patient **18** in step **410**.

(281) In some embodiments, at step **430**, the positions of the RF transmitters **120** tracking the anatomical target are read by positioning sensors. These transmitters **120** identify the initial position of the anatomical target and

any changes in position during the procedure.

(282) In some embodiments, if any RF transmitters **120** must transmit through a medium that changes the RF signal characteristics, then the system will compensate for these changes when determining the transmitter's **120** position.

(283) In some embodiments, at step **440**, the positions of the transmitters **120** on the anatomy are calibrated relative to the LPS coordinate system. In other words, the LPS provides a reference system, and the location of the anatomical target is calculated relative to the LPS coordinates. To calibrate the anatomy relative to the LPS, the positions of transmitters **120** affixed to the anatomical target are recorded at the same time as positions of temporary transmitters **120** on precisely known landmarks on the anatomy that can also be identified on the anatomical image. This calculation is performed by a computer **100**.

(284) In some embodiments, at step **450**, the positions of the RF transmitters **120** that track the anatomical target are read. Since the locations of the transmitters **120** on the anatomical target have already been calibrated, the system can easily determine if there has been any change in position of the anatomical target.

(285) In some embodiments, at step **460**, the positions of the transmitters **120** on the surgical instrument **35** are read. The transmitters **120** may be located on the surgical instrument **35** itself, and/or there may be transmitters **120** attached to various points of the surgical robot **15**.

(286) In some embodiments, at step **470**, the coordinates of the positions of the transmitters **120** on the surgical instrument **35**, and/or the positions read from the position encoders, are calibrated relative to the anatomical coordinate system. In other words, the position data of the surgical instrument **35** is synchronized into the same coordinate system as the anatomy. This calculation is performed automatically by the computer **100** since the positions of the transmitters **120** on the anatomical target and the positions of the transmitters **120** on the surgical instrument **35** are in the same coordinate system and the positions of the transmitters **120** on the anatomical target are already calibrated relative to the anatomy.

(287) In some embodiments, at step **480**, the computer **100** superimposes a representation of the location calculated in step **470** of the surgical device on the 3D anatomical image of the patient **18** taken in step **410**. In some embodiments, the superimposed image can be displayed to the user.

(288) In some embodiments, at step **490**, the computer **100** determines whether the surgical device attached to the end-effector **30** of the surgical robot **15** is within a specified range of the safe zone boundary (for example, within 1 millimeter of reaching the safe zone boundary). In some embodiments, if the end-effector **30** is almost to the boundary, then step **493** is reached. In some embodiments, if it is well within the safe zone boundary, then step **495** is reached.

(289) In some embodiments, at step **493**, the computer **100** stiffens the arm of the surgical robot **15** in any direction that would allow the user to move the surgical device closer to the safe zone boundary.

(290) In some embodiments, at step **495**, the computer **100** determines whether the anatomy needs to be recalibrated. In some embodiments, the user may choose to recalibrate the anatomy, in which case the computer **100** responds to user input. Alternatively, in some embodiments, the computer **100** may be programmed to recalibrate the anatomy in response to certain events. For instance, in some embodiments, the computer **100** may be programmed to recalibrate the anatomy if the RF transmitters **120** on the anatomical target indicate that the location of the anatomical target has shifted relative to the RF transmitters **120** (i.e. this spatial relationship should be fixed.) In some embodiments, an indicator that the anatomical target location has shifted relative to the transmitters **120** is if the computer **100** calculates that the surgical instrument **35** appears to be inside bone when no drilling or penetration is actually occurring.

(291) In some embodiments, if the anatomy needs to be calibrated, then the process beginning at step **430** is repeated. In some embodiments, if the anatomy does not need to be recalibrated, then the process beginning at step **450** is repeated.

(292) In some embodiments, at any time during the procedure, certain fault conditions may cause the computer **100** to interrupt the program and respond accordingly. For instance, in some embodiments, if the signal from the RF transmitters **120** cannot be read, then the computer **100** may be programmed to stop the movement of the robot **15** or remove the surgical instrument **35** from the patient **18**. Another example of a fault condition is if the robot **15** encounters a resistance above a preprogrammed tolerance level.

(293) Referring now to FIG. **14**, a flow chart diagram **1400** for a conventional flexible catheter or wire insertion procedure according to an embodiment of the invention is shown. Catheters are used in a variety of medical procedures to deliver medicaments to a specific site in a patient's body. Often, delivery to a specific location is needed so a targeted diseased area can then be treated. Sometimes instead of inserting the catheter directly, a flexible wire is first inserted, over which the flexible catheter can be slid.

(294) In some embodiments, the distance between each of the calibrating transmitters **120** relative to each other is measured prior to calibration step **500**. In some embodiments, each calibrating transmitter **120** transmits RF

signals on a different frequency so the positioning sensors **12, 110** can determine which transmitter **120** emitted a particular RF signal. In some embodiments, the signal from each of these transmitters **120** is received by positioning sensors **12, 110**. Since the distance between each of the calibrating transmitters **120** is known, and the sensors can identify the signals from each of the calibrating transmitters **120** based on the known frequency, in some embodiments, using time of flight calculation, the positioning sensors **12, 110** are able to calculate the spatial distance of each of the positioning sensors **12, 110** relative to each other. The system is now calibrated. As a result, in some embodiments, the positioning sensors **12, 110** can now determine the spatial position of any new RF transmitter **120** introduced into the room **10** relative to the positioning sensors **12, 110**.

(295) In some embodiments, at step **510**, reference needles that contain the RF transmitters **120** are inserted into the body. The purpose of these needles is to track movement of key regions of soft tissue that will deform during the procedure or with movement of the patient **18**.

(296) In some embodiments, at step **520**, a 3D anatomical image scan (such as a CT scan) is taken of the anatomical target. Any 3D anatomical image scan may be used with the surgical robot **15** and is within the scope of the present invention. In some embodiments, the anatomical image capture area includes the tips of the reference needles so that their transmitters' **120** positions can be determined relative to the anatomy.

(297) In some embodiments, at step **530**, the RF signals from the catheter tip and reference needles are read.

(298) In some embodiments, at step **540**, the position of the catheter tip is calculated. Because the position of the catheter tip relative to the reference needles and the positions of the reference needles relative to the anatomy are known, the computer **100** can calculate the position of the catheter tip relative to the anatomy.

(299) In some embodiments, at step **550**, the superimposed catheter tip and the shaft representation is displayed on the anatomical image taken in step **520**.

(300) In some embodiments, at step **560**, the computer **100** determines whether the catheter tip is advancing toward the anatomical target. If it is not moving to the anatomical target, then step **563** is reached. If it is correctly moving, then step **570** is reached.

(301) In some embodiments, at step **563**, the robot **15** arm is adjusted to guide the catheter tip in the desired direction. If the anatomy needs to be calibrated, then in some embodiments, the process beginning at step **520** is repeated. If the anatomy does not need to be recalibrated, then the process beginning at step **540** is repeated.

(302) In some embodiments, at step **570**, the computer **100** determines whether the procedure has been completed. If the procedure has not been completed, then the process beginning at step **540** is repeated.

(303) In some embodiments, at any time during the procedure, certain fault conditions may cause the computer **100** to interrupt the program and respond accordingly. For instance, in some embodiments, if the signal from the RF transmitter's **120** cannot be read, then the computer **100** may be programmed to stop the movement of the robot **15** or remove the flexible catheter from the patient **18**. Another example of a fault condition is if the robot **15** encounters a resistance above a preprogrammed tolerance level. A further example of a fault condition is if the RF transmitter's **120** on the anatomical target indicate the location of the anatomical target shift so that actual and calculated positions of the anatomy no longer match. In some embodiments, one indicator that the anatomical target location has shifted relative to the transmitter's **120** is if the computer **100** calculates that the surgical instrument **35** appears to be inside bone when no drilling or penetration is actually occurring.

(304) In some embodiments, the proper response to each condition may be programmed into the system, or a specific response may be user-initiated. For example, in some embodiments, the computer **100** may determine that in response to an anatomy shift, the anatomy would have to be recalibrated, and the process beginning at step **520** should be repeated. Alternatively, in some embodiments, a fault condition may require the flowchart to repeat from step **500**. In other embodiments, the user may decide that recalibration from step **520** is desired, and initiate that step himself.

(305) Referring now to FIGS. **15A & 15B**, screenshots of software for use with the described system is provided in accordance with some embodiments of the invention. The software provides the method to select the target area of surgery, plan the surgical path, check the planned trajectory of the surgical path, synchronize the medical images to the positioning system and precisely control the positioning system during surgery. The surgical positioning system and navigation software includes an optical guidance system or RF Local Positioning System (RF-LPS), which are in communication with the positioning system.

(306) FIG. **15A** shows a screen shot **600** of the selection step for a user using a software program as described herein in accordance with some embodiments of the invention. Screen shot **600** includes windows **615, 625, and 635**, which show a 3D anatomical image of surgical target **630** on different planes. In this step, the user selects the appropriate 3D image corresponding to anatomical location of where the procedure will occur. In some embodiments, the user uses a graphic control to change the perspective of the image in order to more easily view the image from different angles. In some embodiments, the user can view the surgical target **630** within separate coordinated views for each of the x-axis, y-axis and z-axis coordinates for each anatomical location in the

database in each window **615**, **625** and **635**, respectively.

(307) In some embodiments, after selecting the desired 3D image of the surgical target **630**, the user will plan the appropriate trajectory on the selected image. In some embodiments, an input control is used with the software in order to plan the trajectory of the surgical instrument **35**. In one embodiment of the invention, the input control is in the shape of a biopsy needle **8110** for which the user can plan a trajectory.

(308) FIG. **15B** shows a screen shot **650** during the medical procedure in accordance with some embodiments of the invention. In some embodiments, the user can still view the anatomical target **630** in different x-axis, y-axis and z-axis coordinate views on windows **615**, **625**, and **635**. As shown in screen shot **650**, the user can see the planned trajectory line **670** in multiple windows **615** and **625**. The actual trajectory and location of the surgical instrument **35** is superimposed on the image (shown as line segment **660**). In some embodiments, the actual trajectory and location of the surgical instrument **35** is dynamically updated and displayed, and is shown as a line segment **660**. In some other embodiments, the actual trajectory and location of the surgical instrument **35** could be shown as a trapezoid or a solid central line surrounded by a blurred or semi-transparent fringe to represent the region of uncertainty. In some embodiments (under perfect conditions with no bending of the surgical instrument as it enters tissues) the tracking system **3417** and robot **15** encoders calculate that the surgical instrument **35** should be located at the solid line or center of the trapezoid. In some embodiments, due to bending of the instrument **35** that might occur if tissues of different densities are crossed, there might be bending, with the amount of reasonably expected bending displayed as the edges of the trapezoid or fringe. In some embodiments, the size of this edge could be estimated knowing the stiffness and tolerance of the surgical instrument **35** within the guide tube **50**, and by using experimental data collected for the same instrument **35** under previous controlled conditions. In some embodiments, displaying this region of uncertainty helps prevent the user from expecting the system to deliver a tool to a target trajectory with a physically impossible level of precision.

(309) As described earlier, in some embodiments, the surgical robot **15** can be used with alternate guidance systems other than an LPS. In some embodiments, the surgical robot system **1** can comprise a targeting fixture **690** for use with a guidance system. In some embodiments, one targeting fixture **690** comprises a calibration frame **700**, as shown in FIGS. **20A-20E**. A calibration frame **700** can be used in connection with many invasive procedures; for example, it can be used in thoracolumbar pedicle screw insertion in order to help achieve a more accurate trajectory position. In some embodiments, the use of the calibration frame **700** can simplify the calibration procedure. In some embodiments of the invention, the calibration frame **700** can be temporarily affixed to the skin of a patient **18** surrounding a selected site for a medical procedure, and then the medical procedure can be performed through a window defined by the calibration frame.

(310) As shown in FIGS. **20A** and **20B**, in some embodiments of the invention, the calibration frame **700** can comprise a combination of radio-opaque markers **730** and infrared, or "active," markers **720**. In some embodiments, the radio-opaque markers **730** can be located within the CT scan region **710**, and the active markers **720** can be located outside of the CT scan region **710**. In some embodiments, a surgical field **17** (i.e., the area where the invasive procedure will occur) can be located within the perimeter created by radio-opaque markers **730**. In some embodiments, the actual distances of the radio-opaque **730** and active markers **720** relative to each other can be measured from a high-precision laser scan of the calibration frame. Additionally or alternatively, in some embodiments, the actual relative distances can be measured by actively measuring the positions of active markers **720** while nearly simultaneously or simultaneously pointing with a pointing device, such as a conventional digitizing probe, to one or more locations on the surface of the radio-opaque markers **730**. In certain embodiments, digitizing probes can comprise active markers **720** embedded in a rigid body **690** and a tip extending from the rigid body.

(311) In some embodiments, through factory calibration or other calibration method(s), such as pivoting calibration, the location of the probe tip relative to the rigid body of the probe can be established. In some embodiments, it can then be possible to calculate the location of the probe's tip from the probe's active markers **720**. In some embodiments, for a probe with a concave tip that is calibrated as previously described, the point in space returned during operation of the probe can represent a point distal to the tip of the probe at the center of the tip's concavity. Therefore, in some embodiments, when a probe (configured with a concave tip and calibrated to marker **730** of the same or nearly the same diameter as the targeting fixture's radio-opaque marker **730**) is touched to the radio-opaque marker **730**, the probe can register the center of the sphere. In some embodiments, active markers **720** can also be placed on the robot in order to monitor a position of the robot **15** and calibration frame **700** simultaneously or nearly simultaneously.

(312) In some embodiments, the calibration frame **700** is mounted on the patient's skin before surgery/biopsy, and will stay mounted during the entire procedure. Surgery/biopsy takes place through the center of the frame **700**.

(313) In some embodiments, when the region of the plate with the radio-opaque markers **730** is scanned intra-operatively or prior to surgery (for example, using a CT scanner), the CT scan contains both the medical images of

the patient's bony anatomy, and spherical representations of the radio-opaque markers **730**. In some embodiments, software is used to determine the locations of the centers of the markers **730** relative to the trajectories defined by the surgeon on the medical images. Because the pixel spacing of the CT scan can be conveyed within encoded headers in DICOM images, or can be otherwise available to a tracking software (for example, the robotic guidance software **3406**), it can, in some embodiments, be possible to register locations of the centers of the markers **730** in Cartesian coordinates (in millimeters, for example, or other length units). In some embodiments, it can be possible to register the Cartesian coordinates of the tip and tail of each trajectory in the same length units. (314) In some embodiments, because the system knows the positions of the trajectories relative to the radio-opaque markers **730**, the positions of the radio-opaque markers **730** relative to the active markers **720**, and the positions of the active markers **720** on the calibration frame **700** relative to the active markers on the robot **15** (not shown), the system has all information necessary to position the robot's end-effector **30** relative to the defined trajectories.

(315) In some other embodiments of the invention, the calibration frame **700** can comprise at least three radio-opaque markers **730** embedded in the periphery of the calibration frame **700**. In some embodiments, the at least three radio-opaque markers **730** can be positioned asymmetrically about the periphery of the calibration frame **700** such that the software, as described herein, can sort the at least three radio-opaque markers **730** based only on the geometric coordinates of each marker **730**. In some embodiments, the calibration frame **700** can comprise at least one bank of active markers **720**. In some embodiments, each bank of the at least one bank can comprise at least three active markers **720**. In some embodiments, the at least one bank of active markers **720** can comprise four banks of active markers **720**. In yet another aspect, the calibration frame **700** can comprise a plurality of leveling posts **77** coupled to respective corner regions of the calibration frame **700**. In some embodiments, the corner regions of the calibration frame **700** can include leveling posts **77** that can comprise radiolucent materials. In some embodiments, the plurality of leveling posts **77** can be configured to promote uniform, rigid contact between the calibration frame **700** and the skin of the patient **18**. In some embodiments, a surgical-grade adhesive film, such as, for example and without limitation, Ioban™ from 3M™, can be used to temporarily adhere the calibration frame **700** to the skin of the patient **18**. 3M™ and Ioban™ are registered trademarks of 3M Company. In some further embodiments, the calibration frame **700** can comprise a plurality of upright posts **75** that are angled away from the frame **700** (see FIG. **20B**). In some embodiments, the plurality of active markers **720** can be mounted on the plurality of upright posts **75**.

(316) As shown in FIG. **20B**, in some embodiments, there are four radio-opaque markers **730** (non-metallic BBs from an air gun) embedded in the periphery of the frame, labeled OP1, OP2, OP3, OP4. In some embodiments, only three markers **730** are needed for determining the orientation of a rigid body in space (the 4th marker is there for added accuracy).

(317) In some embodiments, the radio-opaque markers **730** are placed in an asymmetrical configuration (notice how OP1 and OP2 are separated from each other by more distance than OP3 and OP4, and OP1 and OP4 are aligned with each other across the gap, however OP3 is positioned more toward the center than OP2). The reason for this arrangement is so that a computer algorithm can automatically sort the markers to determine which is which if only given the raw coordinates of the four markers and not their identification.

(318) In some embodiments, there are four banks of active markers **720** (three markers **720** per bank). Only one bank of three markers **720** is needed (redundancy is for added accuracy and so that the system will still work if the surgeon, tools, or robot are blocking some of the markers).

(319) In some embodiments, despite the horizontal orientation of the patient **18**, the angulation of the upright posts can permit the active markers **720** to face toward the cameras (for example cameras **8200** shown in FIG. **81**). or detection devices of the tracking system (for example, the tracking system **3417**). In some embodiments, the upright posts can be angled away from the calibration frame by about 10°.

(320) In some applications, to establish the spatial relationship between the active **720** and radio-opaque markers **730**, a conventional digitizing probe, such as a 6-marker probe, embedded with active markers **720** in a known relationship to the probe's tip (see for example FIG. **20C**) can be used to point to each of the radio-opaque markers **730**. In some embodiments, the probe can point to locations on two opposite surfaces of the spherical radio-opaque markers **730** while recording the position of the probe tip and the active markers **720** on the frame **700** simultaneously. Then, the average position of the two surface coordinates can be taken, corresponding to the center of the sphere. An image of the robot **15** used with this targeting fixture **690** is shown in FIG. **20D**. For placement of conventional surgical screws, a biopsy, injection, or other procedures, in some embodiments, the robot **15** can work through the window formed by the frame **700**. During a surgical procedure, in some embodiments, the working portal is kept on the interior of the frame **700** and the markers **720** on the exterior of the frame **700** can improve accuracy over a system where fiducials are mounted away from the area where surgery is being performed. Without wishing to be bound by theory, simulation, and/or modeling, it is believed that a

reason for improved accuracy is that optimal accuracy of tracking markers **720** can be achieved if tracking markers **720** are placed around the perimeter of the frame **700** being tracked.

(321) Further embodiments of the invention are shown in FIG. **20E** illustrating a calibration frame **700**. This fixture **690** is simplified to make it less obstructive to the surgeon. In some embodiments, the calibration frame **700** can comprise four active markers **720** having a lower profile than the active markers **720** described above and depicted in FIGS. **20A-20D**. For example, the calibration frame **700** can comprise a plurality of upright posts **75** that are angled away from the calibration frame by about 10° . In some embodiments, the active markers **720** are mounted on the posts **75** that are angled back by 10° , and this angulation keeps the markers **720** facing toward the cameras despite the patient being horizontal.

(322) Moreover, in some embodiments, the front markers **720** can have less chance of obscuring the rear markers **720**. For example, posts **75** that are farthest away from the camera or farthest from a detection device of the tracking system **3417** can be taller and spaced farther laterally than the posts **75** closest to the camera.

(323) In some further embodiments of the invention, the calibration frame **700** can comprise markers **730** that are both radio-opaque for detection by a medical imaging scanner, and visible by the cameras or otherwise detectable by the real-time tracking system **3417**. In some embodiments, the relationship between radio-opaque **730** and active markers (**730**, **720**) does not need to be measured or established because they are one in the same.

Therefore, in some embodiments, as soon as the position is determined from the CT scan (or other imaging scan), the spatial relationship between the robot **15** and anatomy of the patient **18** can be defined.

(324) In other embodiments, the targeting fixture **690** can comprise a flexible roll configuration. In some embodiments, the targeting fixture **690** can comprise three or more radio-opaque markers **730** that define a rigid outer frame and nine or more active markers **720** embedded in a flexible roll of material (for example, the flexible roll **705** in FIG. **21A**). As described earlier, radio-opaque markers **730** are visible on CT scans and/or other medical diagnostic images, such as MRI, or reconstructions from O-arm or Iso-C scans, and their centroids can be determined from the 3D image. Active markers **720** include tracked markers **720** that have 3D coordinates that are detectable in real-time using cameras or other means. Some embodiments can utilize active marker systems based on reflective optical systems such as Motion Analysis Inc., or Peak Performance. Other suitable technologies include infrared-emitting marker systems such as Optotrak, electromagnetic systems such as Medtronic's Axiem®, or Flock of Birds®, or a local positioning system ("LPS") described by Smith et al. in U.S. Patent Publication No. 2007/0238985.

(325) Flock Of Birds® is a registered trademark of Ascension Technology Corporation.

(326) Axiem is a trademark of Medtronic, Inc., and its affiliated companies.

(327) Medtronic® is a registered trademark used for Surgical and Medical Apparatus, Appliances and Instruments.

(328) In some embodiments of the invention, at least a portion of the flexible roll **705** can comprise self-adhering film, such as, for example and without limitation, 3M™Ioban™ adhesive film (iodine-impregnated transparent surgical drape) similar to routinely used operating room product model 6651 EZ (3M, St. Paul, MN). Ioban™ is a trademark of 3M company.

(329) In some embodiments, within the flexible roll **705**, the radio-opaque and active markers (**730**, **720**) can be rigidly coupled to each other, with each radio-opaque marker **730** coupled to three or more active markers **720**. Alternatively, in some embodiments, the markers can simultaneously serve as radio-opaque and active markers (for example, an active marker **720** whose position can be detected from cameras or other sensors), and the position determined from the 3D medical image can substantially exactly correspond to the center of the marker **720**. In some embodiments, as few as three such markers **720** could be embedded in the flexible roll **705** and still permit determination of the spatial relationship between the robot **15** and the anatomy of the patient **18**. If radio-opaque markers **730** and active markers **720** are not one in the same, in some embodiments the at least three active markers **720** must be rigidly connected to each radio-opaque marker **730** because three separate non-collinear points are needed to unambiguously define the relative positions of points on a rigid body. That is, if only one or 2 active markers **720** are viewed, there is more than one possible calculated position where a rigidly coupled radio-opaque marker could be.

(330) In some embodiments of the invention, other considerations can be used to permit the use of two active markers **720** per radio-opaque marker **730**. For example, in some embodiments, if two active markers **720** and one radio-opaque marker **730** are intentionally positioned collinearly, with the radio-opaque marker **730** exactly at the midpoint between the two active markers **720**, the location of the radio-opaque marker **730** can be determined as the mean location of the two active markers **720**. Alternatively, in some embodiments, if the two active markers **720** and the radio-opaque marker **730** are intentionally positioned collinearly but with the radio-opaque marker **730** closer to one active marker **720** than the other (see for example FIG. **21B**), then in some embodiments, the radio-opaque marker **730** must be at one of two possible positions along the line in space formed by the two active

markers **720** (see FIG. **21B**). In this case, in some embodiments, if the flexible roll **705** is configured so that each pair of active markers **720** is oriented (when in its final position) with one marker **720** more toward the center of the flexible roll **705**, then it can be determined from the orientations of all markers or certain combinations of markers from different regions which of the two possible positions within each region is the correct position for the radio-opaque marker **730** (see FIG. **21C** showing flexible roll **705** showed rolled on a torso and shown unrolled on a torso with markers **720**, **730** in place). As shown, the radio-opaque markers **730** can be positioned toward the inside of the frame **705**), with marker groups nearer to the top of the figure having the radio-opaque marker **730** positioned below the active markers **720** and marker groups near the bottom of the figure having the radio-opaque marker positioned above the active markers **720**.

(331) In some embodiments, the flexible roll **705** can be positioned across the patient's back or other area, and adhered to the skin of the patient **18** as it is unrolled. In some embodiments, knowing the spatial relationship between each triad of active markers **720** and the rigidly coupled radio-opaque marker **730**, it is possible to establish the relationship between the robot **15** (position established by its own active markers **720**) and the anatomy (visualized together with radio-opaque markers **730** on MRI, CT, or other 3D scan). In some embodiments, the flexible roll **705** can be completely disposable. Alternatively, in some other embodiments, the flexible roll **705** can comprise reusable marker groups integrated with a disposable roll with medical grade adhesive on each side to adhere to the patient **18** and the marker groups **720**, **730**. In some further embodiments, the flexible roll **705** can comprise a drape incorporated into the flexible roll **705** for covering the patient **18**, with the drape configured to fold outwardly from the roll **705**.

(332) In some embodiments, after the roll **705** has been unrolled, the roll **705** can have a desired stiffness such that the roll **705** does not substantially change its position relative to the bony anatomy of the patient **18**. In some embodiments of the invention, a conventional radiolucent wire can be embedded in the perimeter of the frame **700**. In some embodiments, it a chain of plastic beads, such as the commercially available tripods shown in FIG. **21D**, or a commercially available "snake light" type fixture, can be employed to provide desired stiffness to the unrolled fixture such that it maintains its position after unrolling occurs. For example, in some embodiments, the beads of the chain of plastic beads as shown can be affixed to each other with a high friction so that they hold their position once shifted. Further, in some embodiments, chains of beads can be incorporated into, and define, a perimeter of the frame **700**. In some embodiments, this type of frame could be loaded with conventional chemicals that mix at the time of application. For example, in some embodiments, components of a conventional two-part epoxy could be held in separate fragile baggies within the frame that pop open when the user first starts to manipulate the beads. In some embodiments, the user would attach the frame to the patient **18**, and mold it to the contours of the patient's body. After a short period of time, the frame **700** would solidify to form a very rigid frame, locking the beads in their current orientation.

(333) In some embodiments of the invention, the targeting fixture **690** can be an adherable fixture, configured for temporary attachment to the skin of a patient **18**. For example, in some embodiments, the targeting fixture **690** can be temporarily adhered to the patient **18** during imaging, removed, and then subsequently reattached during a follow-up medical procedure, such as a surgery. In some embodiments, the targeting fixture **690** can be applied to the skull of a patient **18** for use in placement of electrodes for deep brain stimulation. In some embodiments, this method can use a single fixture **690**, or two related fixtures. In this instance, the two related fixtures can share the same surface shape. However, one fixture **690** can be temporarily attached at the time of medical image scanning, and can include radio-opaque markers **730** (but not active markers **720**), and the second fixture **690** can be attached at the time of surgery, and can include active markers **720** (but not radio-opaque markers **730**).

(334) In some embodiments, the first fixture (for scanning) can comprise a frame **690** with three or more embedded radio-opaque markers **730**, and two or more openings **740** for application of markings (the markings shown as **750** in FIG. **22B**). In some embodiments, the device **690** can be adhered to the scalp of a patient **18**, and the openings **740** can be used to paint marks on the scalp with, for example, henna or dye (shown as "+" marks **750** in FIG. **22B**). With this fixture **690** in place, in some embodiments, the patient **18** can receive a 3D scan (for example an MRI or CT scan) in which the radio-opaque markers **730** are captured. As illustrated by FIG. **22B**, in some embodiments, the fixture **690** can then be removed, leaving the dye marks **750** on the scalp. In some embodiments, on a later date (before the dye marks **750** wear off), the patient **18** can return, and the surgeon or technician can attach the 2nd fixture (for surgery) containing active markers **720** for intraoperative tracking (see FIG. **22C-22D**). In some embodiments, the fixture **690** shown in FIG. **22C** can be mounted to the scalp, spatially positioned and oriented in the same position as the previously adhered first fixture (shown in FIG. **22A**) by ensuring that the previously placed dye marks **750** line up with holes in the second fixture **690** (see the alignment arrows depicted in FIG. **22C**). Optionally, in some embodiments, the fixture **690** can have a transparent frame for good visualization. The above-described method assumes that the locations of the marks **750** do not change over the period of time between the scan and the return of the patient **18** for surgery. Since the relative positions

between the radio-opaque markers **730** from the temporary (first) fixture **690** (which appear in the scan) and the active markers **720** on the second applied fixture **690** are known through a calibration and/or by careful manufacturing of the fixtures **690**, the coordinate system of the anatomy and the coordinate system of the active markers **720** can be synchronized so that the robot **15** can target any planned trajectory on the 3D image as described further herein. Further, in some embodiments, this method can enable image guidance with only one pre-op scan and without requiring the patient **18** to go home after a pre-op scan. This circumvents the need for a patient **18** to take care of wounds from targeting screws that are invasively drilled into the skull of the patient **18**. (335) In some embodiments of the invention, the targeting fixture **690** can comprise a conventional clamping mechanism for securely attaching the targeting fixture **690** to the patient **18**. For example, in some embodiments, the targeting fixture **690** can be configured to clamp to the spinous process **6301** of a patient **18** after the surgeon has surgically exposed the spinous process. FIG. 23 shows a dynamic tracking device **2300** mounted to the spinous process **2310** in the lumbar spine of a patient **18** in accordance with some embodiments of the invention. This targeting fixture is used with Medtronic's StealthStation. This figure is reprinted from Bartolomei J, Henn J S, Lemole G M Jr., Lynch J, Dickman C A, Sonntag V K H, Application of frameless stereotaxy to spinal surgery, Barrow Quarterly 17(1), 35-43 (2001).

(336) StealthStation® is a trademark of Medtronic, Inc., and its affiliated companies.

(337) In some embodiments, during use of a targeting fixture **690** having a conventional clamping mechanism with image guidance, the relationship between the markers **720**, **730** and the bony anatomy of the patient **18** can be established using a registration process wherein known landmarks are touched with a digitizing probe at the same time that the markers on the tracker are visible. In some embodiments of the invention, the probe itself can have a shaft protruding from a group of markers **720**, **730**, thereby permitting the tracking system **3417** to calculate the coordinates of the probe tip relative to the markers **720**, **730**.

(338) In some embodiments, the clamping mechanism of the targeting fixture **690** can be configured for clamping to the spinous process **2310**, or can be configured for anchoring to bone of the patient **18** such that the fixture **690** is substantially stationary and not easily moved. In some further embodiments, the targeting fixture **690** can comprise at least three active markers **720** and distinct radio-opaque markers **730** that are detected on the CT or other 3D image, preferably near the clamp (to be close to bone). In some alternative embodiments, the active markers **720** themselves must be configured to be visualized accurately on CT or other 3D image. In certain embodiments, the portion of the fixture **690** containing a radio-opaque marker **730** can be made to be detachable to enable removal from the fixture after the 3D image is obtained. In some further embodiments, a combination of radio-opaque **730** and active markers **720** can allow tracking with the robot **15** in the same way that is possible with the frame-type targeting fixtures **690** described above.

(339) In some embodiments, one aspect of the software and/or firmware disclosed herein is a unique process for locating the center of the above-described markers **730** that takes advantage of the fact that a CT scan can comprise slices, typically spaced 1.5 mm or more apart in the z direction, and sampled with about 0.3 mm resolution in the x-axis and y-axis directions. In some embodiments, since the diameter of the radio-opaque markers **730** is several times larger than this slice spacing, different z slices of the sphere will appear as circles of different diameters on each successive x-y planar slice. In some embodiments, since the diameter of the sphere is defined beforehand, the necessary z position of the center of the sphere relative to the slices can be calculated to provide the given set of circles of various diameters. Stated similarly, in some embodiments, a z slice substantially exactly through the center of the sphere can yield a circle with a radius R that is substantially the same as that of the sphere. In some embodiments, a z slice through a point at the top or bottom of the sphere can yield a circle with a radius R approximating zero. In some other embodiments, a z slice through a z-axis coordinate Z1 between the center and top or bottom of the sphere can yield a circle with a radius $R1 = R \cos(\arcsin(Z1/R))$.

(340) In some embodiments of the invention, the observed radii of circles on z slices of known inter-slice spacing can be analyzed using the equation defined by $R1 = R \cos(\arcsin(Z1/R))$. This provides a unique mathematical solution permitting the determination of the distance of each slice away from the center of the sphere. In cases in which a sphere has a diameter small enough that only a few slices through the sphere appear on a medical image, this process can provide a more precise the center of a sphere.

(341) Some embodiments of the use of the calibration frame **700** are described to further clarify the methods of use. For example, some embodiments include the steps of a conventional closed screw or conventional needle (for example, a biopsy needle **8110**) insertion procedure utilizing a calibration frame **700** as follows. In some embodiments, a calibration frame **700** is attached to the patient's **18** skin, substantially within the region at which surgery/biopsy is to take place. In some embodiments, the patient **18** receives a CT scan either supine or prone, whichever positioning orients the calibration frame **700** upward. In some embodiments, the surgeon subsequently manipulates three planar views of the patient's **18** CT images with rotations and translations. In some embodiments, the surgeon then draws trajectories on the images that define the desired position, and strike angle

of the end-effectuator **30**. In some embodiments, automatic calibration can be performed in order to obtain the centers of radio-opaque makers **730** of the calibration frame **700**, and to utilize the stored relationship between the active markers **720** and radio-opaque markers **730**. This procedure permits the robot **15** to move in the coordinate system of the anatomy and/or drawn trajectories.

(342) In some embodiments, the robot **15** then will move to the desired position. In some embodiments, if forceful resistance beyond a pre-set tolerance is exceeded, the robot **15** will halt. In some further embodiments, the robot **15** can hold the guide tube **50** at the desired position and strike angle to allow the surgeon to insert a conventional screw or needle (for example, needle **7405**, **7410** or biopsy needle **8110**). In some embodiments, if tissues move in response to applied force or due to breathing, the movement will be tracked by optical markers **720**, and the robot's position will automatically be adjusted.

(343) As a further illustration of a procedure using an alternate guidance system, in some embodiments, the steps of an open screw insertion procedure utilizing an optical guidance system is described. In some embodiments, after surgical exposure, a targeting fixture **690** comprising a small tree of optical markers, for example, can be attached to a bony prominence in the area of interest. In some embodiments, conventional calibration procedures for image guidance can be utilized to establish the anatomy relative to the optical tracking system **3417** and medical images. For another example, the targeting fixture **690** can contain rigidly mounted, substantially permanent or detachable radio-opaque markers **730** that can be imaged with a CT scan. In some embodiments, the calibration procedures consistent with those stated for the calibration frame **700** can be utilized to establish the anatomy relative to the robot **15** and the medical image.

(344) In some embodiments, the surgeon manipulates three planar views of the patient's CT images with rotations and translations. In some embodiments, the surgeon then draws trajectories on the images that define the desired position and strike angle of the end-effectuator **30**. In some embodiments, the robot **15** moves to the desired position. In some embodiments, if forceful resistance beyond a pre-set tolerance is exceeded, the robot **15** will halt. In some embodiments, the robot **15** holds the guide tube **50** at the desired position and strike angle to allow the surgeon to insert a conventional screw. In some embodiments, if tissues move in response to applied force or due to breathing, the movement will be tracked by optical markers **720**, and the robot's position will automatically be adjusted.

(345) FIG. **36** illustrates an example embodiment **3600** of surgical robot system **1** that utilizes a surveillance marker **710** in accordance with one or more aspects of the invention. As illustrated, the example embodiment **3600** comprises a 4-marker tracker array **3610** attached to the patient **18** and having a surveillance marker, and a 4-marker tracker array **3620** on the robot **15**. In some embodiments, during usage, it may possible that a tracker array, or tracker (**3610** in FIG. **36**), on a patient **18** inadvertently shifts. For example, a conventional clamp positioned on a patient's **18** spinous process **2310** where the tracker **3610** is attached can be bumped by the surgeon's arm and move (i.e., bend or translate) to a new position relative to the spinous process **2310**.

Alternatively, a tracker **3610** that is mounted to the skin of the patient **18** can move gradually with the skin, as the skin settles or stretches over time. In this instance, the accuracy of the robot **15** movement can be lost because the tracker **3610** can reference bony anatomy from a medical image that no longer is in the same position relative to the tracker as it had been during the medical image scan. To overcome such problems, some embodiments of the invention provide a surveillance marker **710** as illustrated in FIG. **36**. As shown, in some embodiments, the surveillance marker **710** can be embodied or can comprise one or more markers **710** rigidly affixed to a patient **18** in a location different than the location in which a primary tracker array **3610** is affixed; for example, a different spinous process **2310**, on the skin, or on a small post drilled into the ilium. Accordingly, in some embodiments, the surveillance marker **710** can be located on the same rigid body as the primary tracker array **3610** but at a different location on the rigid body.

(346) In one embodiment, in response to placement of the surveillance marker **710**, execution of a control software application (e.g., robotic guidance software **3406**) can permit an agent (e.g., a surgeon, a nurse, a diagnostician) to select "set surveillance marker". At this time, the vector (3D) distances between the surveillance marker **710**, and each of the markers **3611**, **3612**, **3613**, and **3614** on the primary tracker array **3610** can be acquired and retained in computer **100** memory (such as a memory of a computing device **3401** executing the control software application). In an embodiment in which a 4-marker tracker array **3610** is utilized (FIG. **36**), four distances **3611a**, **3612a**, **3613a**, and **3614a** can be acquired and retained, representing the distances between the surveillance marker **710** and markers **3611**, **3612**, **3613**, and **3614**. In such embodiment, at each frame of real-time data during a procedure, the surgical robot system **1** disclosed herein can calculate updated distances between each of the markers **3611**, **3612**, **3613**, and **3614** on the primary tracker array **3610** and the surveillance marker **710**. The system **1** can then compare the updated distances or a metric thereof (for example, the sum of the magnitude of each distance) to the available values (for example, values retained in the computer **100** memory). In some embodiments, in view that the surveillance marker **710** and tracker array **3610** can be on the same rigid

body, the updated distances and/or the metric thereof (such as their sum) can remain substantially fixed unless one or more of the tracker array **3610** or the surveillance marker **710** shifts. In some embodiments, in response to a shift of the tracker array **3610** or the surveillance marker **710**, or both, a notification can be issued to alert an agent of a loss in movement accuracy. In some embodiments, if the surveillance marker **710** offset exceeds a pre-set amount, operation of the surgical robot system **1** can be halted. In some embodiments, in response to a user intentionally shifting the tracker array **3610** or the surveillance marker **710** to a new position, execution of the control software application can permit overwriting a set of one or more stored distances with new values for comparison to subsequent frames.

(347) In some embodiments, as illustrated in FIG. **36**, embodiment **3600** of surgical robot system **1** utilizes a surveillance marker **710**, a 4-marker tracker array **3610** attached to the patient **18**, and a 4-marker tracker array **3620** on the robot **15**. It should be appreciated that in some embodiments, the 4-marker tracker array **3620** on the robot **15** can experience an unintentional shift in a manner similar to that for the 4-marker array tracker **3610** on the patient **18**. Consequently, in certain embodiments, a surveillance marker (not shown) can be attached to a different position on the robot **15** arm than the robot's 4-marker tracker array **3620** to control, at least in part, such unintentional shift. In some embodiments, a surveillance marker on the robot **15** may provide lesser efficiencies than a surveillance marker **710** on the patient **18** in view that the robot **15** arm can be manufactured with negligible or minimal likelihood of the robot's tracker array **3620** or surveillance marker (not shown) shifting. In addition or in the alternative, other embodiments can include means for registering whether the tracker **3620** has shifted can be contemplated for the robot's tracker array **3620**. For instance, in some embodiments, the means for registering may not include a surveillance marker, but may comprise the extant robot **15** tracking system **3417** and one or more of the available conventional encoders. In some embodiments, the system **1** and encoder(s) can compare movement registered from the tracker **3620** to movement registered from counts of encoders (not shown) on each robot **15** axis. For example, in some embodiments where the robot's tracker array **3620** is mounted on the housing **27** that rotates with the roll **62** axis (which can be farther away from the base **25** than the z-axis **70**, x-axis **66**, y-axis **68**, and roll **62** axis) then changes in z-axis **70**, x-axis **66**, y-axis **68**, and roll **62** axis encoder counts can provide highly predictable changes in the position of the robot's tracker array **3620** in the coordinate systems of the tracking system **3417** and robot **15**. In some embodiments, the predicted movement based on encoder counts and tracked 3D position of the tracker array **3620** after application of known counts can be compared and, if the values differ substantially (or values are above a predetermined threshold), the agent can be alerted to the existence of that an operational issue or malfunction. The operational issue can originate from one or more of a malfunction in the registration of counts (i.e., electromechanical problem), malfunction in registration of the tracker's markers **3621**, **3622**, **3623**, **3624** (for example, outside of tracking system's optimum volume), or shift in the position of the tracker **3620** on the robot's **15** surface during the move.

(348) It should be appreciated that other techniques (for example, methods, systems, and combinations thereof, or the like) can be implemented in order to respond to operational issues that may prevent tracking of the movement of a robot **15** in the surgical robot system **1**. In one embodiment, marker reconstruction can be implemented for steadier tracking. In some embodiments, marker reconstruction can maintain the robot end-effector **30** steady even if an agent partially blocks markers during operation of the disclosed surgical robot system **1**.

(349) As described herein, in some embodiments, at least some features of tracking movement of the robot's end-effector **30** can comprise tracking a virtual point on a rigid body utilizing an array of one or more markers **720**, such tracking comprising one or more sequences of translations and rotations. As an illustration, an example methodology for tracking a visual point on a rigid body using an array of three attached markers is described in greater detail herein, such methodology can be utilized to implement marker reconstruction technique in accordance with one or more aspects of the invention. FIG. **37**, for example, illustrates an example of a methodology for tracking a visual point **4010** on a rigid body using an array of three attached markers **4001**, **4002**, **4003**. In some embodiments, the method includes contemplating a reference data-frame. In some embodiments, the reference data-frame can be associated with a set of reproducible conditions for the relative positions of the markers **4001**, **4002**, **4003**, but not necessarily defined locations.

(350) FIG. **38** illustrates a procedure for monitoring the location of a point of interest **4010** relative to three markers **4001**, **4001**, **4003** based on images received from the methodology illustrated in FIG. **37** in accordance with some embodiments of the invention. As shown, the method includes translation and rotating the markers **4001**, **4002**, **4003** and point of interest **4010** with the conditions as shown. In some embodiments, the method can include saving the x-axis, y-axis, and z-axis coordinates of the point of interest **4010** in this reference frame for future use. In some embodiments, for each subsequent data-frame the method can include the steps of; 1). transform the markers **4001**, **4002**, **4003** using the conditions defined for the reference frame (keeping track of the rotations and translations), 2). add the point of interest **4010** (which was saved after establishing the reference frame) and 3). transform the point of interest **4010** back to the current location of the markers **4001**, **4002**, **4003**

using inverses of the saved translations and rotations from step 1. In some embodiments, upon or after completing step 1 above, the actual proximity of the markers **4001**, **4002**, **4003** to their original reference data-frame is dictated by marker noise and rigid body rigidity. In some embodiments, the markers will never overlay perfectly with their counterparts that were stored when establishing the reference frame. In some embodiments, the disclosed method can permit the markers **4001**, **4002**, **4003** to get as close as possible to their original relative spacing.

(351) FIGS. **39A-F** illustrate examples of tracking methodology based on an array of three attached markers **4001**, **4002**, and **4003** in accordance with some embodiments of the invention. In some embodiments, the goal can be marker **4001** on the origin, marker **4002** on the positive x-axis, and marker **4003** in the x-y plan in a positive y direction (shown in FIG. **39A**). Assuming a starting configuration as shown in FIG. **39B**, in some embodiments, the method can include translating the rigid body so that marker **4001** is at the origin as shown in FIG. **39C**. In some embodiments, the method can then include rotation about the y-axis so that marker **4002** is in the x-y plane (i.e., $z=0$) (see FIG. **39D**). In some embodiments, the method can then include rotating the z-axis so that marker **4002** is at $y=0$, x coordinate positive (as shown in FIG. **39E**). Finally, in some embodiments, the method can include rotating about the x-axis so that marker **4003** is at $z=0$, y coordinate positive. In some embodiments, a record of the translations and rotations can be retained in order to utilize the negative values to transform position(s) back after adding the point of interest. In some embodiments, when translating the rigid body so that the marker **4001** moves to the origin, the vector to add to each marker's position vector is simply the negative **4001** position vector. In some embodiments, to determine the values of 0 to plug into the rotation matrices in steps 2, 3, and 4, use the arctangent. For example, rotate marker **4002** about the y-axis to $z=0$ where:

$$(352) \quad M2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

(353) FIG. **40** illustrates an example of a two dimensional representation for rotation about the y-axis in accordance with some embodiments of the invention. As shown, FIG. **40** illustrates one embodiment showing a two-dimensional representation looking down the axis about which rotation occurs (e.g., y-axis is going into the page). In this example, a position rotation of $0=56.3^\circ$ about the y-axis is needed to bring **4002** to $z=0$. It should be appreciated that the appropriate direction (+ or -) of the rotation angle to plug into the rotation matrix can be confusing. In some embodiments, it is beneficial to draw the plane of the rotation with the rotation axis coming out of the plane contained in the page surface (for example, the right-hand rule can provide a suitable orientation), then a counterclockwise rotation is positive and a clockwise rotation is negative. In the foregoing example, the axis was going into the page surface, thus a clockwise rotation was positive.

(354) FIGS. **41A-C** illustrates an alternative representation of two dimensional representations for rotations about the axis, depicting how each plane can be drawn for counterclockwise positive and clockwise negative in accordance with some embodiments of the invention. As shown, FIG. **41A** illustrates an alternative representation of a two dimensional representation for rotation about an X-axis. FIG. **41B** illustrates an alternative representation of a two dimensional representation for rotation about a Y-axis. FIG. **41C** illustrates an alternative representation of a two dimensional representation for rotation about a Z-axis

(355) In some embodiments, to rotate the rigid body about the y-axis so that **4002** is in the x-y plane ($z=0$):

$$(356) \quad y = +\tan^{-1} \left(\frac{4002_z}{4002_x} \right)$$

(357) In some embodiments, to rotate the rigid body about the z-axis so that **4002** is at $y=0$,

$$(358) \quad z = -\tan^{-1} \left(\frac{4002_y}{4002_x} \right)$$

(359) In some embodiments, to rotate the rigid body about the x-axis so that **4003** is at $z=0$:

$$(360) \quad x = -\tan^{-1} \left(\frac{4003_z}{4003_y} \right)$$

(361) As described herein, the example method to transform markers **4001**, **4002**, **4003** as close as possible to the reference frame can comprise; 1). translate the rigid body so that **4001** is at the origin (0,0,0), and 2). rotate about the y-axis so that **4002** is in the x-y plane (i.e., $z=0$), and 3). rotate about the z-axis so that **4002** is at $y=0$, x coordinate positive, and 4). rotate about the x-axis so that **4003** is at $z=0$, y coordinate positive. In other embodiments, a method to reach the same reference can comprise: 1). translate the rigid body so that **4001** is at the origin (0,0,0), and 2). rotate about the x-axis so that **4002** is in the x-y plane (i.e., $z=0$), and 3). rotate about the z-axis so that **4002** is at $y=0$, x coordinate positive, 4). rotate about the x-axis so that **4003** is at $z=0$, y coordinate positive. It should be appreciated that there are other possible methods and related actions, both in the reference frame chosen and in how the rigid body is manipulated to get it there. The described method is simple, but does not treat markers equally. The reference frame requires **4001** to be restricted the most (forced to a point), **4002** less (forced to a line), and **4003** the least (forced to a plane). As a result, errors from noise in markers are manifested asymmetrically. For example, consider a case where in a certain frame of data, noise causes each of the three

markers to appear further outward than they actually are or were (represented by **4001a**, **4002a**, and **4003a**) when the reference frame was stored (as depicted in FIG. 42.)

(362) In some embodiments, when the transformations are done to align the apparent markers “as close as possible” to their stored reference position, they will be offset. For example, when the stored point of interest is added, it will be misplaced in a direction on which marker was chosen as **4001** in the algorithm (see **4003**, **4003a** and **4002**, **4002a** for example in FIG. 43).

(363) Some embodiments provide additional or alternative methods for tracking points of interest that can involve more symmetrical ways of overlaying the actual marker positions with the stored reference positions. For example, in some embodiments, for three markers **4001**, **4002**, **4003**, a two-dimensional fitting method typically utilized in zoology can be implemented. (See, e.g., Sneath P. H. A., “Trend-surface analysis of transformation grids,” J. Zoology 151, 65-122 (1967)). The method can include a least squares fitting algorithm for establishing a reference frame and transforming markers to lie as close as possible to the reference. In this case, the reference frame is the same as described earlier except that the common mean point (hereinafter referred to as “CMP”) is at the origin instead of marker **4001**. In some embodiments, the CMP after forcing the markers into the x-y plane is defined in the following equation (and can be represented in FIG. 44):

$$(364) \text{CMP} = \begin{bmatrix} \bar{x} \\ \bar{y} \end{bmatrix} = \begin{bmatrix} (M1_x + M2_x + M3_x) / 3 \\ (M1_y + M2_y + M3_y) / 3 \end{bmatrix}$$

(365) In some embodiments, for the markers to be centered around CMP, the markers can be translated by subtracting the CMP from **4001**, **4002**, and **4003**. It should be noted that the point of interest being tracked is not included in determining CMP.sub.ref.

(366) In some embodiments, the method to transform markers as close as possible to this reference frame can comprise; 1). translating the rigid body so that **4001** is at the origin (0,0,0), and 2). rotating about the y-axis so that **4002** is in the x-y plane (i.e., z=0), and 3). rotating about the z-axis so that **4002** is at y=0, x coordinate positive into the x-ray plane, and 4). rotating about the x-axis so that **4003** is at z=0, y coordinate positive, and finally 5). calculate the CMP for the markers **4001**, **4002**, **4003** and translating the rigid body so that the CMP is at the origin (i.e., subtract the CMP from each point transformed). In some embodiments, steps 1-5 are done for the original set of markers for which the position of the point of interest was known and for the new set for which you are adding the point of interest. A further step can be included for the new set, for example, 6). rotate about the z-axis to best overlay the stored reference markers. In some embodiments, the rotation angle θ is found using the formula from Sneath:

$$(367) \tan \theta = \frac{\text{Math. } x_{\text{pos2}} y_{\text{ref}} - \text{Math. } x_{\text{ref}} y_{\text{pos2}}}{\text{Math. } x_{\text{ref}} x_{\text{pos2}} + \text{Math. } y_{\text{ref}} y_{\text{pos2}}}$$

(368) In some embodiments, if M1, M2, M3 denote the stored reference markers and M'1, M'2, M'3 denote the position being tracked in this data-frame, the equation can be written:

$$(369) \theta \tan^{-1} = \left[\frac{(M'_1 M1_y + M'_2 M2_y + M'_3 M3_y) - (M1_x M'_1 + M2_x M'_2 + M3_x M'_3)}{(M1_x M'_1 + M2_x M'_2 + M3_x M'_3) + (M1_y M'_1 + M2_y M'_2 + M3_y M'_3)} \right]$$

(370) It should be noted that this rotation angle can be small (e.g., smaller than about 1°). In some embodiments, after the markers **4001**, **4002**, **4003** are overlaid, some embodiments of the invention can include adding the point of interest then transforming the point of interest back to its true present location in the current frame of data. In some embodiments, to transform back, negative values saved from the forward transformation steps 1-6 as discussed above can be utilized. That is, for instance, go from step 6 to step 5 by rotating by negative θ , go from step 5 to step 4 by adding the CMP, etc.)

(371) In some embodiments, using this least-squares algorithm, noise is manifested more symmetrically and the point of interest will probably be calculated to be closer to its actual location. This can be illustrated in FIG. 45, which illustrates a depiction of results of applying a least squares fitting algorithm for establishing a reference frame and transforming markers **4001**, **4002**, **4003** as shown in FIG. 44 including noise. Regardless of which method is used, it can be beneficial to monitor the error between the marker locations at a given frame of data and the reference marker locations. In some embodiments, the method can include calculating and sum the vector distances of each marker and report the value in mm.

(372) FIG. 46 for example illustrates a depiction of error calculation for reference frame markers in accordance with some embodiments of the invention. In some embodiments, by continuously displaying this error value, the agent can be alerted if markers **4001**, **4002**, **4003** have become partially obscured, or if a marker **4001**, **4002**, **4003** is no longer securely or rigidly attached to the rigid body. In some embodiments, when performing a best fit on

more than 3 markers, they cannot be forced into a plane, and therefore the problem becomes much more difficult. In some embodiments, one solution is to inspect all or nearly all possible triangles formed by groups of 3 markers **4001**, **4002**, **4003** and evaluate which one gives the least standard deviation of the angles of the vertices. See Chèze L, Fregly B. J., Dimnet J: Technical note: A solidification procedure to facilitate kinematic analyses based on video system data,” Journal of Biomechanics 28(7), 879-884 (1995). In some other embodiments, the method can include calculating a least squares fit of the vertices of the geometric shape, requiring iteration to perform matrix decomposition (i.e., Newton-Raphson method). For example, see Veldpaus F E, Woltring H J, Dortmans L J M G, ‘A least-squares algorithm for the equiform transformation from spatial marker co-ordinates’, Journal of Biomechanics 21(1), 45-54 (1988).

(373) In some embodiments, when tracking 3D movement of a rigid body (for example, a robot **15** end-effector **30** or a targeted bone) using an array of 3 tracking markers **4001**, **4002**, **4003** that are rigidly attached to the rigid body, one example method for quantifying motion can include determining the transformations (translation and rotations) for the movement from a first (neutral) position (defined here as “A”) to second (current frame) position (herein referred to as “B”). In some embodiments, it may be convenient to describe the rotations as a three by three orientation matrix (direction cosines) of the rigid body in the position B, and to treat the three translation values as a 3×1 vector containing the x, y, z coordinates of the origin of the position A coordinate system transformed to position B. In some embodiments, the direction cosine matrix is a 3×3 matrix, the columns of which contain unit vectors that originally were aligned with the x, y, and z axes, respectively, of the neutral coordinate system. In some embodiments, to build a direction cosine matrix, a 3×3 matrix, A, can be defined in a manner that its columns are unit vectors, i, j, and k, aligned with the x, y, and z axes, respectively:

$$(374) A = \begin{bmatrix} i_x & j_x & k_x \\ i_y & j_y & k_y \\ i_z & j_z & k_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(375) Upon or after rotations of the coordinate system occur, in some embodiments, the new matrix (which is the direction cosine matrix, A') is as follows, where the unit vectors i', j', and k' represent the new orientations of the unit vectors that were initially aligned with the coordinate axes:

$$(376) A' = \begin{bmatrix} i'_x & j'_x & k'_x \\ i'_y & j'_y & k'_y \\ i'_z & j'_z & k'_z \end{bmatrix}$$

(377) In some embodiments, to determine the direction cosines and translation vector, the origin and unit vectors can be treated as aligned with the coordinate axes as four tracked points of interest in the manner described herein. For example, if the origin (o) and three unit vectors (i, j, k) are aligned with the coordinate axes, they are treated as virtual tracked points of interest with coordinates of:

$$(378) o = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} . \text{fwdarw.} i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} . \text{fwdarw.} j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} . \text{fwdarw.} k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(379) In some embodiments, these points of interest can provide the direction cosines and translation for the movement when moved along with the three markers from position A to position B. In some embodiments, it may be convenient to implement (for example execute) the method for moving the virtual points to these four points placed into a 3×4 matrix, P.

$$(380) P = \begin{bmatrix} o_x & i_x & j_x & k_x \\ o_y & i_y & j_y & k_y \\ o_z & i_z & j_z & k_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(381) In some embodiments, the matrix is as follows in position A:

$$(382) P' = \begin{bmatrix} a_x & b_x & c_x & d_x \\ a_y & b_y & c_y & d_y \\ a_z & b_z & c_z & d_z \end{bmatrix} = \begin{bmatrix} o'_x & i'_x + o'_x & j'_x + o'_x & k'_x + o'_x \\ o'_y & i'_y + o'_y & j'_y + o'_y & k'_y + o'_y \\ o'_z & i'_z + o'_z & j'_z + o'_z & k'_z + o'_z \end{bmatrix}$$

(383) In some embodiments, after movement, the direction cosine matrix is

$$(384) A' = \begin{bmatrix} i'_x & j'_x & k'_x \\ i'_y & j'_y & k'_y \\ i'_z & j'_z & k'_z \end{bmatrix} = \begin{bmatrix} b_x - a_x & c_x - a_x & d_x - a_x \\ b_y - a_y & c_y - a_y & d_y - a_y \\ b_z - a_z & c_z - a_z & d_z - a_z \end{bmatrix}$$

(385) In some embodiments, the vector o' represents the new position of the origin. In some embodiments, after moving the three markers **4001**, **4002**, **4003** from position A to position B, and bringing the four points (as a 3×4 matrix) along with the three markers **4001**, **4002**, **4003**, the translation of the origin is described by the first column. Further, in some embodiments, the new angular orientation of the axes can be obtained by subtracting the origin from the 2.sup.nd, 3.sup.rd and 4.sup.th columns. These methods should be readily apparent from the following graphic representation in FIG. 47, which illustrates a graphical representation of methods of tracking three dimensional movement of a rigid body.

(386) In some embodiments, if more than three markers **4001**, **4002**, **4003** are utilized for tracking the movement of a rigid body, the same method can be implemented repeatedly for as many triads of markers as are present. For example, in a scenario in which four markers, M1, M2, M3, and M4, are attached to the rigid body, there can be four triads: those formed by {M1, M2, M3}, {M1, M2, M4}, {M1, M3, M4}, and {M2, M3, M4}. In some embodiments, each of these triads can be used independently in the method described hereinbefore in order to calculate the rigid body motion. In some embodiments, the final values of the translations and rotations can then be the average of the values determined using the four triads. In some embodiments, in the alternative or in addition, other methods for achieving a best fit when using more than 3 markers may be used.

(387) In some embodiments, when tracking with four markers, in a scenario in which one of the four markers becomes obscured, it can be desirable to switch to tracking the rigid body with the remaining three markers instead of four. However, this change in tracking modality can cause a sudden variation in the results of one or more calculations utilized for tracking. In some embodiments, the variation can occur because the solution from the one remaining triad may be substantially different than the average of 4 triads. In some embodiments, if using the tracked position of the rigid body in a feedback loop to control the position of a robot **15** end-effector, the sudden variation in results of the calculation can be manifested as a physical sudden shift in the position of the robot **15** end-effector **30**. In some embodiments, this behavior is undesirable because the robot **15** is intended to hold a guide tube **50** steady with very high accuracy.

(388) Some embodiments include an example method for addressing the issue of sudden variation that occurs when one of the four markers M1, M2, M3, M4 is blocked, thereby causing the position to be calculated from a single triad instead of the average of four triads, can include reconstructing the blocked marker as a virtual marker. In some embodiments, to implement such reconstructing step with high accuracy, the most recent frame of data in which all four markers M1, M2, M3, M4 are visible can be retained substantially continuously or nearly continuously (for example in a memory of a computing device implementing the subject example method). In some embodiments, if all four markers M1, M2, M3, M4 are in view, the x-axis, y-axis, and z-axis coordinates of each of the four markers M1, M2, M3, and M4 are stored in computer **100** memory. It should be appreciated that in some embodiments, it may be unnecessary to log all or substantially all frames and is sufficient to overwrite the same memory block with the most recent marker coordinates from a full visible frame. Then, in some embodiments, at a frame of data in which one of the four markers M1, M2, M3, and M4 is lost, the lost marker's position can be calculated based on the remaining triad, using the example method described herein for remaining three markers. That is, the triad (the three visible markers) is transformed to a reference. The stored set of markers is then transformed to the same reference using the corresponding triad with the fourth marker now acting as a virtual landmark. The recovered position of the lost fourth marker can then be transformed back to the current position in space using the inverse of the transformations that took it to the reference position. In some embodiments, after the lost marker's position is reconstructed, calculation of the rigid body movement can be performed as before, based on the average of the four triads, or other best fit method for transforming the rigid body from position A to position B.

(389) In some embodiments, an extension to the methods for reconstructing markers **720** is to use multiple ambiguous synchronized lines of sight via multiple cameras **8200** tracking the same markers **720**. For example, two or more cameras **8200** (such as Optotrak® or Polaris®) could be set up from different perspectives focused on the tracking markers **720** on the targeting fixture **690** or robot **15**. In some embodiments, one camera unit could be placed at the foot of a patient's bed, and another could be attached to the robot **15**. In some embodiments, another camera unit could be mounted to the ceiling. In some embodiments, when all cameras **8200** substantially simultaneously view the markers **720**, coordinates could be transformed to a common coordinate system, and the position of any of the markers **720** would be considered to be the average (mean) of that marker's three dimensional position from all cameras used. In some embodiments, even with extremely accurate cameras, an

average is needed because with system noise, the coordinates from different cameras would not be exactly equal. However, when one line of sight is obscured, the lines of sight from other cameras **8200** (where markers **720** can still be viewed) could be used to track the robot **15** and targeting fixture **690**. In some embodiments, to mitigate twitching movements of the robot **15** when one line of sight is lost, it is possible that the marker **720** positions from the obscured line of sight could be reconstructed using methods as previously described based on an assumed fixed relationship between the last stored positions of the markers **720** relative to the unobstructed lines of sight. Further, in some embodiments, at every frame, the position of a marker **720** from camera **1** relative to its position from camera **2** would be stored; then if camera **1** is obstructed, and until the line of sight is restored, this relative position is recalled from computer memory (for example in memory of a computer platform **3400**) and a reconstruction of the marker **720** from camera **1** would be inserted based on the recorded position of the marker from camera **2**. In some embodiments, the method could compensate for temporary obstructions of line of sight such as a person standing or walking in front of one camera unit.

(390) In certain embodiments, when a marker M1, M2, M3, M4 is lost but is successfully reconstructed in accordance with one or more aspect described herein, the marker that has been reconstructed can be rendered in a display device **3411**. In one example implementation, circles representing each marker can be rendered graphically, coloring the circles for markers M1, M2, M3, M4 that are successfully tracked in green, markers M1, M2, M3, M4 that are successfully reconstructed in blue, and markers M1, M2, M3, M4 that cannot be tracked or reconstructed in red. It should be appreciated that such warning for the agent can serve to indicate that conditions are not optimal for tracking and that it is prudent to make an effort for all four tracking markers to be made fully visible, for example, by repositioning the cameras or standing in a different position where the marker is not blocked. Other formats and/or indicia can be utilized to render a virtual marker and/or distinguish such marker from successfully tracked markers. In some embodiments, it is possible to extend the method described herein to situations relying on more than four markers. For example, in embodiments in which five markers are utilized on one rigid body, and one of the five markers is blocked, it is possible to reconstruct the blocked marker from the average of the four remaining triads or from another method for best fit of the 4 remaining markers on the stored last visible position of all 5 markers. In some embodiments, once reconstructed, the average position of the rigid body is calculated from the average of the 10 possible triads, {M1,M2,M3}, {M1,M2,M4}, {M1,M2,M5}, {M1,M3,M4}, {M1,M3,M5}, {M1,M4,M5}, {M2,M3,M4}, {M2,M3,M5}, {M2,M4,M5}, and {M3,M4,M5} or from another method for best fit of 5 markers from position A to position B.

(391) As discussed above, in some embodiments, the end-effectuator **30** can be operatively coupled to the surgical instrument **35**. This operative coupling can be accomplished in a wide variety of manners using a wide variety of structures. In some embodiments, a bayonet mount **5000** is used to removably couple the surgical instrument **35** to the end-effectuator **30** as shown in FIG. **48**. For example, FIG. **48** shows a perspective view illustrating a bayonet mount **5000** used to removably couple the surgical instrument **35** to the end-effectuator **30**. In some embodiments, the bayonet mount **5000** securely holds the surgical instrument **35** in place with respect to the end-effectuator **30**, enabling repeatable and predictable location of operational edges or tips of the surgical instrument **35**.

(392) In some embodiments, the bayonet mount **5000** can include ramps **5010** which allow identification of the surgical instrument **35** and ensure compatible connections as well. In some embodiments, the ramps **5010** can be sized consistently or differently around a circumference of the bayonet mount **5000** coupled to or integral with the surgical instrument **35**. In some embodiments, the differently sized ramps **5010** can engage complementary slots **5020** coupled to or integral with the end-effectuator **30** as shown in FIG. **48**.

(393) In some embodiments, different surgical instruments **35** can include different ramps **5010** and complementary slots **5020** to uniquely identify the particular surgical instrument **35** being installed. Additionally, in some embodiments, the different ramps **5010** and slots **5020** configurations can help ensure that only the correct surgical instruments **35** are installed for a particular procedure.

(394) In some embodiments, conventional axial projections (such as those shown in U.S. Pat. No. 6,949,189 which is incorporated herein as needed to show details of the interface) can be mounted to or adjacent the ramps **5010** in order to provide automatic identification of the surgical instruments **35**. In some embodiments, other additional structures can be mounted to or adjacent the ramps **5010** in order to provide automatic identification of the surgical instruments **35**. In some embodiments, the axial projections can contact microswitches or a wide variety of other conventional proximity sensors in order to communicate the identity of the particular surgical instrument **35** to the computing device **3401** or other desired user interface. Alternatively, in some other embodiments of the invention, the identity of the particular surgical instrument **35** can be entered manually into the computing device **3401** or other desired user interface.

(395) In some embodiments, instead of a targeting fixture **690** consisting of a combination of radio-opaque **730** and active markers **720**, it is possible to register the targeting fixture **690** through an intermediate calibration. For example, in some embodiments, an example of such a calibration method could include attaching a temporary

rigid plate **780** that contains radio-opaque markers **730**, open mounts **785** (such as snaps, magnets, Velcro, or other features) to which active markers **720** can later be attached in a known position. For example, see FIGS. **49A-F** which depict illustrations of targeting fixtures **690** coupled to a spine portion **19** of a patient **18** in accordance with one embodiment of the invention). The method can then include scanning the subject (using for example CT, MRI, etc.), followed by attaching a percutaneous tracker **795** such as those described earlier or other array of 3 or more active markers **720** rigidly affixed to the anatomy **19** as for example in FIG. **49B**, and then attaching active markers **720** to the temporary plate **780** in the known positions dictated by the snaps, magnets, velcro, etc., as illustrated in FIG. **49C**. In some embodiments, a further step can include activating the cameras **8200** to read the position of the tracker **795** rigidly affixed to the anatomy **19** at the same time as the active markers **720** on the temporary plate **780**. This step establishes the position of the active markers **720** on the temporary plate **780** relative to the radio-opaque markers **730** on the temporary plate **780** as well as the positions of the active markers **720** on the tracker **795** relative to the active markers **720** on the temporary plate **780**, and therefore establishes the position of the anatomy relative to the active markers **720** on the tracker **795**. The temporary plate **780** can be removed (as illustrated in FIG. **49D**), including the active markers **720** and radio-opaque markers **730**. These markers are no longer needed because registration has been performed relative to the active markers on the rigidly affixed tracker **795**.

(396) In some alternative embodiments, variants of the order of the above described steps may also be used. For instance, the active markers **720** could already be attached at the time of the scan. This method has advantage that the radio-opaque markers **730** can be positioned close to the anatomy of interest without concern about how they are attached to the tracker **795** with active markers **720**. However, it has the disadvantage that an extra step is required in the registration process. In some embodiments, a variant of this method can also be used for improved accuracy in which two trackers of active markers **720** are attached above and below the region of interest. For example, a tracker rostral to the region of interest (shown as **795**) could be a spinous process **2310** clamp in the upper lumbar spine and a tracker caudal to the region of interest (shown as **800**) could be a rigid array of active markers **720** screwed into the sacrum (see for example FIG. **49E**). After calibration, the temporary plate **780** is removed and the area between the two trackers (within the region **805**) is registered (see for example FIG. **49F**).

(397) Some embodiments can include methods for transferring registration. For example, a registration performed to establish the transformations in order to transpose from a medical image coordinate system (such as the CT-scanned spine) to the coordinate system of the cameras, can later be transferred to a different reference. In the example described in the above related to FIGS. **49A-F**, a temporary fixture **780** with radio-opaque markers **730** and active markers **720** is placed on the patient **18** and registered. Then, a different fixture **795** is attached to the patient with active markers **720** only. Then the cameras (for example, camera **8200**) are activated, and the active markers **720** on the temporary plate **780** are viewed simultaneously with the active markers **720** on the new tracking fixture **795**. The necessary transformations to get from the temporary markers (those on the temporary plate **780**) to the new markers (i.e. the markers on fixture **795**) are established, after which the temporary plate **780** can be removed. In other words, the registration was transferred to a new reference (fixture **795**). In some embodiments, it should be possible to repeat this transferal any number of times. Importantly, in some embodiments, one registration can also be duplicated and transferred to multiple references. In some embodiments, transferal of registration to multiple references would provide a means for tracking relative motion of two rigid bodies. For example, a temporary targeting fixture may be used to register the anatomy to the cameras **8200**. Then, two new targeting fixtures may be placed on separate bones that are both included in the medical image used for registration. If the registration from the temporary targeting fixture is transferred to both of the new targeting fixtures, both of these bones may be tracked simultaneously, and the position of the robot end effector **30** or any other tracked probe or tool relative to both bones can be visualized. If one bone then moves relative to the other, the end effector's position would be located differently relative to the two trackers and the two medical images (for example, see FIGS. **50B-50D** showing the two trackers **796** and **797** positioned on a portion of spine **19**).

(398) In some embodiments, after registration is transferred to both trackers **796**, **797**, the robot end effector **30** may be perceived by both trackers **796**, **797** to be positioned as shown. In some embodiments, it is possible that one of the bones to which a tracker is mounted moves relative to the other, as shown in exaggerated fashion in FIG. **50C**. In some embodiments, if the end effector **30** is considered fixed, the perception by the tracking system **3417** and software would be that the spine **19** was positioned in two possible relative locations, depending on which tracker is followed (see for example, the representation in FIG. **50D**). Therefore, in some embodiments, by overlaying representations of both medical images, it becomes possible to visualize the relative movement of the bones on which the trackers **796**, **797** are attached. For example, instead of displaying the re-sliced medical image on the screen and showing the position of the robot end effector **30** relative to that image, two re-sliced medical images could be overlapped (each allowing some transparency) and simultaneously displayed, showing

one position where the robot end effectuator currently is positioned relative to both images (see FIGS. 50E-50F). However, the duplication of bones would make the representation cluttered, and therefore in some embodiments, it can be possible to automatically or manually segment the medical image such that only bones that do not move relative to a particular tracker **796**, **797** are represented on the image (shown in FIG. 50E with the bones **19a** highlighted as in relation to bone regions meaningful to tracker **796** with regions **19b** faded, and FIG. 50F with the bones **19a** highlighted in relation to bone regions meaningful to tracker **797**, with regions **19b** faded). Segmenting would mean hiding, fading, or cropping out the portion of the 3D medical image volume that the user does not want to see (represented as the faded regions **19b** in FIGS. 50E and 50F).

(399) In some embodiments, segmentation could involve identifying bordering walls on the 3D image volume or bordering curves on 2D slices comprising the medical image. In some embodiments, by segmenting simple six-sided volumes, enough separation of critical elements could be visualized for the task. In some embodiments, bones on the slice from a CT scans depicted in FIGS. 50G and 50H are shown with segmentation into region **20a**, corresponding to the bone regions **19a** referred to in FIGS. 50E and 50F, and **20b**, corresponding to region **19b**. In some embodiments, the regions **20a** and **20b** can be represented in different shades of color (for example, blue for **20a** and yellow for **20b**). Furthermore, as shown, the segmentation as displayed is depicted to proceed in and out of the page to include the entire CT volume. Moreover, although it goes right through the disc space, this segmentation cuts through one spinous process **2310** in the image in FIG. 50G, and does not follow the facet joint articulations to segment independently moving bones, as shown in a different slice represented in FIG. 50H. However, the re-sliced images of these overlapped volumes should still be useful when placing, for example, pedicle screws since the pedicles are properly segmented in the images.

(400) An example of transferal of registration to multiple trackers includes conventional pedicle screw placement followed by compression or distraction of the vertebrae. For example, if pedicle screws are being placed at lumbar vertebrae L4 and L5, a tracker could be placed on L3 and registered. In some embodiments, conventional pedicle screws could then be placed at L4 and L5, with extensions coming off of each screw head remaining after placement. In some embodiments, two new trackers (for example, trackers substantially similar to **796**, **797**) could then be attached to the extensions on the screw heads, one at L4 and one at L5. Then, the registration could be transferred to both of these new trackers and a tracker at L3 could be removed or thereafter ignored. In some embodiments, if the medical image is segmented so that L4 and rostral anatomy is shown relative to the tracker on L4 (while L5 and caudal anatomy is shown relative to the tracker on L5), then it can be possible to see how the L4 and L5 vertebrae move relative to one another, as compressive or distractive forces are applied across that joint. In some embodiments, such compression or distraction might be applied by the surgeon when preparing the disc space for an inter-body spacer, or inserting the spacer, or when compressing the vertebrae together using a surgical tool after the inter-body spacer is in place, and before locking the pedicle screw interconnecting rod.

(401) In some embodiments, if there is snaking of the spine, for example, when conventional screws are driven in place or the surgeon applies a focal force on one portion of the spine, the two marker trees will move (illustrated as **795a** for tracker **795** and **800a** for tracker **800**) by different amounts and to different orientations (illustrated in FIG. 50A). The altered orientations and positions of the trackers **795**, **800** can be used to calculate how the spine has snaked and adjust the perceived position of the robot **15** or probe to compensate. In some embodiments, because there are multiple degrees of freedom of the vertebrae, knowledge of how the two trackers' orientations shift does not allow a single unique solution. However, it can be assumed that the bending is symmetrical among all the vertebrae to calculate the new position, and even if this assumption is not perfect, it should provide a reasonably accurate solution. In some embodiments, experiments tracking how cadaveric spines respond to focal forces can be used to collect data that will help to predict how the two ends of the lumbar spine would respond during particular types of external loading.

(402) In some embodiments, it is possible to use the same surgical robot **15** already described for navigation with 3D imaging in a different setting where only 2 fluoroscopic views are obtained. In this instance, the surgical robot **15** will be able to accurately move to a desired position that is pre-planned on these two fluoroscopic views. Since the two fluoroscopic views can represent views to which the surgeon or radiologist is already accustomed, planning trajectories on these views should be straightforward. In obtaining the fluoroscopic views, a method is needed to establish the position of the coordinate system of the anatomy relative to the robot's **15** coordinate system. In some embodiments, a way to fulfill this registration is to obtain the fluoroscopic views while a targeting fixture **690** that includes features that are identifiable on the fluoroscopic images is attached to the patient **18**. For example, FIG. 51 shows an example of a fixture for use with fluoroscopic views in accordance with one embodiment of the invention. In some embodiments, the targeting fixture **690** as shown can include features that will appear on 2 fluoroscopic views and active markers **720** for real-time tracking. This targeting fixture **690** has properties that will aid in the ability to set up the coordinate system of the anatomy from the two fluoroscopic images. For example, in some embodiments, the posts **75** as shown are symmetrically spaced around

the frame **700** so that posts **75** and/or their embedded markers **730** would overlay on an x-ray image. That is, if there is no parallax, two posts **75** in an aligned position would appear as a single line segment instead of two, or two posts **75**, each with two embedded radio-opaque markers **730**, would appear as two dots instead of four dots on an x-ray image. These features allow and facilitate the patient **18** or fluoroscopy machine's position to be adjusted until such overlapping is achieved. Similarly, from top or bottom view, the posts **75** and/or their embedded markers **730** would overlap, with a single post appearing as a dot instead of a line segment or two embedded markers **730** in one post appearing as one dot instead of two once the fluoroscopy machine and patient **18** are adjusted to be aligned as desired. In some embodiments, the posts **75** may be designed to be temporarily inserted (i.e., they are present during the scan but are later unplugged from the frame during the procedure so they are not in the way of the user). In some embodiments, the active markers **720** are necessary for later tracking but do not necessarily need to be present during the scan as long as they can be attached with precision to a known position on the frame **700** relative to the radio-opaque markers **730**. For example, in some embodiments, conventional sockets on the fixture **690** could later allow the active markers **720** to be snapped in to a location dictated by the manufacturing of the frame **700** or calibrated using a digitizing probe. Furthermore, note that the goal is not necessarily to get perfect lateral and anteroposterior anatomical views of the spine or other anatomy. The goal is to get alignment of the fixture **700** on the x-ray view. Although it may be beneficial in understanding what it being visualized to also achieve alignment with the anatomical planes, it is unnecessary for registration. An example of how the targeting fixture **690** might appear on anteroposterior or "A-P" and lateral x-rays when affixed to the patient's back but not yet aligned with the x-ray projection is shown in FIGS. 52A-52B.

(403) In some embodiments, after adjusting the position of the patient **18** and fluoroscopy unit, an overlay with good certainty may be obtained for images with radio-opaque markers **730**. FIGS. 53A-B for example illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine in accordance with one embodiment of the invention. As shown, the fluoroscopic images do not need the frame **700** to be positioned exactly aligned with the anatomy or rotated to be vertical and horizontal. In some embodiments, the fluoroscopically obtained images are not required to have the correct aspect ratio such that the image properly represents a calibrated coordinate system. In some embodiments, it is possible to rescale the image to adjust the aspect ratio using known distances between posts **75** or between markers **730**, x-ray visible lengths of posts **75**, or assuming the image should be perfectly circular or square. These distances are known in advance of obtaining the images by the manufacturing process, or by calibration using a digitizing probe or other means. In some embodiments, provided parallax is considered, the ratio of known inter-marker distances can be compared to the ratio of inter-marker distances measured on planar images and used to scale the planar image to achieve the correct aspect ratio. In some embodiments, it is not necessary to rescale the image, but it may help the user to better visualize the image and anatomy when it is displayed in the appropriate aspect ratio. In some embodiments, the comparison can also be used to determine the number of pixels per mm on the image for use in determining relative position of radio-opaque markers **730** and planned trajectory tip and tail. In some embodiments, rescaling facilitates equations for mapping between 2D and 3D space because the pixels per mm in the x and y direction are the same value.

(404) In some embodiments, after obtaining two images, the two images can be used to construct a 3D Cartesian coordinate system because they represent images of the same thing (the fixture) from two orthogonal views. For example, the A-P image could be used to represent the X-Z plane, and the lateral image could be used to represent the Y-Z plane. Radio-opaque markers **730** on the A-P image have known x-axis and z-axis coordinates (as recorded from the manufacturing process or by calibration using a digitizing probe or other means), and the same radio-opaque markers **730** have known y-axis and z-axis coordinates on the lateral image. Therefore, in some embodiments, the x-axis, y-axis, and z-axis coordinates of the markers **730** can be found on the two images, and the positions of the anatomy and planned trajectories relative to these reference points can be related to these reference positions. In some embodiments, the mapping of a point from 3D space to the 2D image and vice versa can be performed knowing the constant mm per pixel, C, on coronal or sagittal images, and multiplying or dividing points by these constants if the center of the image and coordinate system have been shifted to overlap.

(405) FIGS. 54A-B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine in accordance with one embodiment of the invention. As shown, FIGS. 54A-B include overlaid computer-generated graphical images showing the planned trajectory (red **6001**) and the current actual position of the robot **15** end-effector **30** (light blue **6003**). The red circle in **6001** is provided for the user to identify the tail of the planned trajectory (as opposed to the tip). In other embodiments, the line segment could have different colored ends or different shapes on each end (pointed vs. blunt) for distinguishing tip from tail.

(406) In some embodiments, assuming the A-P x-ray represents the X-Z plane and the lateral x-ray represents the Y-Z plane, the algorithm for planning a trajectory and relating this planned trajectory to the robot **15** coordinate

system can include the following steps; 1). Draw a line on the A-P and lateral x-ray views representing where the desired trajectory should be positioned (see for example FIGS. 54A-54B). In some embodiments, the next step can include; 2). from the A-P view, find the X and Z coordinates of the reference opaque markers and of the tip and tail of the desired trajectory, and 3). from the lateral view, find the Y and Z coordinates of the reference opaque markers and of the tip and tail of the desired trajectory, and 4). based on the known coordinates of the active markers relative to the opaque markers, transform the X,Y,Z coordinates of the tip/tail into the coordinate system of the active markers. In some embodiments, the method can include store the locations of tip and tail in this coordinate system in computer **100** memory for later retrieval. In some embodiments, the next steps of the method can include; 5). at any frame in real time, retrieve the active marker **720** locations in the coordinate system of the cameras **8200**, and 6). based on the stored coordinates of the tip and tail relative to the active markers **720** and the current location of the active markers **720** in the coordinate system of the cameras **8200**, calculate the current location of the desired tip and tail in the coordinate system of the cameras **8200**. In some embodiments, the next steps of the method can include; 7). transform the active marker **720** locations and the trajectory tip/tail locations into the coordinate system of the robot **15** using methods described before in which markers on the robot **15** are utilized as references, and 8). send the robot **15** to the desired tip/tail locations using methods described previously.

(407) In some embodiments, while the robot **15** moves to position itself in the desired orientation and position, it is possible to overlay a graphical representation of the current location of the robot **15** on the fluoroscopic images by a method that can include; 1). retrieve current location of the robot **15** guide tube **50** in the coordinate system of the cameras **8200** based on active markers **720** attached to the robot, and 2). transform the guide tip and tail to the coordinate system of the medical images based on the locations of active markers on the targeting fixture **690**, and 3). represent the current positions of tip/tail of the guide tube **50** on the A-P image by a line segment (or other suitable graphical representation) connecting the X,Z coordinates of the tip to the X,Z coordinates of the tail (see for example FIGS. 54A-B), and 4). represent the current positions of tip/tail of the guide tube **50** on the lateral image by a line segment (or other suitable graphical representation) connecting the Y,Z coordinates of the tip to the Y,Z coordinates of the tail.

(408) In some embodiments, in constructing the Cartesian coordinate system based on the two images, it is important to consider directionality. That is, in some embodiments, an x-ray image of the X-Z plane could show positive X to the right and negative X to the left or vice versa. In some embodiments, it could show positive Z upward and negative Z downward or vice versa. In some embodiments, an x-ray image of the Y-Z plane could show positive Y to the right and negative Y to the left or vice versa. In some embodiments, it could show positive Z upward and negative Z downward or vice versa. In some embodiments, if an incorrect assumption is made about the directionality of one of the axes, it would mean that the constructed 3D coordinate system has one or more of its axes pointing in the wrong direction. In some embodiments, this may send the robot **15** to an incorrect position. In some embodiments, one way of ensuring the correct directionality is to query to the user requesting verification of directionality on the images and/or allowing them to flip (mirror) the images on the display **29**, **150**, **3401**. In some embodiments, another way of ensuring the correct directionality is to design the targeting fixture **690** so that the radio-opaque markers **730** are spaced asymmetrically. In some other embodiments, another way of ensuring the correct directionality is to design the targeting fixture **690** with additional radio-opaque features that unambiguously identify top, bottom, left, right, front and rear on images. For example, FIGS. 55A-55B illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine. As shown, the targeting fixture **690** illustrated in FIGS. 55A-B includes a feature **755** designed to substantially eliminate ambiguity about directionality in accordance with one embodiment of the invention. As shown, the feature **755** could reveal a “L”, “T”, or other symbol on the x-ray when the view is correct so as to substantially eliminate directional ambiguity. Furthermore, the “T” feature could be drawn in script (e.g., T) or other asymmetric letter or symbol used so that if an inverted or mirrored x-ray image is presented, the inverted nature is clear and can be compensated.

(409) In some embodiments, the algorithm described here provides the user with two perpendicular x-ray views from which to plan a trajectory, and provides a visual feedback of the current location of a probe. Typically, these two views might be lateral and anteroposterior (A-P) views. In some embodiments, it might also be desirable for the user to see a third plane (for example, an axial plane). Based on knowledge of the anatomy and landmarks visible on the x-rays, in some embodiments, it is possible to create a rough “cartoon” showing an axial view. In some embodiments, the cartoon may help the user understand the approximate current location of the robot **15** or probe. FIG. 56 shows how such a cartoon can be generated from the x-rays. Note that the cartoon will be imperfect with respect to details such as the curvature of the vertebral body, but key landmarks such as the pedicle boundaries should be reasonably well defined. Such an approach would be based on how a typical vertebra is shaped. For example, FIG. 56 illustrates an axial view of a spine showing how a cartoonish axial approximation of

the spine **5601** can be constructed based on a lateral x-ray **5602** and an anteroposterior x-ray **5603** in accordance with one embodiment of the invention. As shown, locations where key landmarks on the adjacent x-ray views intersect the cartoon can be identified with horizontal or vertical lines overlapping the cartoon and x-rays. The user positions these lines using a software interface or software automatically recognizes these features on the x-rays so that the lines intersect the key landmarks, such as a line just tangent to the vertebral body left border **5604**, vertebral body right border **5605**, vertebral body anterior wall **5606**, vertebral body posterior wall **5607**, posterior spinal canal **5608**, left inner pedicle border **5609**, left outer pedicle border **5610**, tip of spinous process **5611**, etc. After using the software to move these lines so that they intersect correct locations on the x-rays, the software can then stretch and morph the cartoon as needed to fit these anatomical limits. A view on the computer display of the axial plane showing this cartoon and the planned trajectory and current position of robot or probe can be generated by the software to provide additional visual feedback for the user.

(410) In some embodiments, in order to achieve well aligned x-rays like those shown in FIGS. **53A-B**, one possible method is trial and error. For example, the user can try to get the x-ray machine aligned to the targeting fixture **690**, attempt to assess alignment by eye, then shoot an x-ray and see how it looks. In some embodiments, if dots are misaligned (for example as shown in FIGS. **52A-52B**), adjustments would be made and a new x-ray image can be prepared. This method can be effective but can be dependent on the skill of the operator in assessing alignment and making corrections, and therefore can result in x-ray exposure to the patient **18** and staff. In some embodiments, it is possible to create a tool to assist in the alignment of the targeting fixture **690**. In some embodiments, the tool could be a conventional laser that can be attached to the emitter or collector panel of the x-ray machine, capable of passing a laser beam parallel to the direction that the x-rays will travel. In some embodiments, the laser could be attached temporarily (using a conventional magnet or adhesive) or permanently, connected to an arm extending from the x-ray machine and oriented in the correct direction, enabling the directed beam to shine down toward the fixture **690**. In some embodiments, if the fixture **690** has a geometric, electronic, or other features capable of visual or other feedback to the user regarding the vector direction of this laser light, it would allow alignment of the x-ray arm without taking any x-rays. An example of such a feature is shown in FIG. **57A** and FIG. **57B**. FIGS. **57A-B** illustrates examples of targeting fixtures **690** that facilitate desired alignment of the targeting fixture **690** relative to the x-ray image plane in accordance with one embodiment of the invention. As shown, some embodiments include a feature **765** temporarily added to the targeting fixture **690**. In some embodiments, feature **765** facilitates desired alignment of the targeting fixture **690** relative to the x-ray image plane from an AP view when a laser (attached to the face of the x-ray emitter or collector) is directed through the opening **766** and toward the crosshairs **767** at the base. In some embodiments, if the laser light does not strike the crosshairs **767** dead center, further adjustment of the x-ray unit's orientation is needed. The temporarily added feature **765** that facilitates desired alignment of the targeting fixture **690** relative to the x-ray image plane from a lateral view when a laser (attached to the face of the x-ray emitter or collector) is directed through the opening and toward the crosshairs at the opposite face. In some embodiments, if the laser light does not strike the crosshairs dead center, further adjustment of the x-ray unit's orientation is needed.

(411) In some embodiments, this method for aligning the radio-opaque markers **730** would have the advantage over trial-and-error methods that are affected by parallax effects, and as described below, do not confound the ability to align markers as needed. For example, with parallax, it may not be clear to the user when good alignment of the markers **730** is achieved, depending on how symmetrically spaced the markers **730** are about the center of the image.

(412) With parallax error, the x-rays may not pass through the subject in a straight line and instead travel from emitter to receiver in a conical pattern. This conical path can produce an image where the details of anatomy on the 2D x-ray that are closer to the emitter of the x-rays will appear farther apart laterally than details of the anatomy that are closer to the receiver plate. In the case of x-ray images in FIGS. **53A-B**, instead of the radio-opaque markers **730** appearing overlaid, they may appear as shown in FIGS. **58A-B**. For example, FIGS. **58A-B** illustrates expected images on anteroposterior and lateral x-rays of the spine with a well aligned fluoroscopy (x-ray) machine when parallax is present in accordance with one embodiment of the invention. As shown, parallax affects spacing symmetrically about the x,y center of the image, with locations of markers **730** closer to the receiver plate of the x-ray unit appearing closer to the center of the image.

(413) Further, in the description, two terms used are “near plane” and “far plane”—these terms refer to markers in the 2D views that appear farther apart or closer together because of parallax. The reason markers are farther apart or closer together is because of their proximity to the emitter or collector of the x-ray machine, with markers nearer the emitter appearing farther apart and markers nearer the collector closer together. However, rather than referencing distance from emitter and collector, “near plane” refers to markers that appear magnified (nearer to the eye) and “far plane” refers to markers that appear more distant.

(414) Parallax will affect the image symmetrically about the center of the image. For example, in some

embodiments, two markers **730** (one in near plane and one in far plane) that are in the same projected position, and are at the center of the image, may appear to be exactly on top of each other, whereas markers **730** in the near plane and far plane that are in the same projected position, but are close to the edge of the image may appear separated by a substantial distance.

(415) FIG. **59A** illustrates two parallel plates with identically positioned radio-opaque markers **730** in accordance with one embodiment of the invention. As shown, this illustrates possible marker **730** separation on an x-ray from markers **730** on two plates that are in the same projected line of sight. FIG. **59B** illustrates resulting expected x-ray demonstrating how marker overlay is affected due to parallax using the two parallel plates as shown in FIG. **59A** in accordance with one embodiment of the invention. By comparing the two parallel plates with identically positioned radio-opaque markers **730** shown in FIG. **59A**, with the resulting expected x-ray in FIG. **59B** demonstrates how marker overlay is affected due to parallax. In some embodiments, an algorithm can be implemented to account for this parallax effect. By doing so, the graphical image indicating the position of the probe or robot **15** can be adjusted to more accurately account for the perceived shift caused by parallax.

(416) In some embodiments, the algorithm requires information to be gathered on the near and far plane positions of the markers **730** on the image. That is, the user can indicate, using software or an automatic scan of the image, the spacing between markers **730**, as shown in FIG. **60**, which shows a representation of the rendering of a computer screen with an x-ray image that is affected by parallax overlaid by graphical markers **732**, **734** over the radio-opaque markers **730** on two plates that have the same geometry in accordance with one embodiment of the invention. In some embodiments, the spacing between near plane and far plane markers **730** is known because of earlier calibration of the plates **700** in which the markers **730** are embedded, and the horizontal and vertical positions of the markers **730** are detectable relative to the center of the image. Therefore, in some embodiments, the parallax shift of the markers **730** can be calculated and applied to the mapping of any calculated three dimensional points on to the two dimensional image, and application of any necessary positional shift. For example, in some embodiments, it might be of interest to display a line segment on the two dimensional image representing how the shaft of a probe or robot **15** guide tube **50** (that is being tracked using optical tracking) would appear following x-ray imaging. In some embodiments, the x-axis, y-axis, and z-axis location of each end of the line segment (which has been calculated from optical tracking data) can be shifted based on the known parallax. Further, in some embodiments, a new line segment can be displayed that better represents how this projected object should appear on the 2D x-ray image.

(417) In some embodiments, a method of implementing this system of two orthogonal fluoroscopy images to control a robot **15** can involve combining a robot **15** and fluoroscopy unit into a single interconnected device. There could be some advantages of this combination. For example, a conventional rotating turntable mechanism could be incorporated that could swing the fluoro arm into place, while at the same time swinging the robot arm **23** out of place (since the robot **15** would typically not be in the surgical field **17** at the same time as the fluoro arm). Furthermore, in some embodiments, the size of the robot arm **23** could be reduced compared to the stand-alone robot **15** because the fluoro arm's mass would serve as a counter-balance weight to help stabilize the robot arm **23**. Moreover, in some embodiments, with integration, the fluoroscopy unit can more quickly transfer the image to the computer **100** and overlay with a graphical plot, for instance, as line segments starting at the center of the image and extending radially (similar to pie slices) around the image to facilitate appropriate marker **730** overlay. In some embodiments, overlaid near and far plane markers **730** should always fall on the same ray if the plates **690** with embedded markers **730** on the subject are aligned substantially parallel (see for example FIG. **61** which shows a graphical overlay for the x-ray image screen intended to help the user physically line up the x-ray machine). In some embodiments, the graphical overlay for the x-ray image screen can help the user physically line up the x-ray machine to avoid parallax. With parallax, any pair of corresponding markers on the 2 plates should lie on the same radial line, although the one in the far plane will lie closer to the middle of the image. In some embodiments, this overlay could be a physical object such as transparent film, or a computer-generated graphical image. In some embodiments, lines are spaced radially by 10 degrees, but actual spacing (frequency of lines) and regions in which lines are drawn could be user selectable.

(418) Some embodiments can include mapping a 3D anatomical coordinate system on to two 2D orthogonal views (and vice versa) while considering parallax. For example, in some embodiments, a rigid frame is mounted to the patient and two perpendicular x-rays are taken to create a 3D coordinate system. To define this 3D coordinate system, a method is needed to map points from the 2D views (each with parallax) to the 3D volume and vice versa. The 3D coordinate system has coordinates x, y, z while the two 2D coordinate systems have coordinates x.sub.AP, z.sub.AP and x.sub.Lat, z.sub.Lat ("AP" for "anteroposterior" and "Lat" for "lateral" views).

(419) In some embodiments, it can be assumed that the x-ray path from emitter to receiver is conical, and therefore linear interpolation/extrapolation can be used to adjust the positions of represented points. In some embodiments, software can calculate the distance of each landmark from the center of the image (indicated by

(dashed or dotted arrows). These distances, together with the known distance between near plane and far plane plates, can provide the necessary information to account for the parallax shift when mapping graphical objects whose positions are known in 3D back on to this 2D image.

(420) Some embodiments can include solving to map x, y, z onto $x_{\text{sub.AP}}, z_{\text{sub.AP}}$ and $x_{\text{sub.Lat}}, z_{\text{sub.Lat}}$. For example, consider two intermediate 2D AP and lateral views represented as follows:

$$(421) x_{\text{ta}} = (x - x_{\text{oa}})s_{\text{AP}} z_{\text{ta}} = (z - z_{\text{oa}})s_{\text{AP}} y_{\text{tl}} = (y - y_{\text{ol}})s_{\text{Lat}} z_{\text{tl}} = (z - z_{\text{ol}})s_{\text{Lat}}$$

(422) Where $x_{\text{sub.ta}}$ and $z_{\text{sub.ta}}$ can be called temporary scaled values of x and z in the AP plane, $y_{\text{sub.tl}}$ and $z_{\text{sub.tl}}$ are temporary scaled values of y and z in the Lat plane, $s_{\text{sub.AP}}$ is the scaling factor in the AP plane, determined from the known near plane.sup.1 marker spacing. $s_{\text{sub.Lat}}$ is the scaling factor in the Lat plane, determined from the known near plane marker spacing of the lateral markers, and $x_{\text{sub.oe}}, z_{\text{sub.oe}}, y_{\text{sub.ol}}$, and $z_{\text{sub.ol}}$ are offsets in AP and Lat planes that position the markers such that they are as they appear centered about the image determined from registered positions of the markers on the images. In other words, $(x_{\text{sub.ta}}, z_{\text{sub.ta}}) = (0,0)$ represents the center of the AP image and $(y_{\text{sub.tl}}, z_{\text{sub.tl}}) = (0,0)$ represents the center of the lateral image. These planar values would be enough to display a 2D representation if no parallax were present or near plane markers were only being displayed.

(423) In some embodiments, to find $x_{\text{sub.oe}}, z_{\text{sub.oe}}, y_{\text{sub.ol}}$, and $z_{\text{sub.ol}}$ consider pairs of points on the x-rays, because the ratio of distance from center on the x-ray is the same as the ratio of distance from center on the temporary scaled values. For example:

$$(424) \frac{x_{\text{AP1}}}{x_{\text{AP2}}} = \frac{x_{\text{ta1}}}{x_{\text{ta2}}} = \frac{(x_1 - x_{\text{oa}})s_{\text{AP}}}{(x_2 - x_{\text{oa}})s_{\text{AP}}} \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} \right) (x_2 - x_{\text{oa}}) = x_1 - x_{\text{oa}} x_2 \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} \right) - x_1 = x_{\text{oa}} \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} \right) - x_{\text{oa}} x_{\text{oa}} \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} - 1 \right) = x_2 \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} \right) - x_1 x_{\text{oa}} = \frac{x_2 \left(\frac{x_{\text{AP1}}}{x_{\text{AP2}}} \right) - x_1}{\frac{x_{\text{AP1}}}{x_{\text{AP2}}} - 1}$$

(425) In some embodiments, it can be seen from this equation that it is important to stay away from points where $x_{\text{sub.AP1}} \sim x_{\text{sub.AP2}}$ because it would result in a divide by zero error. Similar equations can be written for $z_{\text{sub.oe}}, y_{\text{sub.ol}}$, and $z_{\text{sub.ol}}$ as follows:

$$(426) z_{\text{oa}} = \frac{z_2 \left(\frac{z_{\text{AP1}}}{z_{\text{AP2}}} \right) - z_1}{\frac{z_{\text{AP1}}}{z_{\text{AP2}}} - 1} y_{\text{ol}} = \frac{y_2 \left(\frac{y_{\text{Lat1}}}{y_{\text{Lat2}}} \right) - y_1}{\frac{y_{\text{Lat1}}}{y_{\text{Lat2}}} - 1} z_{\text{ol}} = \frac{z_2 \left(\frac{z_{\text{Lat1}}}{z_{\text{Lat2}}} \right) - z_1}{\frac{z_{\text{Lat1}}}{z_{\text{Lat2}}} - 1}$$

(427) This mapping to temporary scaled values gets the near plane markers mapped correctly, but adjustment is needed to account for any position other than near plane as follows:

$$(428) 0x_{\text{AP}} = x_{\text{ta}} k_a(y) z_{\text{AP}} = z_{\text{ta}} k_a(y) y_{\text{Lat}} = y_{\text{tl}} k_l(x) z_{\text{Lat}} = z_{\text{tl}} k_l(x)$$

(429) As specified, $k_{\text{sub.a}}$ is a function of y and $k_{\text{sub.l}}$ is a function of x . For $k_{\text{sub.a}}$, this function is a linear interpolation function, in which if y is the y position of the near plane ($y_{\text{sub.n}}$), then $k_{\text{sub.a}}=1$ and if y is the y position of the far plane ($y_{\text{sub.f}}$), then $k_{\text{sub.a}}$ is the ratio of far plane spacing to near plane spacing, $r_{\text{sub.a}}$. For $k_{\text{sub.l}}$, this function is a linear interpolation function, in which if x is the x position of the near plane ($x_{\text{sub.n}}$), then $k_{\text{sub.l}}=1$ and if x is the x position of the far plane ($x_{\text{sub.f}}$), then $k_{\text{sub.l}}$ is the ratio of far plane spacing to near plane spacing, $r_{\text{sub.l}}$. Note that $y_{\text{sub.n}}, y_{\text{sub.f}}, x_{\text{sub.l}}$, and $x_{\text{sub.f}}$ are in a coordinate system with the origin at the center of the image.

$$(430) k_a = 1 - \left(\frac{y_{\text{tl}} - y_{\text{n}}}{y_{\text{f}} - y_{\text{n}}} \right) (1 - r_a) k_l = 1 - \left(\frac{x_{\text{ta}} - x_{\text{n}}}{x_{\text{f}} - x_{\text{n}}} \right) (1 - r_l)$$

(431) Combining equations,

$$(432) x_{\text{AP}} = x_{\text{ta}} \left[1 - \left(\frac{y_{\text{tl}} - y_{\text{n}}}{y_{\text{f}} - y_{\text{n}}} \right) (1 - r_a) \right] z_{\text{AP}} = z_{\text{ta}} \left[1 - \left(\frac{y_{\text{tl}} - y_{\text{n}}}{y_{\text{f}} - y_{\text{n}}} \right) (1 - r_a) \right] y_{\text{Lat}} = y_{\text{tl}} \left[1 - \left(\frac{x_{\text{ta}} - x_{\text{n}}}{x_{\text{f}} - x_{\text{n}}} \right) (1 - r_l) \right] z_{\text{Lat}} = z_{\text{tl}} \left[1 - \left(\frac{x_{\text{ta}} - x_{\text{n}}}{x_{\text{f}} - x_{\text{n}}} \right) (1 - r_l) \right]$$

(433) It should also be possible to map $x_{\text{sub.AP}}, z_{\text{sub.AP}}, y_{\text{sub.Lat}}$, and $z_{\text{sub.Lat}}$ onto x, y, z . Having 4 equations and 4 unknowns:

$$(434) x_{\text{AP}} = x_{\text{ta}} \left[1 - \left(\frac{y_{\text{tl}} - y_{\text{n}}}{y_{\text{f}} - y_{\text{n}}} \right) (1 - r_a) \right] 1 - \frac{x_{\text{AP}}}{x_{\text{ta}}} = \left(\frac{y_{\text{tl}} - y_{\text{n}}}{y_{\text{f}} - y_{\text{n}}} \right) (1 - r_a) \left(1 - \frac{x_{\text{AP}}}{x_{\text{ta}}} \right) \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) = y_{\text{tl}} - y_{\text{n}} y_{\text{tl}} = \left(1 - \frac{x_{\text{AP}}}{x_{\text{ta}}} \right) \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}}$$

(435) Then substitute into this equation:

$$(436) y_{\text{Lat}} = y_{\text{tl}} \left[1 - \left(\frac{x_{\text{ta}} - x_{\text{n}}}{x_{\text{f}} - x_{\text{n}}} \right) (1 - r_l) \right] y_{\text{Lat}} = \left[\left(1 - \frac{x_{\text{AP}}}{x_{\text{ta}}} \right) \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}} \right] \left[1 - \left(\frac{x_{\text{ta}} - x_{\text{n}}}{x_{\text{f}} - x_{\text{n}}} \right) (1 - r_l) \right]$$

(437) And solve for $x_{\text{sub.ta}}$:

$$(438) y_{\text{Lat}} = \left[\left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) - \frac{x_{\text{AP}}}{x_{\text{ta}}} \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}} \right] \left[1 - \left(\frac{1 - r_l}{x_{\text{f}} - x_{\text{n}}} \right) (x_{\text{ta}} - x_{\text{n}}) \right]$$

$$y_{\text{Lat}} = \left[\left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}} \right] - \left[\frac{x_{\text{AP}}}{x_{\text{ta}}} \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) \right] \left[\left[1 + x_{\text{n}} \left(\frac{1 - r_l}{x_{\text{f}} - x_{\text{n}}} \right) \right] - \left[x_{\text{ta}} \left(\frac{1 - r_l}{x_{\text{f}} - x_{\text{n}}} \right) \right] \right] y_{\text{Lat}} = \left[A - \frac{B}{x_{\text{ta}}} \right] \left[C - D x_{\text{ta}} \right]$$

$$y_{\text{Lat}} = AC - \frac{BC}{x_{\text{ta}}} - AD x_{\text{ta}} + BD AC + BD - y_{\text{Lat}} = \frac{BC}{x_{\text{ta}}} + AD x_{\text{ta}} (AD) x_{\text{ta}}^2 + (y_{\text{Lat}} - AC - BD) x_{\text{ta}} + (BC) = 0$$

(439) Quadratic Formula:

$$(440) x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} x_{\text{ta}} = \frac{-(y_{\text{Lat}} - AC - BD) \pm \sqrt{(y_{\text{Lat}} - AC - BD)^2 - 4(AD)(BC)}}{2(AD)}$$

(441) Where:

$$(442) A = \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}} B = x_{\text{AP}} \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) C = 1 + \left(\frac{1 - r_l}{x_{\text{f}} - x_{\text{n}}} \right) x_{\text{n}} D = \frac{1 - r_l}{x_{\text{f}} - x_{\text{n}}}$$

(443) Then plug into this equation to solve for $y_{\text{sub.tl}}$:

$$(444) y_{\text{tl}} = \left(1 - \frac{x_{\text{AP}}}{x_{\text{ta}}} \right) \left(\frac{y_{\text{f}} - y_{\text{n}}}{1 - r_a} \right) + y_{\text{n}}$$

(445) Then plug into this equation to solve for z.sub.ta:

$$(446) z_{tl} = z_{Lat} / [1 - (\frac{x_{ta} - x_n}{x_f - x_n})(1 - r_l)]$$

$$(447) 0z_{ta} = z_{AP} / [1 - (\frac{y_{tl} - y_n}{y_f - y_n})(1 - r_a)]$$

(448) Solve differently to give another option for z:

$$(449) y_{Lat} = y_{tl} [1 - (\frac{x_{ta} - x_n}{x_f - x_n})(1 - r_l)] (\frac{x_{ta} - x_n}{x_f - x_n})(1 - r_l) = 1 - \frac{y_{Lat}}{y_{tl}} (x_{ta} - x_n) (\frac{1 - r_l}{x_f - x_n}) = 1 - \frac{y_{Lat}}{y_{tl}}$$

$$x_{ta} (\frac{1 - r_l}{x_f - x_n}) - x_n (\frac{1 - r_l}{x_f - x_n}) = 1 - \frac{y_{Lat}}{y_{tl}} x_{ta} (\frac{1 - r_l}{x_f - x_n}) = 1 - \frac{y_{Lat}}{y_{tl}} + x_n (\frac{1 - r_l}{x_f - x_n}) x_{ta} = (1 - \frac{y_{Lat}}{y_{tl}}) (\frac{x_f - x_n}{1 - r_l}) + x_n$$

(450) Substitute into:

$$(451) x_{AP} = x_{ta} [1 - (\frac{y_{tl} - y_n}{y_f - y_n})(1 - r_a)] x_{AP} = [(1 - \frac{y_{Lat}}{y_{tl}}) (\frac{x_f - x_n}{1 - r_l}) + x_n] [1 - (\frac{y_{tl} - y_n}{y_f - y_n})(1 - r_a)]$$

(452) And solve for y.sub.tl:

$$(453) x_{AP} = [(\frac{x_f - x_n}{1 - r_l}) - \frac{y_{Lat}}{y_{tl}} (\frac{x_f - x_n}{1 - r_l}) + x_n] [1 - (\frac{y_{tl} - y_n}{y_f - y_n})(\frac{1 - r_a}{y_f - y_n})]$$

$$x_{AP} = [(\frac{x_f - x_n}{1 - r_l}) + x_n] - [\frac{y_{Lat}}{y_{tl}} (\frac{x_f - x_n}{1 - r_l})] [[1 + y_n (\frac{1 - r_a}{y_f - y_n})] - [y_{tl} (\frac{1 - r_a}{y_f - y_n})]] x_{AP} = [A - \frac{B}{y_{tl}}] [C - D y_{tl}]$$

$$x_{AP} = AC - \frac{BC}{y_{tl}} - AD y_{tl} + BD AC + BD - x_{AP} = \frac{BC}{y_{tl}} + AD y_{tl} (AD) y_{tl}^2 + (x_{AP} - AC - BD) y_{tl} + (BC) = 0$$

(454) Quadratic formula:

$$(455) x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} y_{tl} = \frac{-(x_{AP} - AC - BD) \pm \sqrt{(x_{AP} - AC - BD)^2 - 4(AD)(BC)}}{2(AD)}$$

(456) Where:

$$(457) A = (\frac{x_f - x_n}{1 - r_l}) + x_n B = y_{Lat} (\frac{x_f - x_n}{1 - r_l}) C = 1 + y_n (\frac{1 - r_a}{y_f - y_n}) D = \frac{1 - r_a}{y_f - y_n}$$

(458) From these equations, it is possible to go from a known x,y,z coordinate to the perceived x.sub.AP,z.sub.AP and x.sub.Lat,z.sub.Lat coordinates on the two views, or to go from known x.sub.AP,z.sub.AP and x.sub.Lat,z.sub.Lat coordinates on the two views to an x,y,z coordinate in the 3D coordinate system. It is therefore possible to plan a trajectory on the x.sub.AP,z.sub.AP and x.sub.Lat,z.sub.Lat views and determine what the tip and tail of this trajectory are, and it is also possible to display on the x.sub.AP,z.sub.AP and x.sub.Lat,z.sub.Lat views the current location of the robot's end effector.

(459) In some embodiments, additional measurement hardware (for example, conventional ultrasound, laser, optical tracking, or a physical extension like a tape measure) can be attached to the fluoro unit to measure distance to the attached plates, or other points on the anatomy to ensure that plates are parallel when fluoro images are obtained.

(460) In some embodiments, the identity of the surgical instrument **35** can be used by the control system for the computing device **3401** or other controller for the surgical robot system **1**. In some embodiments, the control system **3401** can automatically adjust axial insertion and/or forces and applied torques depending upon the identity of the surgical instrument **35**.

(461) In some embodiments, when performing a typical procedure for needle **7405**, **7410** or probe insertion (for biopsy, facet injection, tumor ablation, deep brain stimulation, etc.) a targeting fixture **690** is first attached by the surgeon or technician to the patient **18**. The targeting fixture **690** is either clamped to bone (open or percutaneously), adhered as a rigid object to the skin, or unrolled and adhered to the skin (for example using the flexible roll shown as **705** in FIG. **21A**). In some embodiments, the roll **705** could have a disposable drape incorporated. If a flexible roll **705** is used, reflective markers **720** will then be snapped into place in some embodiments.

(462) In some embodiments, once a targeting fixture **690** is attached, the patient **18** can receive an intraoperative 3D image (Iso-C, O-Arm, or intraoperative CT) with radio-opaque markers **730** included in the field of view along with the region of interest. In some embodiments, for best accuracy and resolution, a fine-slice image is preferred (CT slice spacing=1 mm or less). The 3D scan has to include the radio-opaque markers **730** and the anatomy of interest; not including both would disallow calibration to the robot **15**.

(463) In some embodiments, the 3D image series is transferred to (or acquired directly to) the computer **100** of the robot **15**. The 3D image has to be calibrated to the robot's position in space using the locations on the 3D image of the radio-opaque markers **730** that are embedded in the targeting fixture **690**. In some embodiments, this calibration can be done by the technician scrolling through image slices and marking them using the software, or by an algorithm that automatically checks each slice of the medical image, finds the markers **730**, verifying that they are the markers **730** of interest based on their physical spacing (the algorithm is documented herein). In some embodiments, to ensure accuracy, limit subjectivity, and to speed up the process, image thresholding is used to help define the edges of the radio-opaque marker **730**, and then to find the center of the marker **730** (the program is documented herein). Some embodiments of the software can do the necessary spatial transformations to determine the location in the room of the robot's markers relative to anatomy through standard rigid body calculations. For example, by knowing the locations of the radio-opaque markers **730** in the coordinate system of the medical image, and knowing the locations of the active markers **720** on the calibration frame **700** relative to these radio-opaque markers **730**, and monitoring the locations of the active markers on the robot **15** and targeting

fixture **690**.

(464) Some embodiments allow the surgeon to use the software to plan the trajectories for needles/probes **7405**, **7410**. In some embodiments, the software will allow any number of trajectories to be stored for use during the procedure, with each trajectory accompanied by a descriptor.

(465) In some embodiments, the robot **15** is moved next to the procedure table and cameras **8200** for tracking robot **15** and patient **18** are activated. The cameras **8200** and robot **15** are positioned wherever is convenient for the surgeon to access the site of interest. The marker mounts on the robot **15** have adjustable positions to allow the markers **720** to face toward the cameras **8200** in each possible configuration. In some embodiments, a screen can be accessed to show where the robot **15** is located for the current Z-frame **72** position, relative to all the trajectories that are planned. In some embodiments, the use of this screen can confirm that the trajectories planned are within the range of the robot's reach. In some embodiments, repositioning of the robot **15** is performed at this time to a location that is within range of all trajectories. Alternately or additionally, in some embodiments, the surgeon can adjust the Z-frame **72** position, which will affect the range of trajectories that the robot **15** is capable of reaching (converging trajectories require less x-y reach the lower the robot **15** is in the z-axis **70**). During this time, substantially simultaneously, a screen shows whether markers **720**, **730** on the patient **18** and robot **15** are in view of the cameras **8200**. Repositioning of the cameras **8200**, if necessary, is also performed at this time for good visibility.

(466) In some embodiments, the surgeon then selects the first planned trajectory and he/she (or assistant) presses "go". The robot **15** moves in the x-y (horizontal) plane and angulates roll **62** and pitch **60** until the end-effector **30** tube intersects the trajectory vector (see FIGS. 3A and 3B). In some embodiments, during the process of driving to this location, a small laser light will indicate end-effector **30** position by projecting a beam down the trajectory vector toward the patient **18**. This laser simply snaps into the top of the end-effector **30** tube. In some embodiments, when the robot's end-effector **30** tube coincides with the trajectory vector to within the specified tolerance, auditory feedback is provided to indicate that the desired trajectory has been achieved and is being held. Alternately or additionally, in some embodiments, light of a meaningful color is projected on the surgical field **17**. For example, in some embodiments, movement of the patient **18** or robot **15** is detected by optical markers **720** and the necessary x-axis **66**, y-axis **68**, roll **62**, and pitch **60** axes are adjusted to maintain alignment.

(467) In some embodiments, the surgeon then drives Z-frame **72** down until the tip of the end-effector **30** reaches the desired distance from the probe's or needle's target (typically the skin surface). While moving, the projected laser beam point should remain at a fixed location since movement is occurring along the trajectory vector. Once at the desired Z-frame **72** location, in some embodiments, the surgeon or other user can select an option to lock the Z-tube **50** position to remain at the fixed distance from the skin during breathing or other movement. At this point, the surgeon is ready to insert the probe or needle **7405**, **7410**. If the length of the guide tube **50** has been specified and a stop on the needle **7405**, **7410** or probe is present to limit the guide tube **50** after some length has been passed, the ultimate location of the tip of the probe/needle **7405**, **7410** can be calculated and displayed on the medical image in some embodiments. As described earlier, Additionally, in some embodiments, it is possible to incorporate a mechanism at the entry of the guide tube **50** that is comprised of a spring-loaded plunger **54** with a through-hole, and measures electronically the depth of depression of the plunger **54**, corresponding to the amount by which the probe or needle **7405**, **7410** currently protrudes from the tip of the guide tube **50**.

(468) In some embodiments, at any time during the procedure, if there is an emergency and the robot **15** is in the way of the surgeon, the "E-stop" button can be pressed on the robot **15**, at which point all axes except the Z-frame axis **72** become free-floating and the robot's end-effector **30** can be manually removed from the field by pushing against the end-effector **30**.

(469) Some embodiments can include a bone screw or hardware procedure. For example, during a typical procedure for conventional screw or hardware insertion in the spine, the patient **18** is positioned prone (or other position) on the procedure table, and is supported. In some embodiments, a targeting fixture **690** is attached to the patient's spine by the surgeon or technician. In some embodiments, the targeting fixture **690** is either clamped to bone (open or percutaneously) or unrolled and adhered to the skin (for example using roll **705**). The roll **705** could have a disposable drape incorporated. If a flexible roll **705** is used, reflective markers **720** will then be snapped into place in some embodiments.

(470) In some embodiments, once a targeting fixture **690** is attached, the patient **18** can undergo an intraoperative 3D image (Iso-C, O-Arm, or intraoperative CT) with radio-opaque markers **730** included in the field of view along with the bony region of interest. In some embodiments, for best accuracy and resolution, a fine-slice image is preferred (where the CT slice spacing=1 mm or less). The 3D scan in some embodiments has to include the radio-opaque markers **730** and the bony anatomy; not including both would disallow calibration to the robot **15**.

(471) In some embodiments, the 3D image series is transferred to (or acquired directly to) the computer **100** of the

robot **15**, and the 3D image is calibrated in the same way as described above for needle **7405**, **7410** or probe insertion. The surgeon then uses the software to plan the trajectories for hardware instrumentation (e.g., pedicle screw, facet screw). Some embodiments of the software will allow any number of trajectories to be stored for use during the procedure, with each trajectory accompanied by a descriptor that may just be the level and side of the spine where screw insertion is planned.

(472) In some embodiments, the robot **15** is moved next to the table and cameras **8200** for tracking robot **15** and patient **18** are activated. The cameras **8200** are positioned near the patient's head. In some embodiments, the markers for the robot **15** are facing toward the cameras **8200**, typically in the positive y-axis **68** direction of the robot's coordinate system. In some embodiments, a screen can be accessed to show where the robot **15** is located relative to all the trajectories that are planned for the current Z-frame **72** position. Using this screen it can be confirmed that the trajectories planned are within the range of the robot's reach. In some embodiments, repositioning of the robot **15** to a location that is within range of all trajectories is performed at this time. Alternately or additionally, in some embodiments, the surgeon can adjust the Z-frame **72** position, which will affect the range of trajectories that the robot **15** is capable of reaching (converging trajectories require less x-y reach the lower the robot **15** is in Z). During this time, simultaneously in some embodiments, a screen shows whether markers **720** on the patient **18** and robot **15** are in view of the cameras **8200**. Repositioning of the cameras **8200**, if necessary, is also performed at this time for good visibility.

(473) In some embodiments, the surgeon then selects the first planned trajectory and he/she (or assistant) presses "go". The robot **15** moves in the x-y (horizontal) plane and angulates roll **62** and pitch **60** until the end-effector **30** tube intersects the trajectory vector. During the process of driving to this location, in some embodiments, a small laser light will indicate end-effector **30** position by projecting a beam down the trajectory vector toward the patient **18**. This laser simply snaps into the top of the end-effector guide tube **50**. When the robot's end-effector guide tube **50** coincides with the trajectory vector to within the specified tolerance, auditory feedback is provided in some embodiments to indicate that the desired trajectory has been achieved and is being held. In some embodiments, movement of the patient **18** or robot **15** is detected by optical markers **720** and the necessary x-axis **66**, y-axis **68**, roll **62**, and pitch **60** axes are adjusted to maintain alignment.

(474) In some embodiments of the invention, the surgeon then drives Z-frame **72** down until the tip of the end-effector **30** reaches a reasonable starting distance from the site of operation, typically just proximal to the skin surface or the first tissues encountered within the surgical field **17**. While moving, the projected laser beam point should remain at a fixed location since movement is occurring along the trajectory vector. Once at the desired location, the user may or may not select an option to lock the Z-tube **50** position to remain at the fixed distance from the anatomy during breathing or other movement.

(475) One problem with inserting conventional guide-wires and screws into bone through any amount of soft tissue is that the screw or wire may sometimes deflect, wander, or "skive" off of the bone in a trajectory that is not desired if it does not meet the bone with a trajectory orthogonal to the bone surface. To overcome this difficulty, some embodiments can use a specially designed and coated screw specifically intended for percutaneous insertion. Some other embodiments can use an end-effector **30** tip fitted with a guide tube **50** or dilator, capable of being driven all the way down to the bone. In this instance, the guide tube **50** needs to have a sharp (beveled) leading edge **30b**, and may need teeth or another feature to secure it well to the bone once in contact. This beveled tube **50** (i.e. guide tube **50** that includes beveled leading edge **30b**) is driven through soft tissue and next to bone through one of two different methods using the surgical robot system **1** as described.

(476) In applications where conventional screws are to be driven into bone, the surgeon may want to move the end-effector tip **30**, fitted with a guide tube **50** or a conventional dilator, all the way down to the bone. Referring to FIG. **62** showing steps **6210**, **6215**, **6220**, **6225**, **6230**, **6235**, **6240**, **6245**, and either **6250**, **6255**, **6260**, and **6260**, or **6270**, **6275**, **6280** and **6285**, two embodiments address this need. In some embodiments, the user can insert the tube **50** and force it down the axis Z-tube axis **64** by hand, or with the robot **15** until a peak in force is registered by tactile feel or by a conventional force sensor on the end-effector **30** (signaling contact with bone). At this point, it is no longer necessary for the tip of the drill bit **42** to be positioned past the tip of the tube **50** (in fact be better to have it slightly retracted). As described earlier, a drill bit **42** can include a drill stop **46**, and the drill bit **42** can be locked and held (see for example FIGS. **17C-17E** and **17F-17J**). In some embodiments, the stop **46** on the drill bit **42** can then be adjusted by pulling one of the releases **48** and slightly adjusting its position. Then, the tube **50** can be brought up against bone and locked there. Now, the stop **46** can be adjusted to show how much the drill bit **42** would protrude beyond the tip. This same value can be used to offset (extrapolate) the tip of the tube **50** on the software, showing the user where the tip of the drill bit **42** will end up.

(477) In some embodiments, the Z-tube axis **64** is fitted with a conventional force sensor with continuous force readings being displayed on the screen (such as display means **29**). In some embodiments, the Z-frame **72** is then driven down into tissue while continuously adjusting the x-axis **66** and y-axis **68** to keep the tube **50** aligned with

the trajectory vector. In some embodiments, the steps of 6210, 6215, 6220, 6230, 6235, 6240, 6245, 6250 and 6255 can be used to drive the tube 50 toward the target. In this instance, roll 62 and pitch 60, defining orientation, should not change while moving x-axis 66, y-axis 68, and the Z-frame 72 as Z-axis 70 along this vector, while holding Z-tube 50 rigidly locked at mid-range. For this procedure, in some embodiments, the Z-tube 50 stiffness must be set very high, and may require a conventional mechanical lock to be implemented. In some embodiments, if Z-tube 50 is not stiff enough, a counter force from the tissues being penetrated may cause it to move back in the opposite direction of Z-frame 72, and the tube 50 will not have any net advancement. In some embodiments, based on the surgeon's previous experience and lab testing, Z-frame 72 is driven down until a force level from the monitored force on Z-tube 50 matches the force typical for collision with bone (step 6260).

(478) In some alternative embodiments, Z-tube 50 is positioned near the top of its range and Z-frame 72 is advanced (while adjusting x-axis 66 and y-axis 68 to stay on the trajectory vector) until the tube 50 tip is near the outermost border of dissected tissue (i.e. skin during percutaneous procedures). In some embodiments, the Z-tube's motor 160 is then deactivated to allow it to move freely while still monitoring its position (step 6270). In some embodiments, the surgeon then pushes the end-effectuator 30 down while x-axis 66, y-axis 68, roll 62, and pitch 60 adjustments can allow the tube 50 to be aligned with the trajectory vector (step 6275). Moreover, since the Z-tube 50 is passive, in some embodiments, the surgeon can manually force the tube 50 to advance until he/she experiences the tactile sense of the tube hitting bone, at which point the Z-tube 50 position is locked (motor 160 activated) by the surgeon or assistant (step 6280, 6285).

(479) At this point, in some embodiments, the guide tube 50 is adjacent to bone and the surgeon may wish to drill into the bone with a conventional guide-wire or drill bit, or insert a screw. For screw prep and insertion, in some embodiments, the surgeon either uses a method that incorporates guide-wires, or a method that does not use guide-wires.

(480) Some embodiments include a guide-wire method. For example, in some embodiments, a guide-wire is drilled into bone through the guide tube 50. After the guide-wire is in place, Z-frame 72 and tube 50 are driven upward along the trajectory vector until outside the body. In some embodiments, the tube is then released with a quick release from the robot's end-effectuator 30 so it can be positioned at the next trajectory. In some embodiments, a cannulated screw, already commonly used in spine surgery, can then be driven in place over the guide-wire.

(481) Some embodiments include a non-guide-wire method. For example, a pilot hole may or may not be drilled first. In some embodiments, a screw is then driven into bone directly through the guide tube 50, which abuts bone. In some embodiments, the tip of the screw may have the special non-skiving design mentioned above.

(482) In some embodiments, if hardware other than a screw is being inserted, the surgeon may wish to dilate soft tissue. In some embodiments, a dilated path would enable larger and/or more tools and implants to be inserted. In some embodiments, dilation is performed by sliding a series of larger and larger diameter tubes over the initial central shaft or tube. In some embodiments, a series of dilators, specially designed to integrate to the robot's end-effectuator 30, sequentially snap on to each other for this purpose.

(483) In some embodiments, after the screw or hardware has been inserted in the first trajectory, the surgeon drives the robot 15 back up the trajectory vector away from the patient 18. In some embodiments, after the end-effectuator 30 is clear of the patient 18 in the Z direction, the next trajectory is selected and the robot 15 repeats the above steps.

(484) In some embodiments, at any time during the procedure, if there is an emergency and the robot 15 is in the way of the surgeon, the "E-stop" button can be pressed on the robot 15, at which point all axes except Z-frame 72 become free-floating, and the robot's end-effectuator 30 can be manually removed from the field by pushing against the end-effectuator.

(485) In some embodiments, for nerve avoidance during medical procedures, a special conventional dilator tube (not shown) that can be used with the robot 15. In some embodiments, the dilator tube can include multiple electrodes at its tip that can be sequentially activated to find not only whether a nerve is nearby, but also to find which radial direction is the nearest direction toward the nerve. Some embodiments incorporate this guide tube 50 and can identify, warn or incorporate automatic algorithms to steer clear of the nerve.

(486) In some embodiments, it is known that pairs of bone screws such as pedicle screws have better resistance to screw pullout if they are oriented so that they converge toward each other. In some embodiments, for the best potential biomechanical stability, a two-screw surgical construct can consist of specially designed conventional screws that would interconnect in the X Z plane (not shown). That is, one screw can have a socket to accept a threaded portion of the other screw so that the screws interconnect at their tips. A procedure such as this requires exceptional accuracy, otherwise the screw tips would not properly intersect, and is therefore especially well-suited for a surgical robot 15. This type of hardware is useful with certain embodiments of the invention.

(487) In some embodiments, instead of only straight lines, the surgeon has several options for trajectory planning

—straight, curved or boundary for safe-zone surgery. For curved pathway planning, in some embodiments, the surgeon can draw a path on the medical image that has curvature of a user-selectable radius. In some embodiments, special conventional needles and housings can be used to execute these curved paths. In safe zone surgery (tumor or trauma), in some embodiments, the surgeon first plans a box or sphere around the region on the medical image within which the probe tip, incorporating a drill or ablation instrument, will be allowed to reside. In some embodiments, the robot **15** is driven down along a trajectory vector either automatically or manually as described above to position the tip of the probe to be in the center of the safe zone. In some embodiments, the surgeon would then be able pick the tool's axis of rotation (orthogonal to the long axis) based on the desired impact he/she would like for the purpose of preserving tissue and maximizing efficiency and effectiveness for the task at hand. For example, in some embodiments, an axis of rotation at the surface of the skin could be selected to minimize the amount by which the tool travels laterally and rips the skin.

(488) In some embodiments, the robot **15** uses optical markers for tracking. Some embodiments are able to provide accurate localization of the robot **15** relative to the patient **18**, and utilize the LPS because of the advantage of not being limited to line-of-sight. Additionally, in some embodiments, probes utilizing RF emitters on the tip (capable of being tracked by the LPS) can be used for steering flexible probes inside the body. In some embodiments, if the LPS is not yet functional for localization, then localization can be performed using an electromagnetic system such as the Aurora by Northern Digital. Aurora® is a registered trademark of Northern Digital Inc. For example, in this instance, an electromagnetic coil and RF emitters are both present in the probe tip. Some embodiments can offer the option of LPS or electromagnetic localization with steerable needles **7600**. In this embodiment of the invention, the surgeon can monitor the current location on the medical image where the probe tip is currently positioned in real-time and activate RF electrodes to advance and steer the probe tip in the desired direction using a joystick.

(489) As discussed earlier, in some embodiments, the end-effectuator **30** can include a bayonet mount **5000** is used to removably couple the surgical instrument **35** to the end-effectuator **30** as shown in FIG. **48**. Some embodiments can include a modification to the mount **5000** allowing the ability to slide a clamping piece **6300** over the spinous process **6301** without full exposure of the spinous process **6301**. See example FIGS. **63A-63B** illustrating various embodiments of an end-effectuator **30** including a modified mount **5000** with a clamping piece **6300** in accordance with at least one embodiment of the invention. As shown, the clamping piece **6300** comprises clamps **6302** including at least one beveled edge **6310**, and clamp teeth **6330**.

(490) In some embodiments, the surgeon would make a stab incision in the midline and then slide the clamps **6302** of the clamping piece **6300** down along the sides of the spinous process **6301**, pushing tissue away as the tip of the clamping piece is advanced. In some embodiments, the leading edge of the clamping mechanism **6300** would be beveled (see the leading edges **6305** of each clamp **6302** of the clamping mechanism **6300**), and have a shape similar to a periosteal elevator. This allows the clamping mechanism **6300** to separate the muscle tissue from the bony spinous process **6301** as it is advanced. In some embodiments, the leading edges **6305** of the clamping mechanism **6300** can be electrified to enable it to more easily slide through muscle and connective tissues to prevent excessive bleeding.

(491) In some embodiments, a mechanism activated from farther back on the shaft (for example a turn screw **6320**, or conventional spring, etc.) can be activated to deploy clamp teeth **6330** on the clamps **6302**. The same mechanism or another mechanism would close and compress the clamps **6302** together to firmly secure the clamping mechanism **6300** to the spinous process **6301** (see FIGS. **63B-63C**). Additionally, in some embodiments, a screw **6340** aligned with the handle **6350** could deploy to thread into the spinous process **6301** (see for example, FIG. **63C**).

(492) The embodiments as described above and shown in FIGS. **63A-63C** would be especially well suited to percutaneous pedicle screw-rod surgery because the hole made for mounting the clamping mechanism **6300** could also be used as the hole for inserting the conventional rod to interconnect the conventional pedicle screw heads. Further, the embodiments as described above and shown in FIGS. **63A-63C** could also be useful for mounting a marker tree (for example marker tree **795** shown in FIG. **50A**) to other bony prominences, such as transverse processes, long bones, skull base, or others.

(493) FIGS. **64** and **65** illustrate embodiments of clamping piece **6300** actuation on a spinous process **6301** in accordance with some embodiments of the invention. In some embodiments, the mechanism for deploying the clamp teeth **6330** could be comprised of a hollow tool tip **6360** containing teeth **6330** that are to one side of the hollow cavity **6370** during insertion, but are forced toward the opposite side when the mechanism is deployed, such that the embedded teeth penetrate the bone (see the illustration of penetrated teeth **6330a** in FIG. **64**).

(494) FIG. **65** shows an alternative embodiment of the clamping piece **6300** actuation on a spinous process **6301**. As shown, the groups of teeth **6330** are attached to rods **6365** that run down the hollow cavities **6360a** of the hollow tool tips **6360**. These rods **6365** pivot farther up the handle **6350** (pivot point not pictured) and the clamp

teeth **6330** are forced together. For example, in some embodiments, rods **6365** are driven into the hollow cavity **6360a** of the hollow tool tip **6360** on the side away from the bone, forcing the clamp teeth **6330** against and into the bone (for example, see the penetrated teeth **6330a** in FIG. 65).

(495) As described above, the opaque markers **730** must be included in a CT scan of the anatomy. However, it is desirable to crop CT scans as close as possible to the spine to improve resolution. In some embodiments, instead of using markers **730** near where the active markers **720** are located, an alternative is to have a rigid extension containing opaque markers **730** that are temporarily attached near the spine when the scan is taken. In some embodiments, the clamping piece **6300** can be coupled with, or otherwise modified with a targeting fixture **690**. For example, FIG. 66A illustrates a clamping piece **6300** modified with a targeting fixture **690** including a temporary marker skirt **6600** in accordance with at least one embodiment of the invention, and FIG. 66B illustrates a clamping piece **6300** modified with a targeting fixture **690** as shown in FIG. 66A with the temporary marker skirt **6600** detached in accordance with at least one embodiment of the invention. As shown, the temporary marker skirt **6600** includes radio-opaque markers **730** in a temporary “skirt” around the base of the clamping device **6300**. The design of the temporary marker skirt **6600** and clamping device **6300** must be such that the markers **730** in the skirt **6600** have known locations relative to the markers **720** for tracking that are farther away. Once the scan is taken, the opaque markers **730** are not needed. Therefore, in some embodiments, by depressing a conventional release, the skirt **6600** can be removed so it will not be in the way of the surgeon (see for example FIG. 66B).

(496) In some embodiments, it may also be desirable to mount the targeting fixture **690** to another piece that is already rigidly attached to the patient **18**. For example, for deep brain stimulation or other brain procedure where the patient **18** is positioned in a Mayfield head holder, the head holder could serve as an attachment point for the targeting fixture **690**. Since the head holder **6700** and skull form a rigid body, it is possible to track the head holder **6700** under the assumption that the skull moves the same amount as the head holder **6700**. Further, in some embodiments of the invention, a surveillance marker (such as surveillance marker **710** as illustrated in FIG. 36) could be used. For this targeting fixture **690**, active **720** and radio-opaque **730** markers would be rigidly attached to the head holder **6700**. The radio-opaque markers **730** need only be in position when the scan (CT, MRI, etc.) is taken and could subsequently be removed. The active markers **720** need not be in position when the scan is taken but could instead be snapped in place when it is necessary to begin tracking. For example, FIG. 67 shows one possible configuration for active **720** and radio-opaque markers **730** attached to a Mayfield frame **6700** in accordance with one embodiment of the invention. As with other targeting fixtures **690**, it is required that three or more radio-opaque markers **730** and three or more active markers **720** are attached to same rigid body.

(497) One problem with some robotic procedures is that the guide tube **50** must be physically rigidly mounted to the robot's end-effector, and therefore mounting one or more dilator tubes can be challenging. To address this problem, in some embodiments, dilators can be placed over the central guide-tube **50** without removing the robot end-effector **30**. For example, some embodiments can include an end-effector **30** that includes at least one dilator tube **6800**, **6810**. For example, FIG. 68 shows end-effector **30** that includes nested dilators **6805** in accordance with at least one embodiment of the invention. As shown, a nested set **6805** of two or more disposable or non-disposable dilators **6800**, **6810** can be mounted onto the robot's end-effector **30**. In some embodiments, each dilator **6800**, **6810** may have its own removable conventional handle that allows a surgeon or an automated mechanism to force the dilator down into soft tissue. Some embodiments could include additional dilators, for example, a nested set of three dilators of 7 mm, 11 mm, and 14 mm diameter (not shown) may be useful for creating a portal for minimally invasive screw insertion or application of a surgical implant. In some embodiments, each dilator **6800**, **6810** can have greater length as it is closer to the central guide tube **50**, allowing the more central tube **50** to be advanced without radially advancing the dilator tubes **6800**, **6810** out further.

(498) In some further embodiments, the system **1** can include an end-effector **30** that is coupled with at least one cylindrical dilator tube **6900**. For example, FIGS. 69A-69C illustrate various embodiments of an end-effector **30** including cylindrical dilator tubes **6900** in accordance with at least one embodiment of the invention. As shown, in some embodiments, the cylindrical dilator tubes **6900** can be formed from two-halves that snap together. In some embodiments, the cylindrical dilator tubes **6900** can be formed from two-halves that snap together, and in some embodiments, the two-halves snap together over a previous dilator tube **6900**. In some embodiments, the tubes **6900** can be fashioned so that they are strong in resisting radial compression, but not necessarily strong in resisting radial expansion (since their opposing force will be the resisting soft tissues). In some embodiments, the tubes **6900** can also benefit from a mechanism for temporarily attaching a conventional handle at the proximal end for easy insertion then removal of the handle following insertion. Moreover, some embodiments include a mechanism for grasping and extracting each tube **6900** or a cluster of tubes **6900**, or for attaching one or more tubes **6900** to the central guide tube **50**. As depicted in FIGS. 69B and 69C, when the robot's end-effector **30** is raised (following the tube **6900** insertion depicted in FIG. 69B), the tube **6900** or

cluster of tubes **6900** is extracted with it, leaving behind the outermost dilator **6910a** and forming a corridor for surgery. Further, in some embodiments, the surgeon can send the robot's end-effector **30** to coincide with the infinite vector defining the desired trajectory, but above the patient **18**. In some embodiments, the surgeon then sends the robot's end-effector **30** down this vector until the tip of the central guide pin or tube **50** is ready to penetrate soft tissue. In some embodiments, a starter incision may be made to help the central guide tube **50** penetrate the tissue surface. In some embodiments, the surgeon continues to send the robot's end-effector **30** down the trajectory vector, penetrating soft tissue, until the target is reached (for example, when the tube **50** abuts bone of a target region). Then, in some embodiments, while the robot **15** holds the central tube **50** steady, each sequential dilator **6900** is slid down the central tube **50** over the previous dilator **6900**. When desired dilation is complete, in some embodiments, the proximal end of the dilator tube **6900** may be secured to the patient **18** (or external assembly), and the central tube **50** and all but the outermost dilator tube **6910** would be removed.

(499) Some embodiments include tubes **6900** that comprise a polymeric material. In some embodiments, the tubes **6900** can include at least one either radiolucent or radio-opaque material. In some embodiments, dilators **6900** may be radio-opaque so that their position may be easily confirmed by x-ray. Further, in some embodiments, the outermost dilator **6910** may be radiolucent so that the position of pathology drawn out through the tube, or implants, or materials passed into the patient through the tube, may be visualized by x-ray.

(500) As described earlier, in some embodiments, the use of conventional linear pulse motors **160** within the surgical robot **15** can permit establishment of a non-rigid position for the end-effector **30** and/or surgical instrument **35**. In some embodiments, the use of linear pulse motors **160** instead of motors with worm gear drive enables the robot **15** to quickly switch between active and passive modes.

(501) The ability to be able to quickly switch between active and passive modes can be important for various embodiments. For example, if there is a need to position the robot **15** in the operative field, or remove the robot **15** from the operative field. Instead of having to drive the robot **15** in or out of the operative field, in some embodiments, the user can simply deactivate the motors **160**, making the robot **15** passive. The user can then manually drag it where it is needed, and then re-activate the motors **160**.

(502) The ability to be able to quickly switch between active and passive modes can be important for safe zone surgery. In some embodiments, the user can outline a region with pathology (for example a tumor **7300**) on the medical images (see for example FIG. **70** showing the displayed tumor **7310** on display means **29**). In some embodiments, algorithms may then be implemented where the robot **15** switches from active to passive mode when the boundary of the region is encountered. For example, FIG. **70** shows the boundary region **7320** within the patient **18** displayed as region **7325** on the display means. Anywhere outside the boundary **7320**, the robot becomes active and tries to force the end-effector **30** back toward the safe zone (i.e. within the boundary **7320**). Within the boundary **7320**, the robot **15** remains passive, allowing the surgeon to move the tool (such as drill bit **42**) attached to the end-effector **30**.

(503) In some further embodiments, the user can place restrictions (through software) on the range of orientations allowed by the tool within the safe zone (for example, boundary **7320**, and displayed as boundary **7325** in FIG. **70**). In some embodiments, the tool can only pivot about a point along the shaft that is exactly at the level of the skin. In this instance, the robot **15** freely permits the surgeon to move in and out and pivot the end-effector **30**, but does not allow left-right or front-back movement without pivoting. For example, in some embodiments, if the surgeon wants to reach a far left point on the tumor **7300**, the surgeon must pivot the tool about the pivot point and push it to the appropriate depth of insertion to satisfy the boundary **7320** conditions and force the tip (for example, the tip of the drill bit **42**) to that location. This type of limitation can be valuable because it can prevent the surgeon from "ripping" tissue as the drill is moved around to destroy the tumor **7320**. Further, it also allows the surgeon to access a safe zone farther distal while keeping clear of a critical structure farther proximal.

(504) Some embodiments include curved and/or sheathed needles for nonlinear trajectory to a target (for example, such as a tumor **7320** described earlier). In some embodiments, with a curved trajectory, it is possible to approach targets inside the body of a patient **18** that might otherwise be impossible to reach via a straight-line trajectory. For example, FIG. **71A** illustrates a robot end-effector **30** coupled with a curved guide tube **7400** for use with a curved or straight wire or tool **7410** in accordance with at least one embodiment of the invention. In some embodiments, by forcing a curved or straight wire or tool **7410** through the curved guide tube **7400**, at least some curvature will be imparted to the wire or tool **7410**. In some embodiments, the curved or straight wire or tool **7410** may comprise a compliant wire capable of forming to the curvature of the guide tube **7400**. In some other embodiments, the curved or straight wire or tool **7410** may comprise a non-compliant wire, capable of substantially retaining its shape after entering and exiting the guide tube **7400**. A disadvantage of using a very compliant wire is that the tissues that it encounters may easily force it off the desired path. A disadvantage of using a very non-compliant wire is that it would be difficult to achieve a useful amount of curvature. Further, forcing a straight wire of intermediate compliance through a curved guide tube **7400** may produce some curvature

of the wire, but less curvature than that of the guide tube **7400**. It is possible to mathematically or experimentally model the mechanical behavior of the wire **7410** to determine how much curvature will be imparted. For example, by knowing the orientation of the guide tube **7400**, in some embodiments, the robot may be used to accurately guide the curved wire **7410** to a desired target by using computerized planning to predict where the wire **7410** would end up as it traveled through tissue. Further, in some embodiments a very non-compliant wire or tool **7410** can be manufactured in the shape of an arc with a specific radius of curvature, and then fed through a guide tube **7400** with the same radius of curvature. By knowing the orientation of the guide tube **7400** (i.e. substantially the same as wire or tool **7410**), computerized planning can be used to predict where the wire or tool **7410** would end up as it traveled through tissue.

(505) Some other embodiments may use a straight guide tube **50** with a wire or tool **7410** that may be curved or straight. For example, FIG. **71B** illustrates a robot end-effectuator **30** coupled with a straight guide tube **50** for use with a curved or straight wire or tool **7405**, **7410** in accordance with at least one embodiment of the invention. Some surgical methods may use curved needles **7410** that are manually positioned. In general, the needles consist of a rigid, straight outer guide tube through which is forced an inner needle **7405** with tendency to take on a curved shape. In existing manual devices, the inner needle **7405** is comprised of nitinol, a shape memory alloy, and is formed with significant curvature. This curved needle **7410** is flattened and then fed through the outer guide tube. When it exits the other end of the guide tube, it bends with significant force back toward its original curved configuration. Such a system could be adapted for use with the robot **15** if the curvature of the exiting portion of the needle per unit measure exiting is known, if the radial position of the curved needle **7410** relative to the straight housing is known. In some embodiments, the radial position of the curved needle **7410** can be determined by using marks placed on the curved and straight portions, or through a non-circular cross-section of the straight guide tube and curved needle **7410** (for example, square cross-section of each). In this instance, in some embodiments, it would then be possible to preoperatively plan the path to the target (such as a tumor **7300**) and then adjust the robot **15** to guide the curved wire or tool **7410** through this path. In some embodiments, the system **1** can include the ability to electrically stimulate distally while advancing a wire (for example, such as wire **7405**, **7410**) through soft tissue. For example, some embodiments include a guide tube **7500** capable of being coupled to the robot **15** by end-effectuator **30** that is insulated along its entire shaft but has an electrode **7510** on or near the tip (see for example FIG. **72**). In some embodiments, the use of the tube **7500** to perform electromyography (“EMG”) can enable the system **1** to detect whether nerves come in contact with the guide tube **7500** as the guide tube **7500** is advanced. Some alternative embodiments can include a conventional pin (for example, stainless steel pins such as Kirschner-wires) instead of a tube **7500**, insulated along its shaft but not at the tip. In some embodiments, the wire could be connected to a stimulator outside the body and would have the ability to stimulate distally while advancing the pin through soft tissue. In some embodiments, stimulation would allow the ability to identify critical tissue structures (i.e., nerves, plexus).

(506) In some further embodiments, a portion of the leading edge of the guide tube **7500** may be insulated (i.e. comprise a substantially non-electrically conductive area), and a portion of the leading edge may be uninsulated (i.e. the region is inherently electrically conductive area). In this instance, it can be possible to determine the radial direction of the tube **7500** that is closest to the nerve by watching the response as the tube **7500** is rotated. That is, as the tube **7500** is rotated, the EMG nerve detection will have the most pronounced response when the uninsulated portion is nearest the nerve, and the least pronounced response when the uninsulated portion is farthest from the nerve. In some embodiments, it would then be possible for the user to manually steer the robot **15** to automatically steer the tube **7500** farther away from the nerve. In addition, this modified tube **7500** could have a conventional fan-like retractor (not shown) that can be deployed to gently spread the underlying muscle fibers, thereby making an entry point for disk removal, or screw insertion. In some embodiments, the combination of EMG and gentle retraction can enhance the safety and outcomes of robotic assisted spinal surgery.

(507) As described above, one way of taking advantage of the directional electromyographic response is for the user to manually rotate the tube **7500**. In some other embodiments, the tube **7500** can be to continuously oscillated back and forth, rotating about its axis while potentials are monitored. In some embodiments, to achieve the same function without rotating the tube **7500**, the leading edge of the tube **7500** could have conductive sections that could be automatically sequentially activated while monitoring potentials. For example, in some embodiments, an array of two, three, four, or more electrodes **7510** (shown in FIG. **72**) can be positioned around the circumference of the leading edge of the tube **7500**. As shown in FIG. **72**, regions **7511** between the electrodes **7510** are insulated from each other (because the outer surface of **7500** is insulated). In some embodiments, the electrodes **7510** can be sequentially activated at a very high rate while recording potentials, and correlating which electrode produces the greatest response.

(508) Some embodiments can include a steerable needle capable of being tracked inside the body. For example, U.S. Pat. No. 8,010,181, “System utilizing radio frequency signals for tracking and improving navigation of

slender instruments during insertion in the body”, herein incorporated by reference, describes a steerable flexible catheter with two or more RF electrodes on the tip, which are used for steering. According to the method described in U.S. Pat. No. 8,010,181, the side or sides of the tip where the electrodes emit RF have less friction and therefore the probe will steer away from these sides.

(509) In some embodiments of the invention, a steerable needle **7600** can be coupled with the system **1**. In some embodiments, the system **1** can include a steerable needle **7600** coupled with the robot **15** through a coupled end-effector **30**, the steerable needle **7600** capable of being tracked inside the body of a patient **18**. For example, FIG. **73** illustrates a steerable needle **7600** in accordance with at least one embodiment of the invention. In some embodiments, steerable needle **7600** can comprise a plurality of flattened angled bevels **7605** (i.e. facets) on the tip of the probe, with each flat face of each bevel **7605** having an RF electrode **7610**. A magnetic coil sensor **7620** embedded within the needle **7600** can enable localization of the tip adjacent to the electrodes **7610**. In some embodiments, RF (using for example RF transmitters **120** described earlier) can be used for steering, whereas localization would use electrodes **7610** with the magnetic coil sensor **7620**. Some embodiments as described may use off-the-shelf electromagnetic localization system such as the Aurora® from Northern Digital, Inc. (<http://www.ndigital.com>), which has miniature coils capable of fitting inside a catheter.

(510) During surgical procedures, pedicle screws or anterior body screws are inserted in two locations. However, there is a chance of failure due to screw pullout. To enhance resistance to pullout, screws are angled toward each other. For example, some embodiments can include intersecting and interlocking bone screws **7700** such as those illustrated in FIG. **74**, illustrating one embodiment of intersecting and interlocking bone screws **7700** in accordance with at least one embodiment of the invention. As shown, bone screws **7700** can be coupled and can intersect and interlock **7720**. In some embodiments, the intersecting and interlocking bone screws **7700** as shown can be removed without destroying a large area of bone.

(511) Some embodiments of the system **1** can include conventional tracking cameras with dual regions of focus. For example, camera units such as Optotrak® or Polaris® from Northern Digital, Inc., can be mounted in a bar so that their calibration volume and area of focus are set. Optotrak® or Polaris® are registered trademarks of Northern Digital, Inc (see for example FIG. **81** showing camera bar **8200**). In some embodiments, when tracking the robot **15** and targeting fixture **690** with optical trackers (for example, active markers **720**), maintaining markers **720** within the center of the volume can provide the best focus. However, it is not possible for both the targeting fixture's **690** markers and the robot's **15** markers to be substantially centered simultaneously, and therefore both are offset from center by substantially the same distance.

(512) In some embodiments, one solution to this issue is to set up two pairs of cameras **8200** with one camera shared, that is, cameras **1** and **2** form one pair, and cameras **2** and **3** form another pair. This configuration is the same as the Optotrak® system (i.e., three cameras in a single bar), however, the Optotrak® only has one volume and one common focal point. Conversely, some embodiments of the invention would be tuned to have two focal points and two volumes that would allow both the targeting fixture **690** and the robot **15** to be centered at the same time. In some embodiments, the orientations of the lateral cameras can be adjusted by known amounts with predictable impact on the focal point and volume.

(513) In a further embodiment of the invention, two separate camera units (for example, two Polaris® units) can be mounted to a customized conventional bracket fixture including adjustment features (not shown). In some embodiments, this fixture would be calibrated so that the vectors defining the directions of the volumes and distance to focal point can be adjustable by known amounts. In some embodiments, the user could then point one Polaris® unit at the robot's markers, and the other Polaris® unit at the targeting fixture's **690** markers **720**. The position of the adjustment features on the bracket would tell the computer what the transformation is required to go from one camera's coordinate system to the other.

(514) In some further embodiments, the cameras **8200** (such as Optotrak® or Polaris®) focused on a particular region could be further improved by a conventional automated mechanism to direct the cameras **8200** at the center of the target. Such a method would improve accuracy because in general, image quality is better toward the center of focus than toward the fringes. In some embodiments, conventional motorized turrets could be utilized to adjust azimuth and elevation of a conventional bracket assembly for aiming the cameras **8200** (and/or in conjunction with movement of cameras **8200** on camera arm **8210** as shown in FIG. **81**). In some embodiments, feedback from the current location of active markers **720** within the field of view would be used to adjust the azimuth and elevation until the camera **8200** points directly at the target, regardless of whether the target is the center (mean) of the markers **720** on the robot **15**, the center of markers **720** on the targeting fixture **720**, or the center of all markers **720**. In some embodiments, such a method would allow the center of focus of the cameras **8200** to continuously move automatically as the patient **18** or robot move, ensuring the optimal orientation at all times during the procedure.

(515) Some embodiments can include a snap-in end-effector **30** with attached tracking fixtures **690** (including

active markers **720**). For example, some embodiments include snap-in posts **7800** attached to the end-effectuator **30** and tracking fixtures **690**. In some embodiments, the snap-in posts **7800** can facilitate orienting tracking markers **720** to face cameras **8200** in different setups by allowing markers **720** to be mounted to each end-effectuator **30**. FIG. 75A-75B illustrates configurations of a robot **15** for positioning alongside a bed of a patient **18** that includes a targeting fixture **690** coupled to an end-effectuator **30** using a snap-in post **7800**. In some embodiments, with the robot **15** in a typical configuration alongside a bed with the patient's **18** head toward the left, one end-effectuator **30** could have right-facing markers **720** (fixture **690**) (illustrated in FIG. 75A) for cameras **8200** positioned at the foot of the bed. In some embodiments, the same type of end-effectuator **30** could have left-facing markers **720** (fixture **690**) for cameras **8200** positioned at the head of the bed (illustrated in FIG. 75B). In some embodiments, the fixtures **690** are mounted where they would be closer to the cameras **8200** than the end-effectuator **30** so that the surgeon does not block obscure the markers **720** from the camera when using the tube **50**. In some further embodiments, each interchangeable end-effectuator **30** could include conventional identification electronics. For example, in some embodiments, each interchangeable end-effectuator **30** could include an embedded conventional chip and a press-fit electrical connector. In some embodiments, when the system **1** includes a snap-in end-effectuator **30** with attached tracking fixtures **690**, the computer **100** may recognize which end-effectuator is currently attached using the identification electronics. In some embodiments, when the system **1** includes a snap-in end-effectuator **30** with attached tracking fixtures **690**, the computer **100** may recognize which end-effectuator is currently attached using the identification electronics, and apply stored calibration settings.

(516) The robot system **1** contains several unique software algorithms to enable precise movement to a target location without requiring an iterative process. In some embodiments, an initial step includes a calibration of each coordinate axis of the end-effectuator **30**. During the calibration, the robot **15** goes through a sequence of individual moves while recording the movement of active markers **720** that are temporarily attached to the end-effectuator (see FIG. 76). From these individual moves, which do not have to fall in a coordinate system with orthogonal axes, the required combination of necessary moves on all axes is calculated.

(517) In some embodiments, it is possible to mount optical markers **720** for tracking the movement of the robot **15** on the base of the robot **15**, then to calculate the orientation and coordinates of the guide tube **50** based on the movement of sequential axes (see earlier description related to FIG. 16). The advantage of mounting markers **720** on the base of the robot **15** is that they are out of the way and are less likely to be obscured by the surgeon, tools, or parts of the robot. However, the farther away the markers **720** are from the end-effectuator **30**, the more the error is amplified at each joint. At the other extreme, it is possible to mount the optical markers **720** on the end-effectuator **30** (as illustrated in FIG. 76). The advantage of mounting markers **720** on the end-effectuator is that accuracy is maximized because the markers **720** provide feedback on exactly where the end-effectuator **30** is currently positioned. A disadvantage is that the surgeon, tools, or parts of the robot **15** can easily obscure the markers **720** and then the end-effectuator's **30** position in space cannot be determined.

(518) In some embodiments, it is possible to mount markers **720** at either extreme or at an intermediate axis. For example, in some embodiments, the markers **720** can be mounted on the x-axis **66**. Thus, when the x-axis **66** moves, so do the optical markers **720**. In this location, there is less chance that the surgeon will block them from the cameras **8200** or that they would become an obstruction to surgery. Because of the high accuracy in calculating the orientation and position of the end-effectuator **30** based on the encoder outputs from each axis, it is possible to very accurately determine the position of the end-effectuator **30** knowing only the position of the markers on the x-axis **66**.

(519) Some embodiments include an algorithm for automatically detecting the centers of the radio-opaque markers **730** on the medical image. This algorithm scans the medical image in its entirety looking for regions bounded on all sides by a border of sufficient gradient. If further markers **730** are found, they are checked against the stored locations and thrown out if outside tolerance.

(520) Some biopsy procedures can be affected by the breathing process of a patient, for example when performing a lung biopsy. In some procedures, it is difficult for the clinician to obtain a sample during the correct breathing phase. The use of tracking markers **720** coupled to a bone of the patient cannot alone compensate for the breathing induced movement of the target biopsy region. Some embodiments include a method of performing a lung biopsy with breathing correction using the system **1**. Currently, for radiation treatment of lung tumors, breathing is monitored during CT scan acquisition using a "bellows" belt (see for example CT scanner **8000** in FIG. 77, with bellows image **8010**). The bellows monitors the phase of breathing, and when the clinician tells the patient to hold their breath, CT scan of the patient **18** is performed. The bellows output **8010** shows the phase in which the CT was taken. Later, targeted radiation bursts can be applied when the lung is in the right position as monitored by the bellows during the treatment phase. A CT scan is taken while the bellows monitors the breathing phase and when the patient held their breath during the CT scan. Later, radiation bursts are applied instantaneously when that same

phase is reached without requiring the patient **18** to hold their breath again.

(521) Some embodiments include a method of performing a lung biopsy with breathing correction using the system **1**. In some embodiments, a tracking fixture **690** is attached to the patient **18** near biopsy site and bellows belt on the patient's **18** waist. In some embodiments, a CT scan of the patient **18** is performed with the patient holding their breath, and while monitoring the breathing phase. In some embodiments, a clinician locates the target (for example, a tumor) on the CT volume, and configures the robot **15** to the target using at least one of the embodiments as described earlier. In some embodiments, the robot **15** calibrates according to at least one embodiment described earlier. In some embodiments, the robot **15** moves into position above the biopsy site based the location of at least one tracking marker **720**, **730**. In some embodiments, the bellows belt remains in place, whereas in other embodiments, the markers **720**, **730** on the patient **18** can track the breathing phase. In some embodiments, based on the bellows or tracking markers **720**, **730**, the computer **100** of the computing device **3401** within platform **3400** can use robotic guidance software **3406** to send a trigger during the calibrated breathing phase to deploy a biopsy gun to rapidly extract a biopsy of the target (such as a tumor). In some embodiments, a conventional biopsy gun (or tool, such as biopsy gun tip **8100** in FIG. **78**) could be mounted in the robot's end-effector **30** and activated by a conventional mechanism (such as for example, by a toggled digital output port). For example, as shown, the biopsy gun tip **8100** can comprise a biopsy needle **8110** including a stroke length **8120**, a sampling window **8130** and a biopsy tip **8140**. In some embodiments, the biopsy needle **8110** in the biopsy gun tip **8100** can be mounted to the end-effector **30**. In some embodiments, the biopsy needle **8110** can be inserted (under guidance by the robot **15**) at least partially into the superficial tissues near the target (for example, the moving lung tumor). In some embodiments, the biopsy gun tip **8100** can fire as directed by a software **3406** trigger, requiring only a small penetration to retrieve the biopsy.

(522) Deep brain stimulation ("DBS") requires electrodes to be placed precisely at targets in the brain. Current technology allows CT and MRI scans to be merged for visualizing the brain anatomy relative to the bony anatomy (skull). It is therefore possible to plan trajectories for electrodes using a 3D combined CT/MRI volume, or from CT or MRI alone. Some embodiments include robot **15** electrode placement for asleep deep brain stimulation using the system **1** where the acquired volume can then be used to calibrate the robot **15** and move the robot **15** into position to hold a guide **50** for electrode implantation.

(523) In some embodiments, a Mayfield frame **6700** modified including one possible configuration for active and radio-opaque markers (shown in FIG. **67** in accordance with one embodiment of the invention) can be used for electrode placement for asleep deep brain stimulation. In some embodiments, the active markers **720** do not need to be attached at the time of the scan as long as their eventual position is unambiguously fixed. In some embodiments, the radio-opaque markers **730** can be removed after the scan as long as the relative position of the active markers **720** remains unchanged from the time of the scan. In some embodiments, the marker **730** can be a ceramic or metallic sphere, and for MRI, a suitable marker is a spherical vitamin E capsule. In some embodiments, the end-effector **30** can include an interface for feeding in a conventional electrode cannula and securing the electrode housing to the skull of the patient **18** (for example, using a Medtronic StimLoc® lead anchoring device to the skull). StimLoc® is a trademark of Medtronic, Inc., and its affiliated companies.

(524) In some embodiments, the system **1** can perform the method steps **7910-7990** as outlined in FIG. **79** for DBS electrode placement. As show, in some embodiments, the patient **18** can receive an MRI **7910**, and the target and trajectory can be planned **7915**. Surgery can be initiated under general anesthesia **7920**, and the head frame (as shown in FIG. **67**) can be attached to the patient **18** with three screws in the skull **7925**. In some embodiments, a CT scan can be performed **7930**, and the previously obtained MRI **7910** can be merged with the CT scan **7935**. During the CT scan, software can automatically register the anatomy relative to the markers **720**, **730** that are mounted on the head holder. In some embodiments, the robot **15** can direct a laser at the skin of the patient **18** to mark flaps **7940**. In some embodiments, the skin of the patient **18** can be prepared and draped **7945**, and scalp flaps can be prepared **7950**. As shown, in some embodiments, the robot **15** can laser drill entry holes **7955**, and the StimLoc can be secured bilaterally **7960** (permanent implant, 2 screws per electrode). In some embodiments, the robot **15** can auto-position a conventional electrode guide adjacent to entry point at a fixed (known) distance from target **7965**. In some embodiments, the dura can be opened, a cannula and electrode inserted, and a StimLoc clip can be positioned **7970**. In some embodiments, steps **7965**, **7970** are repeated for the other side of the patient's skull **7975**. In some embodiments, a verification CT scan is performed **7980**, a cap is placed over the StimLoc, and the flaps are closed.

(525) In some embodiments, the robot system **1** includes at least one mounted camera. For example, FIG. **81** illustrates a perspective view of a robot system including a camera arm in accordance with one embodiment of the invention. In some embodiments, to overcome issues with line of sight, it is possible to mount cameras for tracking the patient **18** and robot **15** on an arm **8210** extending from the robot. As shown in FIG. **81**, in some embodiments, the arm **8210** is coupled to a camera arm **8200** via a joint **8210a**, and the arm **8210** is coupled to the

system **1** via joint **8210b**. In some embodiments, the camera arm **8200** can be positioned above a patient (for example, above a patient **18** lying on a bed or stretcher as shown in FIG. **81**). In this position, in some embodiments, it might be less likely for the surgeon to block the camera when the system **1** is in use (for example, during a surgery and/or patient examination). Further, in some embodiments, the joints **8210a**, **8210b** can be used to sense the current position of the cameras (i.e. the position of the camera arm **8200**). Moreover, in some embodiments, the exact position of the end-effectuator **30** in the camera's coordinate system can be calculated based on monitored counts on each robot axis **66**, **68**, **70**, **64**, and in some embodiments, the cameras **8200** would therefore only have to track markers **720** on the patient **18**.

(526) Some embodiments include an arm **8210** and camera arm **8200** that can fold into a compact configuration for transportation of the robot system **1**. For example, FIG. **82A** illustrates a front-side perspective view of a robot system including a camera arm in a stored position, and FIG. **82B** illustrates a rear-side perspective view of a robot system including a camera arm in a stored position in accordance with one embodiment of the invention.

(527) Some embodiments can include methods for prostate **8330** immobilization with tracking for imaged-guided therapy. In some embodiments, to enable the insertion of a needle (**7405**, **7410**, **7600**, **8110** for example) into the prostate **8330** utilizing 3D image guidance, a 3D scan of the prostate **8330** relative to reference markers **720**, **730** or other tracking system **3417** is needed. However, the prostate **8330** is relatively mobile and can shift with movement of the patient **18**. In some embodiments, it may be possible to immobilize the prostate **8330** while also positioning and securing tracking markers **720** in close proximity to improve tracking and image guidance in the prostate **8330**.

(528) The prostate **8330** is anatomically positioned adjacent to the bladder **8320**, the pubic bone **8310**, and the rectum **8340** (see for example FIG. **83** showing a lateral illustration of a patient lying supine, depicting the normal relative positions of the prostate **8330**, rectum **8340**, bladder **8320**, and pubic bone **8310**). This position facilitates entrapment of the prostate **8330**, especially when it is enlarged, against the bladder **8320** and pubic bone **8310** via anterior displacement applied within the rectum **8340**. In some embodiments, displacement could be applied using a balloon **8410**, a paddle **8420**, or a combination of the two elements. For example, FIG. **84A** shows a lateral illustration of a patient lying supine, showing how inflation of a balloon can cause anterior displacement of the prostate **8330** toward the pubic bone **8310**, and a controllable amount of compression against the pubic bone **8310** in accordance with one embodiment of the invention. Further, FIG. **84B** shows a lateral illustration of a patient lying supine, showing how shifting of a paddle in the rectum **8340** can cause anterior displacement of the prostate **8330** toward the pubic bone **8310**, and a controllable amount of compression against the pubic bone **8310** in accordance with one embodiment of the invention.

(529) In some embodiments, the balloon **8410** has the advantage that it can be inserted into the rectum **8340** uninflated, and then when inflated. In some embodiments, it will displace the wall of the rectum **8340** and prostate **8330** laterally toward the pubic bone **8310**. In some embodiments, a paddle **8420** can cause lateral displacement of the rectal wall and prostate **8330** if a pivot point near the anus is used.

(530) In some embodiments, it is possible to configure a device consisting of a balloon **8410** and paddle **8420** such that fiducials are embedded in the device, with these fiducials being detectable on the 3D medical image (for instance, such as MRI). For example, FIG. **85** shows a sketch of a targeting fixture and immobilization device to be used for tracking the prostate **8330** during image-guided surgical procedures in accordance with one embodiment of the invention. As shown, active tracking markers **720** can be rigidly interconnected to the paddle element **8420** such that these tracking markers **720** protrude from the rectum **8340** and are visible to tracking cameras (for example, **8200**) during the medical procedure. For example, FIG. **86** shows an illustration of the device as illustrated in FIG. **85**, in place in the rectum **8340** with prostate **8330** compressed and immobilized and tracking markers visible protruding caudal to the rectum **8340** in accordance with one embodiment of the invention.

(531) In some embodiments, in addition to applying lateral force from the side of the rectum **8340**, it is also possible to apply lateral force from the side of the abdomen of the patient **18**. In some embodiments, this secondary lateral force, used in conjunction with the force from the rectal wall, may assist in keeping the prostate **8330** immobilized. Additionally, it can serve as a support to which the tracking markers **720** are attached, and can serve as a support to which the rectal paddle/balloon **8420**, **8410** can be attached for better stabilization. In some embodiments, the abdominal support can consist of a piece that presses from anterior toward posterior/inferior to press against the top of the bladder **8320** region. For example, conventional straps or pieces that encircle the legs can provide additional support. Since the abdominal shape and leg shape varies among patients, some customization would be beneficial. In some embodiments, adjustable straps and supports made of thermoplastic material could be utilized for customization. In some embodiments, commercially available thermoplastic supports (for example, from Aquaplast Inc) can be used. In some embodiments, the supports are formed by first dipping the support material in hot water to soften it, then applying the support to the patient's skin and molding it.

After removing the support material from the hot water, the temperature is low enough that it does not burn the skin, but is warm enough that the support material remains soft for 1-5 minutes. In some embodiments, when the support cools, it maintains the skin contours against which it has been formed. In some embodiments, this type of support could be made for immobilizing the prostate **8330** shaped like moldable briefs. In this instance, the support would be dipped in hot water and then external straps and/or manual pressure would be applied to force the support device to press down toward the prostate **8330**. Further, in some embodiments, the support could be manufactured in two halves, formed so that it is molded while two halves are tied together, and then removed (untied) when cool (so that it can later be reattached in the same configuration during the procedure).

(532) In some embodiments, the combination of the elements as described above (including balloon **8410** and/or paddle **8420**, enables real-time tracking of the prostate **8330**, and manual or robotically assisted insertion of needles (for example, **7405**, **7410**, **7600**, **8110**) into the prostate **8330** based on targeting under image guidance. In some embodiments, the procedure can include the conventional abdominal support device as described above. The device would be prepared by dipping in hot water until soft, then applying to the patient such that gentle pressure is maintained from anterior to posterior/inferior against the bladder **8320** region and prostate **8330**. In some embodiments, under palpation, the tracking device (paddle **8420** with coupled fixture **690** including markers **720** illustrated in FIG. **85**) would be inserted into the rectum **8340** with the paddle **8420** and radio-opaque markers **730** adjacent to the prostate **8330**. In this instance, gentle pressure can be manually applied to the protruding handle by the surgeon to maintain the position of the interior paddle **8420**. In some embodiments, the balloon **8410** is inflated to maintain gentle compression against the prostate **8330**, and to immobilize the prostate **8330** against the pubic bone **8310**. In some embodiments, if the conventional abdominal device is used, the abdominal device is interconnected to the rectal device at this point for additional stability. In some embodiments, an MRI is obtained. During the MRI, the active tracking markers **720** are not attached since they are metallic. In some embodiments, sockets or other conventional quick-connect mechanical device are present in the locations where the markers **720** or marker tree (fixture **690**) will later be inserted. In some embodiments, the MRI captures an image of the prostate **8330**, and the radio-opaque markers **730** embedded in the handle **8425**. In some embodiments, the MRI can be captured with the patient's legs down to allow the patient **18** to fit into the gantry of the scanner. In some embodiments, the patient **18** is positioned on the procedure table with legs raised. Tracking markers **730** are snapped into the sockets on the protruding handle or the marker tree **690** with markers **720** is otherwise fastened. In some embodiments, registration of the markers **730**, **720** is achieved by software (for example, using one or more modules of the software **3406** using the computing device **3401**), which automatically detects the positions of the radio-opaque markers **730** on the medical image. In some embodiments, the known relative positions of the active tracking markers **720** and the radio-opaque marker **730** fiducials synchronizes the coordinate systems of the anatomy of the patient **18**, tracking system **3417** and software **3406**, and robot **15**. In some embodiments, the surgeon plans trajectories for needle **7405** insertion into the prostate **8330** on the medical image, and the robot **15** moves the guide tube **50** to the desired 3D location for a needle **7405** of known length to be inserted to the desired depth.

(533) Some embodiments can use a dual mode prostate **8330** tracking for image-guided therapy. For example, in some embodiments, it is possible to accurately track the prostate **8330** using a combination of two tracking modalities, including fiber optic tracking. For this alternate method to be used, an optical tracker (fiber optic probe **8700**) would first be applied externally. This probe **8700** would be registered to the 3D medical image (for example, using an MRI scan) in substantially the same way as previously described, such as for the spine tracking using CT imaging. In some embodiments, after registering and calibrating so that the coordinate systems of the medical image and cameras **8200** are synchronized, a means of updating and correcting for movement of the prostate **8330** can be used. In some embodiments, the probe **8700** can comprise a fiber optic sensor with a Bragg grating. For example, FIG. **87**. illustrates a demonstration of a fibre Bragg grating ("FBG") interrogation technology with a flexible fiber optic cable in accordance with one embodiment of the invention. As shown the technology is available from Technobis Fibre Technologies, Uitgeest, Holland. As the fiber optic cable is bent by hand, the system accurately senses the position to which the cable deforms. As depicted in FIG. **88**, in some embodiments, the probe **8700** could be inserted into the urethra with the tip of the sensor positioned at the prostate **8330**. Since the prostate **8330** surrounds the urethra, a sensor such as probe **8700** positioned in the urethra should show very accurately how the prostate **8330** moves. As shown, the probe **8700** can be coupled with the fixture **690** including markers **720**, **730**, and coupled to the computer **100** with optical tracker electronics **8810** and fiber optic electronics **800** coupled to the computer **100**, coupled to the robot **15**.

(534) In some embodiments, markings **8910** (gradations) capable of being visualized on MRI can be placed on the outer shaft of the probe **8700** (see for example, FIG. **89**). In some embodiments, if the MRI is obtained while the probe **8700** is in position in the urethra, it is possible to determine which point or points along the length of the probe **8700** represent key landmarks within the prostate **8330** (e.g., distal entry, proximal exit, midpoint). In some

embodiments, these points can then be tracked by the fiber optic electronics **8800** during the procedure. In some embodiments, the points can then be used to adjust the coordinate system of the prostate **8330** so that the local coordinate system remains properly synchronized with the coordinate system of the optical tracking system **3417** even if the surrounding anatomy (specifically the anatomy to which the tracking markers **720**, **730** are attached) shifts relative to the prostate **8330**. In other words, the position of the tracking markers **720**, **740** on the patient's skin surface gives an approximate estimate of where the prostate **8330** is currently located, and the fiber optic probe **8700** (which is rigidly interconnected to the tracking fixture **690**) corrects this position to substantially improve accuracy and account for shifting of the prostate **8330**.

(535) In some embodiments, image-guided therapy can be performed using one or more of the embodiments as described. For example, in some embodiments, the fiber optic probe as depicted in FIG. **88** can include optically visible **8900** and MRI visible **8910**. In some embodiments, the probe **8700** is inserted into the penis and advanced until the tip passes into the bladder **8320** (shown in FIG. **89**). In some embodiments, the marking **8900**, **8910** will provide information about what section of the fiber optic is positioned within the prostate **8330**. In some embodiments, the depth of insertion is recorded based on visible markings **8900** on the proximal end that has not entered the penis is recorded. In some embodiments, this information can be used to check whether the probe **8700** has moved, or to reposition the probe **8700** if it is intentionally moved. In some embodiments, the proximal end may be secured (taped) to the penis to prevent advancement or withdrawal with patient **18** movement. In some embodiments, the distal end may have a feature to prevent it from easily sliding back out of the bladder **8320**. For example, as shown in FIG. **90**, some embodiments include a probe **8700** that comprises an inflatable tip **8920**. In some embodiments, the inflatable tip **8920** can be enlarged or flared in the area near the tip. In some embodiments the tip **8920** comprises a balloon that is inflatable after the tip has passed into the bladder **8320**, whereas in other embodiments, the tip **8920** comprises conventional soft wings that deploy after the tip has passed into the bladder **8320**. As shown in FIG. **90**, in some embodiments, a targeting fixture **690** is attached to the patient **18** in the region of the perineum (or abdomen or other suitable surface). The targeting fixture has embedded radio-opaque fiducial markers **730** that will show up on the MRI (or other 3D scan), and is equipped with a conventional quick-connect interface that will later accept an attachment with active tracking markers **720**. These tracking markers **720** do not need to be present yet, especially if they are not MRI compatible. The targeting fixture **690** can be rigidly interconnected with the proximal end of the fiber optic probe **8700**.

(536) In some embodiments, the patient **18** is positioned outside or in the gantry of the MRI scanner before scanning. In some embodiments, the fiber optic tracking system **9100** is briefly activated to record position of the fiber optic probe **8700** along its entire length for later reference (see FIG. **91**). Once recorded, the electronic interface (**8800**) for the fiber optic tracking system **9100** may be disconnected and removed from the MRI area.

(537) In some embodiments, an MRI scan is obtained. The scan must visualize the prostate **8330**, the radio-opaque fiducials **730** on the targeting fixture **690**, and the markings **8910** that are present along the urethral tube that will be tracked with fiber optic probe **8700**. In some embodiments, the position of the prostate **8330** along the fiber optic probe **8700** at the time of the scan is recorded from the radio-opaque markings **8910** on its surface.

(538) In some embodiments, the patient is positioned on the procedure table, and optical tracking markers **720** are snapped into the targeting fixture (see FIG. **92**) and activated. In some embodiments, registration of the markers **720** is achieved by software (for example by one or more modules within the device **3401**), which automatically detects the positions of the radio-opaque markers **730** on the medical image. The known relative positions of the active tracking markers **720** and the radio-opaque fiducials **730** synchronizes the coordinate systems of the anatomy, tracking system **3417**, and robot **15**. The fiber optic tracking system **9100** is activated.

(539) In some embodiments, the offset of the prostate **8330** from the position recorded on the MRI scan is determined as the offset of the prostate **8330** in the optically sensed position of the probe **8700** relative to the position at the time of the MRI scan. In some embodiments, the surgeon plans trajectories for insertion of the needle **7405** into the prostate **8330** (from the medical image), and the robot **15** moves the guide tube **50** to the desired 3D location for a needle **7405** to be inserted to the desired depth (see FIG. **93**). In some embodiments, an offset necessary to ensure that the correct region of the prostate **8330** is targeted, is determined from the probe **8700** sensed offset, and the position of the guide tube **50**.

(540) In some other embodiments, the probe **8700** could be inserted down the esophagus to track movement of the stomach, intestines, or any portion of the digestive system. In some embodiments, it could be inserted into a blood vessel to track the position of major vessels inside the body. In some embodiments, it could be inserted through the urethra into the bladder, ureters, or kidney. In all cases, it would help localize internal points for better targeting for therapy.

(541) In some further embodiments, the probe **8700** could be combined with a conventional catheter for other uses. For example, fluid could be injected or withdrawn through a hollow conventional catheter that is attached along its length to the probe **8700**. Further, in some embodiments, a conventional balloon catheter could also be

utilized. The balloon could be temporarily inflated to secure a portion of the probe **8700** within the urethra, or other position inside the body, ensuring that the probe **8700** does not move forward or backward once positioned where desired.

(542) A number of technologies for real-time 3D visualization of deforming soft tissue and bony anatomy without the radiation are available and/or are in development. In some embodiments, the surgical robot **15** can use these technologies during surgery, or other image-guided therapy. In some embodiments, the use of real-time 3D visualization, automated non-linear path planning and automated steering and advancement of flexible catheters or wires (for example wires **7405**, **7410**, **8600**, or **8110**) in a non-linear path becomes increasingly important.

(543) In some embodiments, it may be possible to visualize soft tissues in real time by combining MRI (magnetic resonance imaging) and ultrasound or contrast enhanced ultrasound (“CEUS”). For example, in some embodiments, an MRI scan and a baseline ultrasound scan would be obtained of the anatomy of interest. In some embodiments, landmarks visualized on the ultrasound would be correlated to the MRI (for example, borders of organs, blood vessels, bone, etc.). In some embodiments, a discrete set of key landmarks could be correlated such that the movement of other points of interest between these landmarks could be interpolated. In some embodiments, a computerized geometric model (with its unmoved baseline position corresponding to the anatomy seen on the MRI) would be created. Then, when movements of the landmark points are detected on ultrasound, the positions of the corresponding tissues visualized on the model can be adjusted. In some embodiments, the ultrasound would be allowed to run continuously, providing real-time data on the positions of the landmarks. In some embodiments, changes in landmark position would be used to update the model in real time, providing an accurate 3D representation of the soft tissues without exposure to radiation. In some embodiments, optical tracking markers **720** attached to the conventional ultrasound probes could provide data on the movement of the probes relative to the anatomy, which would affect the model calibration. In some embodiments, for accurate 3D positions of the points on the soft tissues, it may be necessary to utilize several conventional ultrasound probes locked in a rigid orientation relative to each other. In other embodiments, the ultrasound probes can be synchronized so that their relative positions are known or can be extracted. In some embodiments, optical markers **720** on multiple conventional ultrasound probes would allow registration of the multiple ultrasound probe orientations in the same coordinate system.

(544) In some further embodiments of the invention, other methods for assessing distance to tissues of interest, such as electrical conductivity, capacitance, or inductance of the tissues as mild electrical current is applied.

(545) In the modeling approach described above for visualizing soft tissues, it should be recognized that tracking a large number of landmarks helps ensure that the model is accurate. However, there is a trade-off that tracking a large number of landmarks may slow down the process, and disallow real-time updating or require a lengthy registration process. In some embodiments, as fewer landmarks are tracked, tissue modeling to predict deformation of the non-tracked parts of the model becomes increasingly important. In some embodiments, for tissue modeling, the elasticity and other mechanical qualities of the tissues are needed. It may be possible to assess the status of the tissues through a mechanism such as spectroscopy, where absorbance of light passed through tissue might provide information on the composition of tissues, electrical conductivity, DEXA scan, MRI scan, CT scan or other means. This information could be provided to the computer model to allow better estimation of soft tissue deformation.

(546) Another possible mechanism for visualizing soft tissues can include injecting a conventional liquid tracer into the patient **18** that causes different tissues to become temporarily detectable by an external scan. For example, the tracer could comprise a radioactive isotope that is attracted more to certain types of cells than others. Then, when the patient is placed near an array of conventional radiation sensors, the sensors could detect the concentrations of the isotope in different spatial locations.

(547) Some embodiments include a mechanism to allow the user to control the advancement and direction of a flexible catheter or wire (for example wire **7405**, **7410**, **7600**, or **8110**) through an interface with the robot **15**. In some embodiments, this mechanism can snap or lock into the robot's end-effectuator **30**. In some embodiments, the guide tube **50** on the robot's end-effectuator **30** provides accurately controlled orientation and position of the catheter or wire at the point where it enters the patient. In some embodiments, the mechanism would then allow the user to control the rate and amount of advancement of the tube **50**, the rate and amount of rotation of the tube **50**, and activation of steering RF energy (for example, as described earlier with regard to steerable needle **7600** in FIG. **73**). In some embodiments, based on assumptions about the condition of the soft tissues, and locations of obstacles such as blood vessels, nerves, or organs between entry into the patient and the target, a non-linear path is planned by the software **3406** with parameters under the user's control. For example, in some embodiments, the method can include a command sequence such as “advance 5 mm, activate steering toward an azimuth of +35°, continue advancing 5 mm while rotating at 1° per second,” etc. In some embodiments, during advancement of the catheter or wire **7600** (or other wire **7405**, **7410**, or **8110**), the real-time location of the tip is tracked using LPS or

other visualization means. In some embodiments, the path plan is recalculated based on divergence from the expected path and advancement continues. In some embodiments, this advancing/turning snap-in mechanism can also be used with beveled needles, taking advantage of the direction of the bevel, and the beveled face deflection force that moves the needle laterally away from the face when advanced. In some embodiments, software **3406** would plan which direction the bevel should be oriented during different phases of needle advancement.

(548) In some embodiments, a mechanism similar to the one described above can also be used for automatic hole preparation and insertion of screws. For example, in some embodiments, the end-effectuator **30** could have a conventional mechanism that would allow a tool to be retrieved from a conventional tool repository located somewhere outside the surgical field **17**. In some embodiments, features on the tool holder would allow easy automated engagement and disengagement of the tool. In some embodiments, after retrieving the tool, the end effectuator **30** would move to the planned screw location and drill a pilot hole by rotating the assembly at an optimal drilling speed while advancing. In some embodiments, the system **1** would then guide the robot **15** to replace the drill in the repository, and retrieve a driver with appropriately sized screw. In some embodiments, the screw would then be automatically positioned and inserted. In some embodiments, during insertion of the screw, thrust and torque should be coordinated to provide good bite of the screw into bone. That is, the appropriate amount of forward thrust should be applied during rotation so the screw will not strip the hole.

(549) Some embodiments of the method also include algorithms for automatically positioning conventional screws. For example, in some embodiments, different considerations may dictate the decision of where the screw should be placed. In some embodiments, it may be desirable to place the screw into the bone such that the screw is surrounded by the thickest, strongest bone. In some embodiments, algorithms can be used to locate the best quality bone from CT or DEXA scans, and to find an optimized trajectory such that the width of bone around the screw is thickest, or remains within cortical instead of cancellous bone for the greatest proportion. In some embodiments, it may be desirable to place the screw into the bone at an entry point that is most perpendicular to the screw, or is at a “valley” instead of a peak or slope on the bony articulations. In some embodiments, by placing the screw in this way, it is less likely to skive or rotate during insertion and therefore likely to end up in a more accurate inserted location. In some embodiments, algorithms can be used to assess the surface and find the best entry point to guide the screw to the target, while penetrating the bone perpendicular to the bone surface. In other embodiments, it may be desirable to place screws in a multi-level case such that all the screw heads line up in a straight line or along a predictable curve. In some embodiments, by aligning screw heads in this way, the amount by which the surgeon must bend the interconnecting rod is minimized, reducing the time of the procedure, and reducing weakening of the metal rod due to repeated bending. In some embodiments, algorithms can be used that keep track of anticipated head locations as they are planned, and suggest adjustments to trajectories that provide comparable bony purchase, but better rod alignment.

(550) Some embodiments of the invention can use an LPS system that uses time-of-flight of RF signals from an emitter to an array of receivers to localize the position of the emitter. In some embodiments, it may be possible to improve the accuracy of the LPS system by combining it with other modalities. For example, in some embodiments, it may be possible use a magnetic field, ultrasound scan, laser scan, CT, MRI or other means to assess the density and position of tissues and other media in the region where the RF will travel. Since RF travels at different rates through different media (air, tissue, metal, etc.), knowledge of the spatial orientation of the media through which the RF will travel will improve the accuracy of the time-of-flight calculations.

(551) In some embodiments, an enhancement to the robot **15** could include inserting a conventional ultrasound probe into the guide tube **50**. In some embodiments, the ultrasound probe could be used as the guide tube **50** penetrates through soft tissue to help visualize what is ahead. As the guide tube **50** advances, penetrating soft tissue and approaching bone, the ultrasound probe would be able to detect contours of the bone being approached. In some embodiments, this information could be used as a visual reference to verify that the actual anatomy being approached is the same as the anatomy currently being shown on the 3D re-sliced medical image over which the robot is navigating. For example, in some embodiments, if a small protrusion of bone is being approached dead center on the probe/guide tube **50** as it is pushed forward, the region in the center of the ultrasound field representing the raised bone should show a short distance to bone, while the regions toward the perimeter should show a longer distance to bone. In some embodiments, if the position of the bony articulation on the re-sliced medical image does not appear to be lined up with the 2D ultrasound view of where the probe is approaching, this misalignment could be used to adjust the registration of the robot **15** relative to the medical image. Similarly, in some embodiments, if the distance of the probe tip to bone does not match the distance perceived on the medical image, the registration could also be adjusted. In some embodiments, where the guide tube **50** is approaching something other than bone, this method may also be useful for indicating when relative movement of internal soft tissues, organs, blood vessels, and nerves occurs.

(552) Some embodiments can include a nerve sensing probe. For example, in some embodiments, for sensing

whether a penetrating probe is near a nerve, an electromyography (“EMG”) response to applied current could be used, enabling the ability of the robot **15** to steer around nerves. For example, as shown in FIG. **94**, a probe **9400** could be used, with 1 or more cannulation offset from the probe's **9400** central axis that would enable a thin wire **9410** to extend from the tip **9405**, ahead and to one side of the tip **9405**. A beveled tip **9405** (or a conical or rounded tip) could be used.

(553) In some embodiments, the probe **9400** could be advanced manually or automatically and stopped, then the stimulating wire **9410** could be extended and current applied. In some embodiments, the EMG could be checked to verify whether a nerve is in proximity. In some embodiments, the stimulating wire **9410** could be retracted, and probe **9400** rotated so that the portal for the stimulating wire **9410** is positioned at a different azimuth index. In some embodiments, the probe **9400** could again be extended to check for the presence of nerves in a different region ahead. In some embodiments, if a nerve is encountered, it would be known which direction the nerve is located, and which direction the probe **9400** would need to be steered to avoid it. In some embodiments, instead of a single wire **9410** extending and checking for a nerve, multiple wires **9410** could simultaneously be extended from several portals around the probe **9400**. In some embodiments, the wires **9410** could be activated in sequence, checking for EMG signals and identifying which wire **9410** caused a response to identify the direction to avoid or steer. In some embodiments, it could be necessary to fully retract the stimulating wires **9410** before attempting to further advance the probe **9400** to avoid blocking progress of the probe **9400**. In some embodiments, the stimulating wires **9410** would have a small enough diameter so as to be able to penetrate a nerve without causing nerve damage.

(554) As noted elsewhere in this application, the robot **15** executed trajectories for paths into a patient **18** are planned using software (for example, at least one module of the software **3406** running on the computing device **3401** including computer **100**) where the desired vectors are defined relative to radio opaque markers **730** on the image and therefore relative to active markers **720** on the targeting fixture **690**. In some embodiments, these trajectories can be planned at any time after the image is acquired, before or after registration is performed. In some embodiments, it is possible that this trajectory planning can be done on another computerized device. For example, in some embodiments, a conventional portable device (such as a tablet computer, or a laptop computer, or a smartphone computer) could be used. In some embodiments, the 3D image volume would be transferred to the portable device, and the user would then plan and save the desired trajectories. In some embodiments, when robotic control is needed, this same image volume could be loaded on the console that controls the robot **15** and the trajectory plan could be transferred from the portable device. In some embodiments, using this algorithm, it would therefore be possible for a series of patients **18** to each to have a targeting fixture **690** applied and an imaging scan, such as a CT scan. In some embodiments, the 3D volume for each patient **18** could be exported to different portable devices, and the same or different surgeons could plan trajectories for each patient **18**. In some embodiments, the same or different robot **15** could then move from room to room. In some embodiments, in each room, the robot **15** would be sterilized (or have sterile draping applied, and would receive the scan and trajectory plan. The robot **15** would then execute the plan, and then move to the next room to repeat the process. Similarly, the portion of the registration process in which the 3D image volume is searched for radio-opaque markers **730** could be performed on the portable device. Then, in some embodiments, when the robot **15** arrives, the registration information and the trajectories are both transferred to the robot **15** console. In some embodiments, by following this procedure, the time of computation of the image search algorithm on the robot **15** console is eliminated, increasing efficiency of the overall process when the robot **15** is required in multiple rooms.

(555) Although several embodiments of the invention have been disclosed in the foregoing specification, it is understood that many modifications and other embodiments of the invention will come to mind to which the invention pertains, having the benefit of the teaching presented in the foregoing description and associated drawings. It is thus understood that the invention is not limited to the specific embodiments disclosed hereinabove, and that many modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although specific terms are employed herein, as well as in the claims which follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the described invention, nor the claims which follow.

(556) It will be appreciated by those skilled in the art that while the invention has been described above in connection with particular embodiments and examples, the invention is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the invention are set forth in the following claims.

Claims

1. A medical robot system, comprising: an end effector including a guide tube, the guide tube being electrically insulated and having at its distal portion at least one electrode; a robot coupled to the end effector, the robot configured for controlled movement and positioning; a position tracker attached to a surgical instrument held by the end effector; a motor assembly coupled to the robot, the motor assembly being configured to move the end effector along an x-axis, a y-axis, and a z-axis; a tracking receiver configured to receive tracking signals from the position tracker attached to the surgical instrument; and a control unit coupled to the motor assembly and the tracking receiver, the control unit configured to supply instruction signals to the motor assembly, the instruction signals configured to cause the motor assembly to selectively move the end effector along the x-axis, y-axis, and z-axis, the control unit being further configured to: (i) calculate a position of the position tracker by analysis of the tracking signals received by the tracking receiver; (ii) display the position of the position tracker with respect to a patient; and (iii) selectively control actuation of the motor assembly in response to the signals received by the tracking receiver.
2. The system of claim 1, wherein the at least one electrode includes a plurality of electrodes circumferentially spaced around an exterior surface of the guide tube.
3. The system of claim 2, wherein the control unit is configured to automatically sequentially activate the electrodes for directional detection of nerves.
4. The system of claim 1, wherein the at least one electrode includes only a portion of a leading edge of the guide tube for directional detection of nerves in a radial direction.
5. The system of claim 1, wherein the end effector comprises a tube having a pitch axis, a roll axis and a tube axis defining an axis of rotation of the tube.
6. The system of claim 1, wherein the end effector comprises a needle.
7. The system of claim 1, wherein the control unit is adapted to receive information from the tracking receiver in an iterative fashion.
8. The system of claim 1, wherein the control unit is adapted to receive information from the tracking receiver in a dynamic fashion.
9. The system of claim 1, wherein the control unit receives a user selection of the desired location within the patient's body and is adapted to dynamically provide information that allows the motor assembly to move the end effector to the desired location.
10. The system of claim 1, wherein the position tracker is operatively coupled to the control unit, and wherein the control unit is configured to cause the position tracker to emit signals at one or more selected times.
11. The system of claim 1, wherein the end effector has an outer surface, and wherein the position tracker includes a plurality of transmitters radially spaced about an outer surface of the end effector.
12. A medical robot system, comprising: an end effector including a guide tube, the guide tube being electrically insulated and having at its distal portion at least one electrode; a robot coupled to the end effector, the robot configured for controlled movement and positioning; a position tracker coupled to the end effector; a motor assembly coupled to the robot, the motor assembly being configured to move the end effector along an x-axis, a y-axis, and a z-axis; a tracking receiver configured to receive tracking signals from the position tracker; and a control unit operatively coupled to the motor assembly and the tracking receiver, the control unit configured to operatively control the motor assembly to selectively move the end effector along the x-axis, y-axis, and z-axis, the control unit being further configured to: (i) calculate a position of the position tracker by analysis of the signals received by the tracking receiver; (ii) display the position of the position tracker with respect to the patient; (iii) receive a desired trajectory of the end effector or an instrument coupled thereto; and (iv) selectively control actuation of the motor assembly based upon the received desired trajectory and in response to the signals received by the tracking receiver.
13. The system of claim 12, wherein the at least one electrode includes a plurality of electrodes circumferentially spaced around an exterior surface of the guide tube.
14. The system of claim 13, wherein the control unit is configured to automatically sequentially activate the electrodes for directional detection of nerves.
15. The system of claim 12, wherein the at least one electrode includes only a portion of a leading edge of the guide tube for directional detection of nerves in a radial direction.
16. The system of claim 12, wherein the control unit is adapted to receive information from the tracking receiver in an iterative fashion.
17. The system of claim 12, wherein the control unit is adapted to receive information from the tracking receiver in a dynamic fashion.

18. The system of claim 12, wherein the control unit receives a user selection of the desired location within the patient's body and is adapted to dynamically provide information that allows the motor assembly to move the end effector to the desired location.

19. The system of claim 12, wherein the position tracker is operatively coupled to the control unit, and wherein the control unit is configured to cause the position tracker to emit signals at one or more selected times.

20. The system of claim 12, wherein the end effector has an outer surface, and wherein the position tracker includes a plurality of transmitters radially spaced about an outer surface of the end effector.
